

Classical Mechanics and Dynamical Systems

Lectures 13–29 — Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

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1 Lecture 13: Dynamical Symmetry – 1

Understanding the relation of symmetry $\Leftrightarrow$ invariance $\Rightarrow$ conservation laws. First step in this regard was taken by Emmy Noether.

(conserved quantities $\vec{P}$, $\vec{L}$, $E(H) \dots$)

Noether's theorem :-

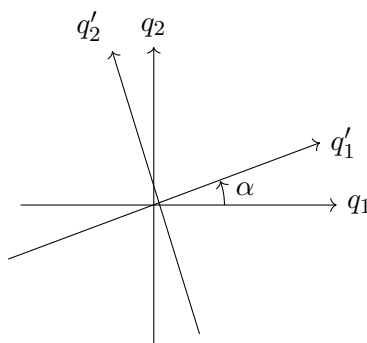
Let us take a case of a particle moving in 2-D :-

$$H(q_1, q_2, p_1, p_2) = \frac{\vec{p}^2}{2m} + V(q_1, q_2)$$

$$p^2 = p_1^2 + p_2^2$$

If the potential has spherical symmetry (in 2-D its rotational) then we know that H is C.O.M.

$$\text{if } V(q_1, q_2) = f(r), \quad r = \sqrt{q_1^2 + q_2^2}$$



In rotated frame of reference what remains constant :-

- 1) H
- 2) EOM don't change
- 3) Solution space/set do not change

Q) Does solution remains same/const?

Lets take simplest case

$$x^2 = 4$$

$$x^2 - 4 = 0$$

$$(x - 2)(x + 2) = 0$$

$$x = 2, -2 \quad \{-2, 2\}$$

after rota transformation

$$x'^2 = 4 \quad x' = -x$$

$$(x' - 2)(x' + 2) = 0$$

$$x' = -2, +2 \quad \{-2, 2\}$$

since $x' = -x$

we deduce that the +2 root changed to -2 and $-2 \rightarrow +2$,

but set of solution remains constant/same

even though the elements of set of solution can convert into one another.

So the defⁿ of dynamical symmetry becomes,

* Dynamical symmetry :-

The dynamical symmetry of a given system is a set of transformations of phase space variables, such that EOM don't change and as a consequence, the set of solutions do not change.

(1) Understanding Dynamical symmetry in Lagrangian formalism :-

$$\begin{array}{ccc}
 L(q, \dot{q}) & \longrightarrow & L'(Q(\alpha), \dot{Q}(\alpha)) \quad \alpha = \text{transformation parameter} \\
 \uparrow & & \downarrow \\
 \text{(Autonomous dyn. system)} & & L' \quad (L' = L \text{ if } \alpha = 0 \text{ "by def"}^n) \\
 & & Q(0) = q
 \end{array}$$

EOM do not change if $L' = L$

or, Invariance $\Rightarrow L' = L$

$$\begin{aligned}
 \frac{dL(Q(\alpha), \dot{Q}(\alpha))}{d\alpha} &= 0 \\
 \Rightarrow \frac{\partial L}{\partial Q} \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d\dot{Q}}{d\alpha} &= 0 \\
 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}} \right) \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d}{d\alpha} \left(\frac{dQ}{dt} \right) &= 0
 \end{aligned}$$

α, t are independent so

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}} \right) \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d}{dt} \left(\frac{dQ}{d\alpha} \right) &= 0 \\
 \Rightarrow \frac{d}{dt} \left(\left(\frac{\partial L}{\partial \dot{Q}} \right) \cdot \left(\frac{dQ}{d\alpha} \right) \right) &= 0 \\
 \Rightarrow \frac{\partial L}{\partial \dot{Q}} \left(\frac{dQ}{d\alpha} \right) &= \text{const of motion}
 \end{aligned}$$

in our frame

$$\frac{\partial L}{\partial \dot{Q}} \left(\frac{dQ}{d\alpha} \right) \equiv \frac{\partial L}{\partial \dot{q}} \left(\frac{dQ}{d\alpha} \right) \Big|_{\alpha=0} = p \frac{dQ}{d\alpha} \Big|_{\alpha=0} = \underline{\text{COM}}$$

Remember, that Lagrangian (L) is not unique, we can add total time derivative of some fun. of q, t , therefore it is not necessary that

$$\text{Invariance} \Rightarrow \underline{L' = L}$$

we can have $L' \neq L$ still EOM don't change.

Note :- The crucial point of Noether's theorem is that set of transformations should be continuous, that's why we differentiated with 'α', and it must be connected with Identity (original $(q, \dot{q})$.) In other words if we have original coordinates and then we make a transformation continuously from no transformation at all.

Parity is not continuous transformation

Continuous Transformation

- Rotation
- shift in origin
- shift in origin in time
- shear, scale
- gauge transformations (vary χ continuously)

↓
(gauge invariance $\Rightarrow$ leads to conservation of charge).

Symmetry :- If Lagrangian is invariant under continuous set of transformations then Noether's theorem tells us $\exists$ a conserved quantity for each transformation parameter.

eg: In shift of origin we $(x, y, z) \longrightarrow (x + \alpha, y + \beta, z + \gamma)$

we have 3 parameters of transformation i.e. α, β, γ

So according to Noethers theorem we will have 3 conserved quantity.

(2) Understanding in Hamiltonian Framework :-

$$H(q, p) : \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad \boxed{\dot{X}_{\sim} = J \nabla H}$$

$\uparrow$ (set of EOM)

$$\exists \quad \left(\begin{array}{l} \{q_i, q_j\} = 0 = \{p_i, p_j\} \\ \{q_i, p_j\} = \delta_{ij} \end{array} \right)$$

Q.) What set of transformations leaves Hamiltons EOM unchanged

Ans) canonical transformations.

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i} \quad \boxed{\dot{\xi}_{\sim} = J \nabla_{\xi} K}$$

$\uparrow$ (EOM)

also

$$\begin{aligned} \{Q_i, P_j\} &= \delta_{ij} \\ \{Q_i, Q_j\} &= 0 = \{P_i, P_j\} \end{aligned}$$

to see set of solutions remain unchanged, we need only those canonical transformations for which $\underline{K = H}$, else it won't be a symmetry. (How?)

Suppose $X_{\sim} = (q, p) \xrightarrow{\text{C.T.}} \xi_{\sim} = (Q, P)$

where $\underline{X}, \underline{\xi}$ are set of variables.

$$\left. \begin{aligned} \{Q_i, P_j\} &= \delta_{ij} = \sum_{k=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k} \frac{\partial P_j}{\partial q_k} \right) \\ \{Q_i, Q_j\} &= 0 = \sum_{k=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial Q_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k} \frac{\partial Q_j}{\partial q_k} \right) \\ \{P_i, P_j\} &= 0 = \sum_{k=1}^n \left(\frac{\partial P_i}{\partial q_k} \frac{\partial P_j}{\partial p_k} - \frac{\partial P_i}{\partial p_k} \frac{\partial P_j}{\partial q_k} \right) \end{aligned} \right\} \Rightarrow \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right)^T J \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right) = J$$

all these relations must be valid to qualify for transformation to be canonical.

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix} \quad I_n, O_n \text{ are } n \times n \text{ matrices.}$$

any $\{A, B\}$ can be written as, $\{A, B\} = (\nabla A)^T J (\nabla B) \quad (M_{p \times q})^T J_{m \times m} M_{p \times q}, q = 2n, p = 2n$

If a matrix (M) follows

$$\text{if, } M^T J M = J,$$

then M is called symplectic matrix

Recall,

$$J^2 = -I, \quad J^T = J^{-1} = -J$$

so symplectic matrix have inverse so canonical transformations Matrix is invertible. $M = \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right) = \text{non-singular } \underline{|M|} \neq 0$

$$\Updownarrow$$

we know, canonical transformation must be inverse transformations,

also, Product of two symplectic matrix is also a symplectic matrix. (as set of symplectic matrices form a group).

$$M_1.M_2 = M_3 \quad M_1, M_2, M_3 \rightarrow \text{symplectic matrices.}$$

physically it implies that two canonical transformations applied serielly are equivalent to a single canonical transformation.

Note:- The inverse, composition law suggests set of canonical transformation for n-d.o.f Hamiltonian system form a group. It's the group of $2n \times 2n$ symplectic matrices. Called

$$Sp(2n, \mathbb{R}).$$

$\hookrightarrow$ Group of C.T.

Note: J is also a symplectic matrix

$$M^T J M = J$$

$$\text{LHS, } \Rightarrow J^T J J \Rightarrow (-J) J^2 = (-J)(-I) = J = \underline{\text{RHS}}$$

Q) what is C.T. corresponding Symplectic matrix 'J' ?

$$\left. \begin{matrix} Q = -p \\ P = q \end{matrix} \right\} \Leftrightarrow (M = J)$$

Note:- We know that group of orthogonal matrices in 3-dimensions have three parameters, (for rotation)

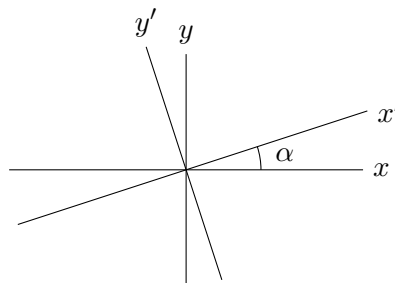
$$SO(n)$$

$\hookrightarrow$ Group of orthogonal matrices in n-dimension with determinant (1)

(Digression:)

Q) How many parameters for n-D ?

Ans: (a) rotation in 2-D about point (origin).



$$SO(2); \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$\downarrow$

orthogonal.

set of transformation is homogeneous (origin unchanged).

2-D rotation is (linear, Homogeneous)

$$(|SO(2)| = \pm 1, \text{ length unchanged.})$$

Defining rotation :-

(for n-dimensions)

- linear
- Homogeneous
- distances unchanged $\Rightarrow$ Matrice should be orthogonal
- $|SO(n)| = \pm 1$ (unimodular)

Note:- In 3-D if we rotate by a single axis, it is not termed as rotation. as from defⁿ we have not defined rotation as rotation about a single axis. For Rotation T should be linear, Homogeneous, orthogonal, and unimodular.

Note:- If the transformation is homogeneous (origin – fixed) to achieve distances unchanged the transformation matrix should be orthogonal. and if we want orientation to be unchanged we

need it to be unimodular. $|M| = 1$. So as we know $\det(\text{orthogonal } m) = \pm 1$. we choose $\underline{+1}$ in rotation. for reflection $\underline{|M| = -1}$.

In 3-D we can do rotation in a plane, where two variables are affected at a time. For two variables affected at a time; no. of rotations are nC_2 . for n-dimension.

$$\text{no. of rotation} \Rightarrow {}^nC_2 = \frac{n(n-1)}{2}$$

$$\text{for } n = 3, \quad \text{no. of rotations} = \frac{3 \cdot 2}{2} = 3$$

So we think (mistakenly) that rotation is occurring about a axis, but really we mean is rotation in xz plane, yz plane, xy plane.

$$\text{Hence, only for 3-D } \left(\frac{n(n-1)}{2} = n \right)$$

So $SO(n)$ has nC_2 generators.

In exactly the same way,

Now we can ask how many generators does this symplectic group $Sp(2n, R)$ have? (or) How many possible parameters we need to specify all possible canonical transformations.

let us make Infinitesimal transformation.

$$M = I + \epsilon G_{(2n \times 2n)} \longrightarrow \text{Generator of Transformation matrix} \\ \longleftrightarrow \text{parameter}$$

$$\begin{aligned} M^T J M &= J \\ \Rightarrow (I + \epsilon G)^T J (I + \epsilon G) &= J \\ \Rightarrow (I + \epsilon G^T) J (I + \epsilon G) &= J \\ \Rightarrow (J + \epsilon G^T J) (I + \epsilon G) &= J \\ \Rightarrow J + \epsilon G^T J + \epsilon J G + \epsilon^2 G^T J G &= J \quad \text{dropping } \underline{\underline{\epsilon^2}} \\ \Rightarrow J + \epsilon G^T J + \epsilon J G &= J \\ \Rightarrow \boxed{G^T J = -J G} \\ \Rightarrow G^T &= -J G J^{-1} \quad (J^{-1} = -J) \\ \Rightarrow \boxed{G^T = J G J} \end{aligned}$$

So Generator of symplectic transformation must satisfy above equation.

so symplectic matrices are generated by these matrices which satisfies $G^T = J G J$

exercise $\rightarrow$ Show that $Sp(2n)$ has $2(2n+1)$ generators.

what are generators? The way we look at it can be interpreted by looking at infinitesimal transformation.

$$\begin{aligned} SO(2) &= \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \xrightarrow{(\alpha \rightarrow \delta \alpha)} \begin{pmatrix} 1 & \delta \alpha \\ -\delta \alpha & 1 \end{pmatrix} \\ &= I + \begin{pmatrix} 0 & \delta \alpha \\ -\delta \alpha & 0 \end{pmatrix} \\ &= I + \delta \alpha \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{aligned}$$

$\hookrightarrow$ Generator.

$\hookrightarrow$ parameter

for $SO(3)$ we will have such 3 generators.

and for $Sp(2n)$ we will have $2(2n + 1)$ generators.

Note:- we actually like to write generators as Hermitian matrix. because we need most of time to exponentiate hermitian matrix. as, e^{iH} = unitary matrix.

Eg 2-D oscillator :-

$$H = \frac{1}{2} (q_1^2 + p_1^2 + w^2 q_2^2 + p_2^2)$$

quasi periodic ($w = \text{irrational}$ [?])

periodic but complicated ($w = \text{rational}$ [?])

$\downarrow$

($w = \text{rational no}$)

if ($w = 1$)

$$H = \frac{1}{2} (q_1^2 + p_1^2 + q_2^2 + p_2^2)$$

isotropic (same k in all directio [?])

$\hookrightarrow$ spring constant [?]

H is invariant under $SO(4)$.

$\hookrightarrow$ set of rotations in 4-D phase space.

Note:- we could have dynamical symmetry arising as a consequence of transformations not just from of potentials (not just of physical coordinate) but it could also involve the momenta (phase-space).

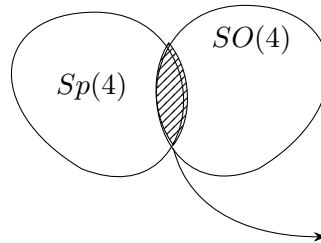
$\rightarrow$ no. of generators $SO(4) = {}^4C_2 = 6$ -(1)

$\rightarrow$ set of canonical transformations ($n = 2$) $\rightarrow Sp(2 \times 2)$

$Sp(4)$, has $2(4 + 1) = 10$ generators. -(2)

From (1) and (2) we deduce that all transformations which leaves hamiltonian unchanged need not be canonical transformations, similarly all C.T need not leave 'H' unchanged.

For symmetry we need transformation to be canonical and which leaves 'H' unchanged.
($K = H$)



Dynamical symmetry group is

$$Sp(4) \cap SO(4). \sim SU(2)$$

$\hookrightarrow$ Isomorphic

$SU(2)$ set of 2×2 matrices which are unitary and unimodular ($|M| = 1$)

$$U \text{ is unitary} \Rightarrow \boxed{U^\dagger U = I}$$

Q. Is it true $U^\dagger U = I = U U^\dagger$

Ans, Only true for finite dimensional matrices (as left inverse becomes equal to right inverse). But for infinitesimal matrices $U^\dagger U = I \not\Rightarrow U U^\dagger$, But we are concerned with finite-dimensional matrices, so it does not matter.

$$\begin{aligned} U^\dagger U = I &\Rightarrow \text{partial isometry} \\ U U^\dagger = I &\Rightarrow \quad \quad \quad \text{“} \quad \quad \text{”} \end{aligned}$$

verify that following J_1, J_2 are c.o.m are true.

$$\begin{aligned} J_2 &= \frac{1}{2} (q_1 q_2 + p_1 p_2) & J_1 &= \frac{1}{4} (q_1^2 + p_1^2 - q_2^2 - p_2^2) \\ J_3 &= \frac{1}{2} (q_1 p_2 - q_2 p_1) \end{aligned}$$

verify, $(\{J_i, H\} = 0, \quad \{J_i, J_j\} = \epsilon_{ijk} J_k)$.

2 Lecture 14: (Finding dyn. symmetry group $SU(n)$)

Recall 2-D isotropic oscillator :-

$$\begin{pmatrix} m = 1 \\ w = 1 \end{pmatrix}$$

$$H = \frac{1}{2} (q_1^2 + p_1^2 + q_2^2 + p_2^2)$$

$$\begin{aligned} F_1 &= H_1 & (q_1^2 + p_1^2) \\ F_2 &= H_2 & (q_2^2 + p_2^2) \end{aligned}$$

its 2-dof and we have 2 c.o.M which are in involution with each other ($\{F_1, F_2\} = 0$. (Integrable)

Note:-

The dynamical symmetry will be that set of transformations of 4 phase space variables which leaves the H, EOM unchanged and hence preserves the solution set.

$Sp(2n) = Sp(4)$ = Group of canonical Transformations for this H.

$SO(4) = SO(4)$ = Group of rotations which ‘leaves’ ‘H’ unchanged

$SU(n) = SU(2)$ = Group of C.T. which leaves ‘H’ unchanged.

(also called as the dyn. symmetry of this H).

Let’s focus on $SU(2)$. :-

$SU(2)$ consists of all 2×2 matrices, $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, such that

($\Rightarrow$ implies)

$$\begin{aligned} M^\dagger M &= I \\ \det(M) &= +1 \end{aligned}$$

There are $4 \times 2 = 8$ parameters as each entry can be complex no.

$$\begin{aligned} a &= a_1 + ia_2 \\ b &= b_1 + ib_2 \\ c &= c_1 + ic_2 \\ d &= d_1 + id_2 \end{aligned} \quad (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2)$$

$\hookrightarrow 8$ parameters.

using, $M^\dagger M = I$ or $M^{-1} = M^\dagger$

$\rightarrow$ we will have 4 conditions (one for each element).

Therefore no. of parameters are reduced to 4.

after imposing $|M| = +1$, one more parameter is lost— so total of three parameters are left.

(Digression) :-

$$M^\dagger = M^{-1} \quad M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

$$M^\dagger = \begin{pmatrix} \alpha^* & \gamma^* \\ \beta^* & \delta^* \end{pmatrix} \quad \text{--- (1)}$$

$$M^{-1} = \frac{1}{|M|} \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix}^T = \frac{1}{(|M|)} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} \quad \text{--- (2)}$$

equating (1) and (2) and $|M| = \alpha\delta - \beta\gamma = 1$

$$\alpha^* = \delta \quad \text{--- (3)}$$

$$\beta^* = -\gamma \quad \text{--- (4)}$$

$$\gamma^* = -\beta \quad \text{--- (5)}$$

$$\delta^* = \alpha \quad \text{--- (6)}$$

$$\alpha^* = \delta, \delta^* = \alpha \Rightarrow \begin{aligned} (\alpha_1 + i\alpha_2)^* &= (\delta_1 + i\delta_2) \Rightarrow \alpha_1 = \delta_1 \\ (\delta_1 + i\delta_2)^* &= \alpha_1 + i\alpha_2 \Rightarrow \alpha_2 = -\delta_2 \end{aligned}$$

so M becomes

$$M = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \quad \begin{aligned} |M| &= 1 \\ \alpha\delta - \beta\gamma &= 1 \\ \alpha\alpha^* + \beta\beta^* &= 1 \\ |\alpha|^2 + |\beta|^2 &= 1 \end{aligned}$$

Hence, any (2×2) unimodular, unitary matrix can be written is

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} \quad \text{with,} \quad (|\alpha|^2 + |\beta|^2 = 1) \quad \begin{cases} \alpha = \alpha_1 + i\alpha_2 \\ \beta = \beta_1 + i\beta_2 \end{cases}$$

The Generators of $SU(2)$ are c.o.M which are not in involution with each other. and turns out these are J_1, J_2, J_3

$$\left. \begin{aligned} J_1 &= \frac{1}{4} (q_1^2 + p_1^2 - q_2^2 - p_2^2) \\ J_2 &= \frac{1}{2} (q_1 q_2 + p_1 p_2) \\ J_3 &= \frac{1}{2} (q_1 p_2 - p_1 q_2) \end{aligned} \right\} \begin{aligned} \{J_1, J_2\} &= J_3 \\ \{J_2, J_3\} &= J_1 \\ \{J_3, J_1\} &= J_2 \end{aligned} \quad \text{(or)} \quad \begin{cases} \{J_i, J_j\} \\ = \epsilon_{ijk} J_k \end{cases}$$

(so 3 parameters require 3 generators)

Q) How many independent parameters in $(n \times n)$ unitary matrix

Ans, $2n^2 - n^2 = n^2$

So, for $SU(n)$ we have $(n^2 - 1)$ independent parameters where 1 parameter got reduced from $|M| = \pm 1$ condition.

Therefore, i.e. ($SU(3)$ has $3^2 - 1 = 8$ generators)

For 3-D Isotropic oscillator :-

$Sp(2n)$. $Sp(6)$ is group of C.T.

$SO(N)$ $SO(6)$ is group of symmetry transformations

dimension of phasespace

$SU(8)$ is one required dynamical symmetry group.

The Kepler problem :-

$$H(\vec{r}, \vec{p}) = \frac{\vec{p}^2}{2m} + \left(-\frac{k}{r}\right) \quad \dots \quad \left(\begin{array}{l} \text{assuming interaction} \\ \text{is attractive.} \end{array} \right)$$

Note:-

$$V(r) = \begin{cases} -k/r \\ r^2 \end{cases} \quad \text{are the only potentials for which bounded orbits are closed curves.}$$

↓

$$\left(\begin{array}{l} \text{periodic} \\ \text{orbits} \end{array} \right).$$

$$V(r) = r^2 \quad \text{3-D oscillator, with } SU(3)$$

Q) what is symmetry group of Kepler's hamiltonian?

C.O.M :- $H, \vec{L}^2, \vec{L} \cdot \hat{n} \rightarrow 3$ independent c.o.m which are in involution with each other.

$$\vec{L} \text{ is c.o.m but } \{L_i, L_j\} = \epsilon_{ijk} L_k \quad \left(\begin{array}{l} \text{not in involution} \\ \text{with each other} \end{array} \right).$$

Note

Any central potential problem will have these c.o.m but there is something special with kepler problem as there exist further C.O.M ($\vec{A}$).

$$\vec{A} = \vec{p} \times \vec{L} - \frac{mk\vec{r}}{r} \quad \left(\begin{array}{l} A = f(L) \text{ so its not} \\ \text{independent} \end{array} \right)$$

(laplace- Runge -lenz vector)

(verifying $\vec{A}$ is C.O.M) :-

($\vec{L}$ is c.o.m)

$$\frac{d\vec{A}}{dt} = \frac{d\vec{p}}{dt} \times \vec{L} + \vec{p} \times \frac{d\vec{L}}{dt} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{mk}{r} \frac{d\vec{r}}{dt}$$

$$\begin{aligned}
&\text{for } V = -k/r \\
&F = -\nabla V \\
&= -\nabla(-k/r) \\
&= \nabla(k/r) \\
&= -k/r^2 \hat{e}_r \\
&\frac{d\vec{p}}{dt} = F = \frac{-k}{r^3} \vec{r}
\end{aligned}$$

$$\begin{aligned}
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} \vec{r} \times (\vec{r} \times \vec{p}) + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{mk}{r} \frac{d\vec{r}}{dt} \\
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} (\vec{r} \times (\vec{r} \times \vec{p})) + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \quad \left(\vec{p} = m \frac{d\vec{r}}{dt} \right)
\end{aligned}$$

$$\begin{aligned}
\bar{a} \times (\bar{b} \times \bar{c}) &= \text{vector in plane of } \bar{b}, \bar{c}. \\
&= \alpha \vec{b} + \beta \vec{c}
\end{aligned}$$

LHS is linear, α, β are scalars formed out of a, b, c as LHS is linear RHS must be linear,

$$\begin{aligned}
&\alpha = \lambda(\vec{a} \cdot \vec{c}) \\
\Rightarrow &\beta = \mu(\vec{a} \cdot \vec{b})
\end{aligned}$$

‘ λ & μ ’ should be universal, it can’t depend on $\vec{A}, \vec{B}, \vec{C}$.

$$\bar{a} \times (\bar{b} \times \bar{c}) = \lambda(\bar{a} \cdot \bar{c}) \vec{b} + \mu(\bar{a} \cdot \bar{b}) \vec{c}$$

interchang b, c .

$$\vec{a} \times (\vec{c} \times \vec{b}) = -\bar{a} \times (\bar{b} \times \bar{c})$$

only way it can happen is, $\mu = -\lambda$

$$\bar{a} \times (\bar{b} \times \bar{c}) = \lambda \left((\bar{a} \cdot \bar{c}) \vec{b} - (\bar{a} \cdot \bar{b}) \vec{c} \right)$$

turns out $\lambda = 1$ using special case $\hat{i} \times (\hat{i} \times \hat{j}) = \hat{i} \times \hat{k} = -\hat{j}$

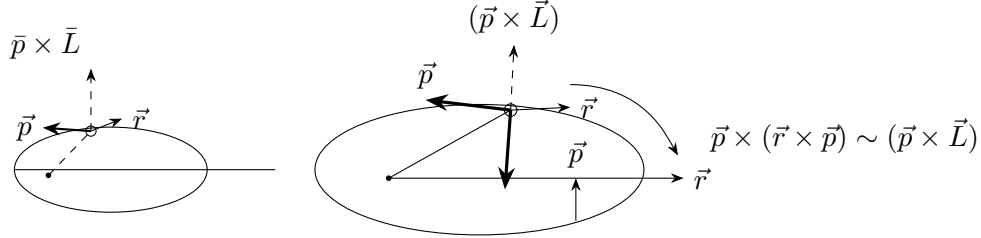
$$\lambda \{ (\hat{i} \cdot \hat{j}) \hat{i} - \hat{j}(\hat{i} \cdot \hat{i}) \} = -\hat{j} \Rightarrow \boxed{\lambda = 1}$$

$$\begin{aligned}
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} [(\vec{r} \cdot \vec{p}) \vec{r} - r^2 \vec{p}] + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \\
&= \frac{-k}{r^3} (\vec{r} \cdot \vec{p}) \vec{r} + \frac{k\vec{p}}{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \\
&= \frac{-k(m)}{r^3} \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) \vec{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} \quad \left(\vec{u} \cdot \frac{d\vec{u}}{dt} = u \frac{du}{dt} \right) \\
&= \frac{-mk}{r^3} r \frac{dr}{dt} \vec{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r}
\end{aligned}$$

$$\frac{d\vec{A}}{dt} = 0 \quad \text{therefore } \vec{A} \text{ is C.O.M.}$$

$\vec{A}$ is c.o.m therefore it remains unchanged in direction and magnitude.

$$\vec{A} = \vec{p} \times (\vec{r} \times \vec{p}) - \frac{mk\vec{r}}{r}$$



(dir. of $\vec{A}$ lies along semi major axis.)

The $\frac{d\vec{A}}{dt} = 0$, implies orbit does not show precession.

$$\left[\begin{array}{l} \text{Binary stars precess } \simeq 4^\circ \text{ per orbit} \\ \text{(Precession of mercury } \simeq 500'' \text{ per century)} \end{array} \right]$$

So in total we got $H, \vec{L}, \vec{A}$ total of $(1+3+3) = 7$ C.O.M but phase space is only 6-dimensional. It turns out that they are not independent of each other. $\left(\begin{array}{l} \vec{A}^2 \text{ is expressed in terms of } H \text{ \& } \vec{L}^2 \\ \vec{A} \cdot \vec{L} = 0. \end{array} \right.$

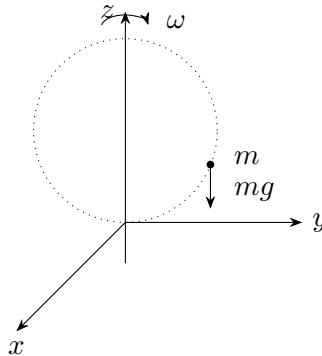
Note:- $\frac{d\vec{A}}{dt} = 0$ independent of sign of 'k' so $\exists \vec{A}$ for repulsive forces also.

* The dynamical symmetry group is $SU(4)$. instead of $SU(3)$ This extra symmetry is seen in terms of degeneracy in quantum mechanics.

This extra symmetry is because of $\vec{A}$.

Problem 1)

Beed on a rotating hoop of wire.



$$V = \begin{cases} 0 & \text{at } z = 0 \\ mgz & z = z \end{cases}$$

$$L = T - V$$

$$L = \frac{1}{2}m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

constraint $\longrightarrow$

$$y^2 + (z - R)^2 = R^2$$

The moment we provided $\vec{w}$ to hoop we need cylindrical symmetry. (s, ϕ, z)

$$\begin{aligned} x &= s \cos \phi, & \dot{x} &= \dot{s} \cos \phi - s \sin \phi \dot{\phi} \\ y &= s \sin \phi, & \dot{y} &= \dot{s} \sin \phi + s \cos \phi \dot{\phi} \\ z &= z, & \dot{z} &= \dot{z} \end{aligned}$$

so L becomes;

$$L = \frac{1}{2}m \left[\dot{s}^2 + s^2 \dot{\phi}^2 + \dot{z}^2 \right] - mgz \quad \text{--- (1)}$$

constraint becomes

$$\begin{aligned} (x^2 + y^2) + (z - R)^2 &= R^2 \\ \Rightarrow s^2 + (z - R)^2 &= R^2 \quad \text{--- (2)} \end{aligned}$$

$$\text{also, } \dot{\phi} = \omega \quad \text{--- (3)}$$

eqⁿ (2) implies, $z = R \pm \sqrt{R^2 - s^2}$

we should choose $z = R - \sqrt{R^2 - s^2}$

as for $s = 0, z \rightarrow 0$

$$\dot{z} = \frac{+(-2s) \dot{s}}{2\sqrt{R^2 - s^2}} = \frac{s \dot{s}}{\sqrt{R^2 - s^2}} \quad \text{--- (4)}$$

using eqn (3) (4) in (1). 'L' becomes

$$\begin{aligned} L &= \frac{m}{2} \left[\dot{\rho}^2 + \rho^2 \omega^2 + \frac{\rho^2 \dot{\rho}^2}{R^2 - \rho^2} \right] - mg \left(R - \sqrt{R^2 - \rho^2} \right) \\ &= \frac{m}{2} \left[\frac{\dot{\rho}^2 R^2 - \dot{\rho}^2 \rho^2 + \rho^2 \omega^2 R^2 - \rho^2 \omega^2 \rho^2 + \rho^2 \dot{\rho}^2}{R^2 - \rho^2} \right] - mgR + mg\sqrt{R^2 - \rho^2} \\ L(\rho, \dot{\rho}) &= \frac{m}{2} \left(\frac{\dot{\rho}^2 R^2}{R^2 - \rho^2} \right) + \frac{1}{2}m\rho^2 \omega^2 - mgR + mg\sqrt{R^2 - \rho^2} \end{aligned}$$

since L only depends on one dynamical variable of space (ρ) it has only one degree of freedom (ρ).

$$\boxed{\frac{\partial L}{\partial \rho} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\rho}} \right)}$$

$$p_\rho = \frac{\partial L}{\partial \dot{\rho}} = \frac{mR^2 \dot{\rho}}{(R^2 - \rho^2)} \quad \Rightarrow \quad \dot{\rho} = \frac{p(R^2 - \rho^2)}{mR^2}$$

$$\begin{aligned} H(\rho, p) &= p\dot{\rho} - L \\ &= p\dot{\rho} \frac{mR^2}{(R^2 - \rho^2)} - \frac{p^2(R^2 - \rho^2)^2}{m^2 R^4} - L \end{aligned}$$

$$H(\rho, p) = \frac{p^2(R^2 - \rho^2)}{mR^2} - \frac{1}{2}m\omega^2\rho^2 + mgR - mg\sqrt{R^2 - \rho^2}$$

$$\dot{\rho} = \frac{\partial H}{\partial p} = \frac{2p(R^2 - \rho^2)}{mR^2}$$

$$\dot{p} = \frac{-\partial H}{\partial \rho} = \frac{+2p^2\rho}{mR^2} + m\omega^2\rho - \frac{mg\rho}{\sqrt{R^2 - \rho^2}}$$

3 Lecture 15: (Randomness in phase-space)

How do we find eqn of phase-trajectory using Hamiltons. eqn of motion.?

$$H(q, p)$$

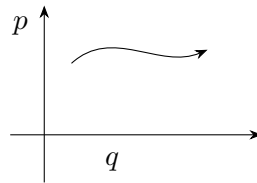
$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Ans, we have to solve these equations; if these eqn are integrable.

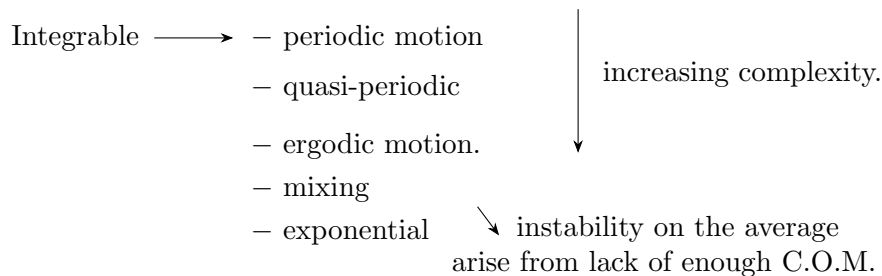
for 1-D.O.f, C.O.M = H

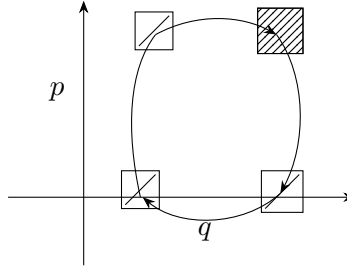
so trajectory is just given by $H = \text{const.}$



for n -D.O.f, we need n -C.O.M and find the C.T. which takes to (I, θ) variables and task is not easy to do.

(If system is integrable, we can predict what will happen at any time 't' in future provided initial 'phase-space point'.)



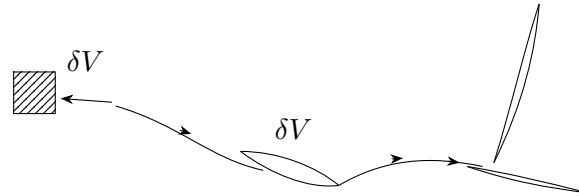


final *pos* phase space volume is closer to initial one

If initial volume element in phase space visits the neighbourhood of every point of part of phase-space, in given sufficient time, then we say motion is ergodic.

Note :-

Worse can happen to initial volume element, even when volume is preserved.

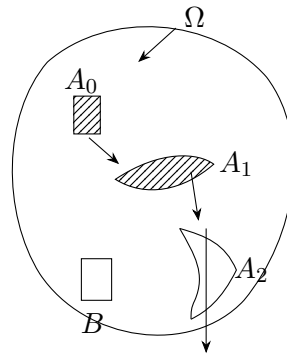


as time goes on such pieces of volume element can go as far as size of system (arbitrary far) such kind of motion is ergodic too but there is a special name to the motion called as mixing.

Mixing :-

Suppose the volume of full phase space is Ω .

A_0 — initial phase space volume



$\mu \rightarrow$ measure

If A_0 is completely mixed up, how much of A_0 is present in B .

$$\frac{\mu(A_n \cap B)}{\mu(A_0)} = \frac{\mu(B)}{\mu(\Omega)}$$

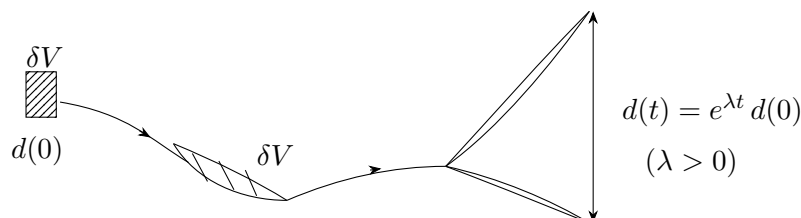
if, $\lim_{n \rightarrow \infty} \mu(A_n \cap B) = \frac{\mu(A_0) \mu(B)}{\mu(\Omega)}$

then the dynamics is said to be strongly mixing.

Note :- mixing implies ergodicity whereas ergodicity does not imply mixing as we can have ergodic motion without mixing (distortion of phase-space element).

• Rate of mixing :-

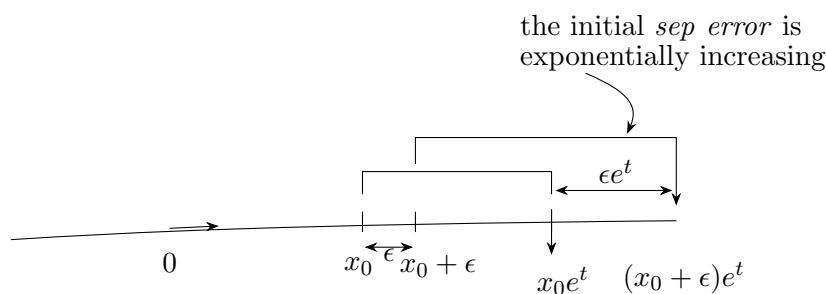
Typically, mixing occurs exponentially fast.



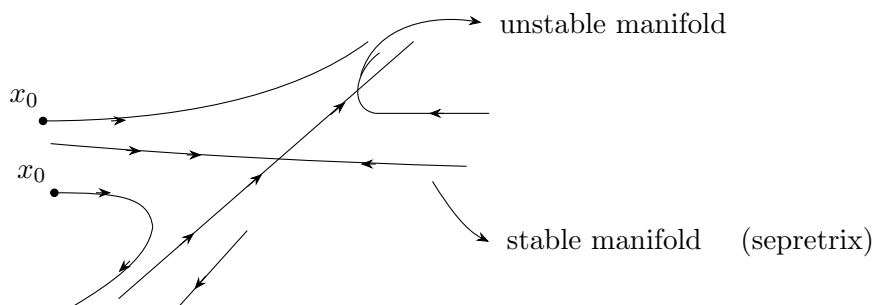
system is said to be in, exponential sensitivity to initial conditions.

example $(1 - D \text{ phase-space}) \rightarrow \text{hypothetical}$

$$\dot{x} = x \quad x(t) = x(0) e^t$$



This is just an analogy to understand but keep in mind $\dot{x} = x$ is integrable and this is not chaos.



we can see that x_0 decides the future of two sm phase space points is very different. which are in neighbourhood.

so it is like if we have sepretrix at every point that would imply chaos.

conditions of having chaos.

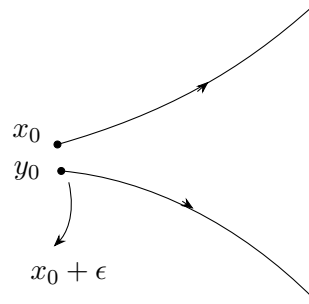
- (1) Bounded –phase space.

- (2) Exponential sensitivity to initial conditions.
- (3) A dense set of unstable periodic orbits.

Exponential sensitivity in bounded –phase space :-

If we have two different initial condⁿ x_0, y_0 with different phase trejectories. Find separation between $x(t), y(t)$.

$$|x_0 - y_0| = \epsilon$$



$$\lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = 0$$

as $t \rightarrow \infty$

$$|x(t) - y(t)| \leq L$$

↓
system
size
(bounded)

to avoid such situations let us take $\epsilon \rightarrow 0$

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = \lambda(x_0) \quad 0$$

$$|x(0) - y(0)| = \epsilon$$

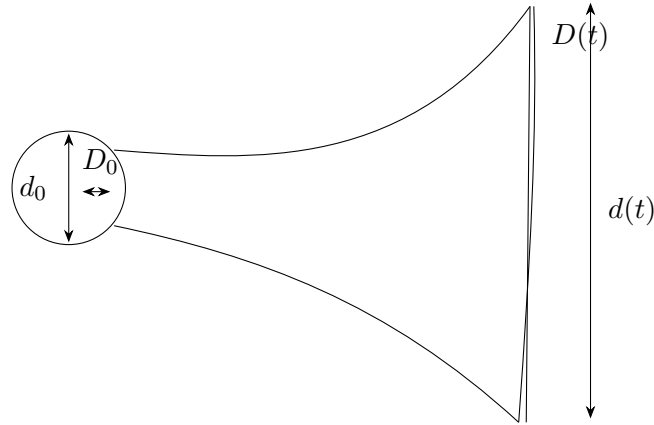
but lets invert the limits,

$$\lim_{t \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = \lambda(x_0) \quad - \textcircled{1}$$

⊥ ‘Liapunov exponent’

For n -diamentional phase space we need n -liapunov exponent, so there is a full spectrum of liapunov exponents.

eqn $\textcircled{1}$ provides maximum Liapunov exponent. as if a circle shrinks to line



(d_0 has exponential increment while D_0 has decreased with time.).

In continous time dynamics, $N = 3$ is minimum required for chaos.

For descrete time dynamics $N = 1$ will do.

$N = \text{d.o.f. Particle}$

Instead of differential eqn we call them difference eqn difference eqn are called maps.

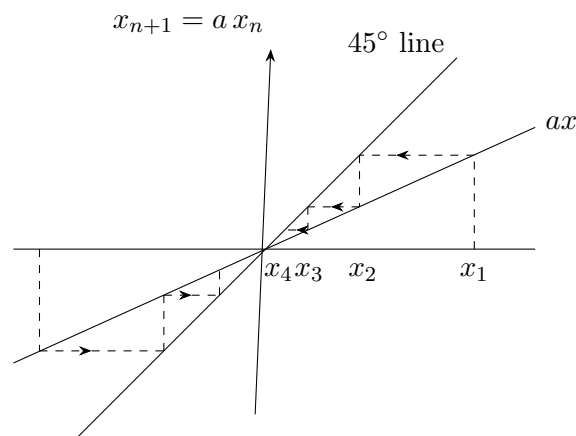
1 - D map :-

$$(\Delta x_n = x_{n+1} - x_n = a.)$$

$$x_{n+1} = f(x_n) \quad n = 0, 1, 2 \dots \text{ discrete time}$$

(i) $(a < 1)$

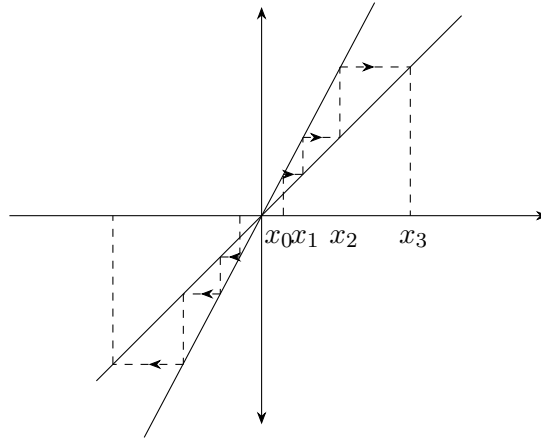
suppose $a < 1$



So, $x = 0$ is a stable fixed point

as we went to $x = 0$ from both side

(ii) $(a > 1)$

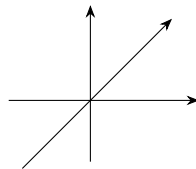


So, $x = 0$ in this case is unstable fixed point.

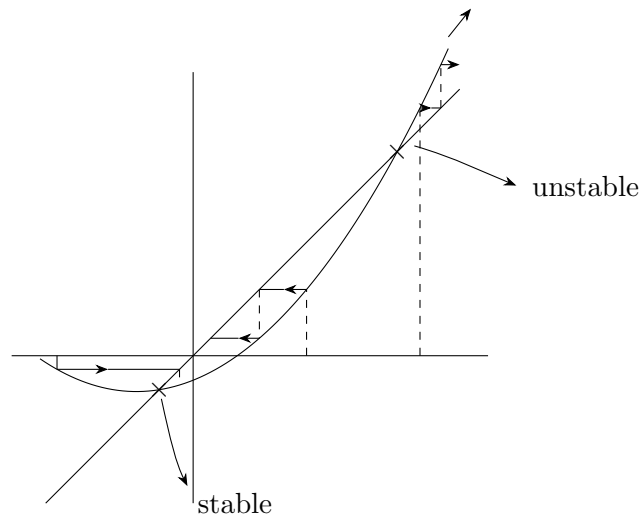
(iii) $a = 1$ map is $x = x$ (identity map)

(trivial degenerate case)

Where ever the particle is, it does not move at all.



(iv) for any arbitrary map.



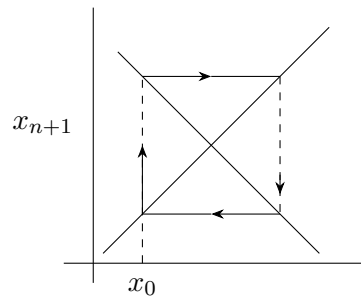
we can also see that slope at stable point is < 1 and slope at unstable fixed point is > 1 .

so, if ' x^* ' is fixed point then,

$$|f'(x^*)| > 1 \Rightarrow x^* \text{ is unstable}$$

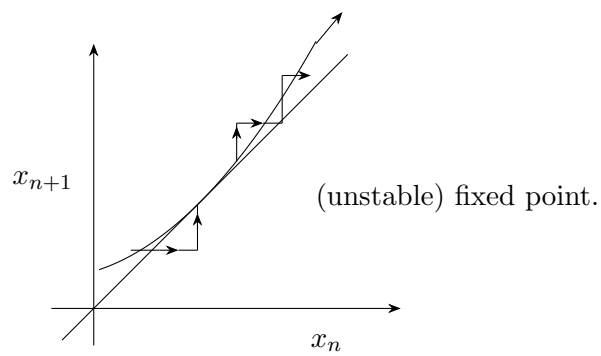
$$|f'(x^*)| < 1 \Rightarrow x^* \text{ is, stable}$$

for $f'(x^*) = -1$ *cyclic* (loop).



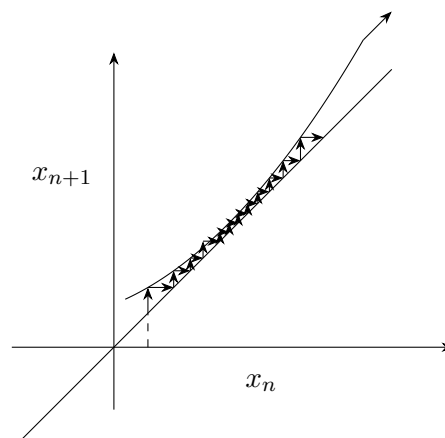
x^* = marginal fixed point (indifferent fixed point).

(v) Tangent case:-



(vi)

(fixed point is missed)



(Intermittency)

(Irregularity followed by long burst of regularity)

4 Lecture 16

Exponential divergence of phase trajectory can be characterised by “liapunov exponent”.

Note :-

we can find finite time “liapunov exponent” but it is not relevent for the purpose we have in mind.

So the reason why we need very long time liapunov exponent is following

If we start with variable x_0

$x_n = x_0$ after ‘n’ time steps.

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \cdots x_n$$

x_n is not computable as error is exponentially increasing

$$\downarrow \quad (x_0, x_n)$$

ex. If we write no. in binary and lets say error doubles each steps then error from 100^{th} decimal *ste* place comes to first decimal place after 100 steps.

If ‘x’ represents any physical quantity, there may be some function of ‘x’ $\rightarrow \phi(x)$ whose avg. can be written as $\langle \phi(x) \rangle$

$$\langle \phi(x) \rangle = \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j)$$

$$n \rightarrow \text{time} \quad \langle \phi(x) \rangle = \text{time average}$$

$\phi(x_0)$	$\phi(x_1)$	$\phi(x_2)$	$\phi(x_n)$
t_0	t_1	t_2	$-t_n$

since x_j is not computable, then $\langle \phi(x) \rangle$ is meaningless. but we would like to know what is long time avg

$$\langle \phi(x) \rangle = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j) \quad - \textcircled{1}$$

$$\downarrow \quad \text{(Replacing time avg by ensamble avg.)}$$

$$\langle \langle \phi(x) \rangle \rangle = \int dx \rho(x) \phi(x) \quad - \textcircled{2}$$

If the system is ergodic; $\textcircled{1} = \textcircled{2}$.

$$\text{time avg.} = \text{ensamble avg.}$$

Note :- $\rho(x)$ should not vary with evolution and to attain this the dynamics has to seen very very long time, thats why we need long time avg also for this reason we need liapunov exponent for very long time

Non linear Map :-

simplest is $\rightarrow$ Bernoulli map/(shift).

defⁿ :-

(Bernoulli map is the map of unit interval to itself.)

x_0 lies in $[0, 1]$

If x_1 is $f(x_0)$; x_1 lies in $[0, 1]$

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \dots \longrightarrow x_n$$

$$x_0 \in [0, 1]$$

$$x_1 \in [0, 1] \Rightarrow x_n \in [0, 1]$$

ex :- $x_{n+1} = 2x_n \bmod 1$

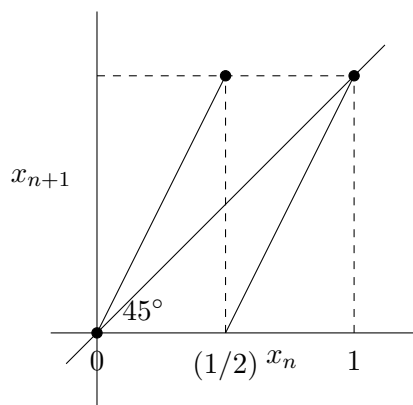
In map of $f(x)$ fixed point x^* satisfies :-

$$x^* = f(x^*)$$

fixed points are given by intersection of x_{n+1} with 45° line. which are $\{0, 1\}$

$$f'(0) = f'(1) = 2 > 1$$

so, $x^* = 0, 1$ are unstable fixed points.



This can also happen

$$f(a) = b$$

$$f(b) = a$$

first iterator of above map.

$$f^{(2)}(x) = 4x \bmod 1$$

now we have four stable points $\{0, 1/3, 2/3, 1\}$

slope at each point is 4. which is greater than 1 so fixed points are unstable

so its the non-linearity which leads to chaos.

“Any rational no is part of periodic orbit”.

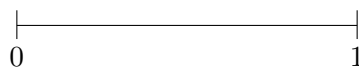
ex $x_0 = 1/3 = 0.01010101 \dots$

$$x_1 = \frac{2}{3} = 0.1010101$$

$$x_2 = \frac{1}{3} = 0.01010101 \dots = x_0 \Rightarrow \text{period '2' cycle}$$

→ (repetitive in pattern or terminating.)

Because every point is unstable, every rational no is unstable, it should act like a seperatrix which throws out no. on each side.



on line from 0 to 1 there are so many irrational no. than rational no.

and if $x_0 =$ irrational no. it wanders every where *and* by filling this interval $[0, 1]$ densely.

→ The system is ergodic.

→ The system is dissipative (How ?) → will prove later

★ Liapunov Exponent for (1-D map) :-

$$\lambda(x_0) = \lim_{n \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{1}{n} \log \frac{|x_n - y_n|}{|x_0 - y_0|} \quad |x_0 - y_0| = \epsilon$$

$$= \lim_{n \rightarrow \infty} \left(\lim_{\epsilon \rightarrow 0} \frac{1}{n} \ln \left| \frac{f^n(x_n) - f^n(x_0 + \epsilon)}{x_0 - (x_0 + \epsilon)} \right| \right)$$

$$f(x) = ax$$

$$f'(x) = a$$

$$f'(x) < 1 \Rightarrow \text{stable fixed points.} \quad (0 < a < 1)$$

$$a \in [0, 1]$$

also for, liapunav exponent $\lambda(x_0) = \ln a < 0$ so for stable fixed point liapunav exponent becomes negative (everything falls to one place).

Q.) what would be liapunav exponent for period 2 cycle?

$$f(a) = b$$

$$f(b) = a$$

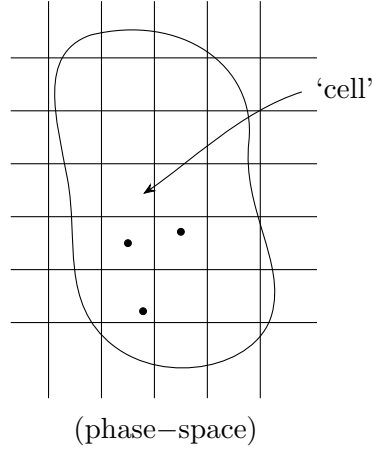
$$\text{for stability} \quad |f'(a) f'(b)| < 1$$

(slope < 1)

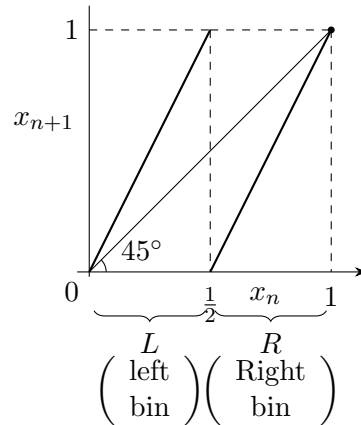
⇒ liapunov exponent will be negative for period 2 cycle.

Note :- The map is random, in the following sense, we agree that the rule is completely deterministic. for each x_0 we have well defined x_1 (with no randomness) and yet this is as random as coin toss.

suppose below is our phase space. In reality we need bins/cells to keep track of phase space points. so we do partition of phase space into cells/bins and numerically calculate how many times the perticular point lies in perticular cell/bin.



For $x_{n+1} = 2x_n \bmod 1$



we just want to know a particular point lies in left bin or right bin.

Any irrational number from 0 to 1 can be written as non-terminating in binary

$$x_i = 0.0110011111001010111000001001 \dots$$

$$x_0 = 0.a_0 a_1 a_2 \dots a_i \dots a_n$$

$$x_1 = a_0.a_1 a_2 a_3 \dots a_{i+1} \dots a_{n+1} \dots a_n$$

Above x_i is equivalent in asking wheather a_i is '0' or '1'

$$a_i = 0 \quad \text{for} \quad \longleftrightarrow L$$

$$a_i = 1 \quad \text{for} \quad \longleftrightarrow R$$

$$\text{for } x_i < \frac{1}{2} \quad a_0 = 0 \quad \longleftrightarrow L$$

$$1 > x_i > \frac{1}{2} \quad a_0 = 1 \quad \longleftrightarrow R$$

So any arbitrary string of irrational no. can be written as

$$L L R L R R R L L \dots L R L \dots L L R \dots$$

which can then also be said to be obtained from coin toss experiment.

$$\rho_2(x_2) = \int \delta(x_2 - f(x_1)) \cdot \delta(x_1 - f(x_0)) dx_1$$

Moral of the story

any given point; $y \xrightarrow{\text{iteration}} f(y)$

after 'n' time steps whats the probability density in x

$$\begin{aligned} \rho_n(x) &= \int dy \delta(x - f(y)) \rho_{n-1}(y) \\ \downarrow \lim(n \rightarrow \infty) \quad (n = \text{time steps}) \\ \left(\begin{array}{l} \text{Frobenius -} \\ \text{Perron equation.} \end{array} \right) \quad \boxed{\rho(x)} &= \int dy \delta(x - f(y)) \rho(y) \\ &\hookrightarrow \text{invariant density.} \end{aligned}$$

$\rho_n(x)$ = density of n^{th} iterate does not depends on initial condition.

—→ singular integral equation with kernel $K(x, y) = \delta(x - f(y))$

Integral eqⁿ :-

$$\phi(x) = \int dy K(x, y) \phi(y)$$

$K(x, y)$ = kernel of the integral operator.

where integral operator is $\int dy K(x, y)$ acting on $\phi(y)$

—→ Eigen value = 1

Normalization condition on $\rho(x)$

$$\int dx \rho(x) = 1$$

[?]

$$\rho(x) \geq 0, \quad x \in I$$

$$f(y) = 2y \bmod 1$$

$$\rho(x) = \int_0^1 dy \delta(x - f(y)) \rho(y) = \begin{cases} 2y & 0 \leq y < \frac{1}{2} \\ 2y - 1 & 1 \geq y > \frac{1}{2} \end{cases}$$

$$\rho(x) = \int_0^{1/2} dy \delta(x - 2y) \rho(y) + \int_{1/2}^1 dy \delta(x - 2y + 1) \rho(y)$$

$$\begin{aligned}
& \left\{ \delta(ax) = \frac{1}{|a|} \delta(x) \right. \\
& = \left(\frac{1}{2} \right) \int_0^{1/2} dy \, \delta\left(y - \frac{x}{2}\right) \rho(y) + \frac{1}{2} \int_{1/2}^1 dy \, \delta\left(y - \frac{x+1}{2}\right) \rho(y) \\
& \rho(x) = \frac{1}{2} \left[\rho\left(\frac{x}{2}\right) + \rho\left(\frac{x+1}{2}\right) \right] \quad \dots \quad \text{functional equation}
\end{aligned}$$

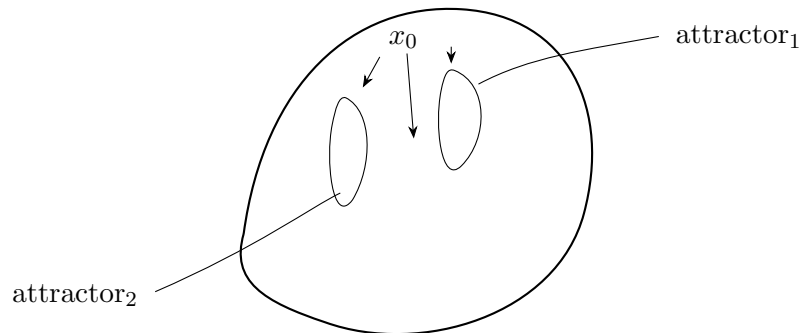
$$\rho(x) = \text{const from normalizing} \quad \int_0^1 c \, dx = 1 \quad (c = 1)$$

$$\boxed{\rho(x) = 1}$$

Therefore iterates of irrational no. uniformly and densely fill up the complete interval.

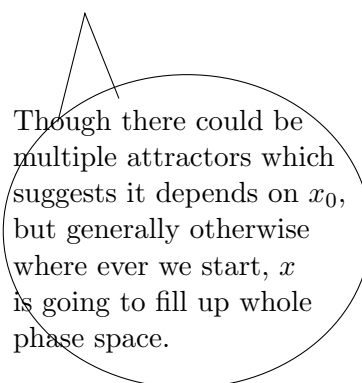
5 Lecture 17

(Q) Does Liapunov exponent depend on the initial condition or not?



If the initial condition is such that the trajectory comes to an arbitrary point in phase space.

Since the Liapunov exponent is calculated by taking time steps to infinity, and the point has explored the whole phase space, therefore in that sense the Liapunov exponent should not depend on the initial condition.



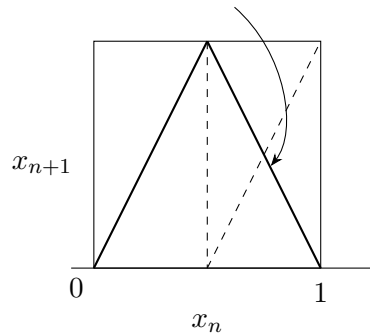
So the Liapunov exponent now becomes the property of the trajectory rather than of a point.

Baker's map :-

(measure (area, volume) is preserved, still system is chaotic), conservative system.

It is like bernoulli map where we have to cut the graph above $x_{n+1} = 1$ and put it back. We could have also folded the peice back rather than cutting to gain non-linearity.

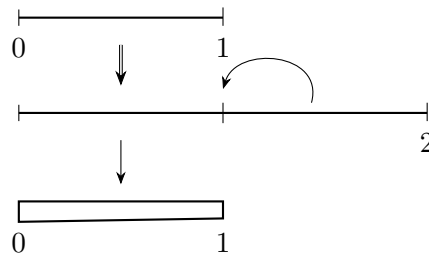
$$x_{n+1} = 2x_n \bmod 1 \quad \longrightarrow \quad (\text{fold just}) \quad (\text{analogy})$$



Tent map (non-linear, contineous)

$\lambda(x_0) = \ln 2$, all rational no are periodic unstable orbits.

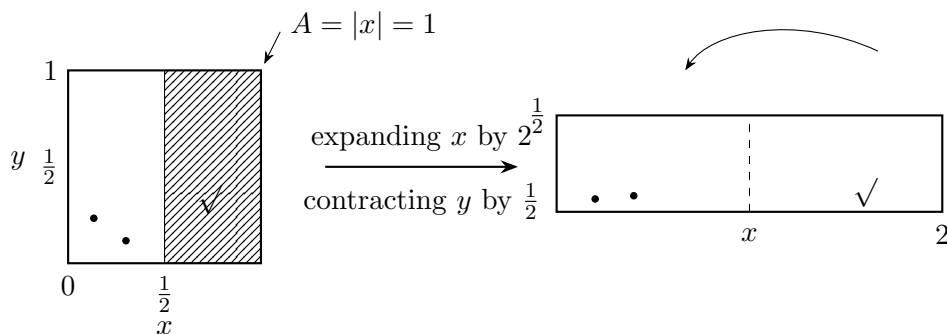
So the idea is to stretch the interval then put the second half part on top of first half map.



Backer's map takes one more diamencion (increasing dimension by one) in such a way that measure is preserved. $x_{n+1} = 2x_n \bmod 1$ [?]

$$y_{n+1} = \begin{cases} \frac{1}{2} y_n & (x_n < \frac{1}{2}) \\ \frac{1}{2} + \frac{1}{2} y_n & (x_n > \frac{1}{2}) \end{cases}$$

lets draw, x_{n+1} vs y_{n+1} ; and x_n vs y_n .

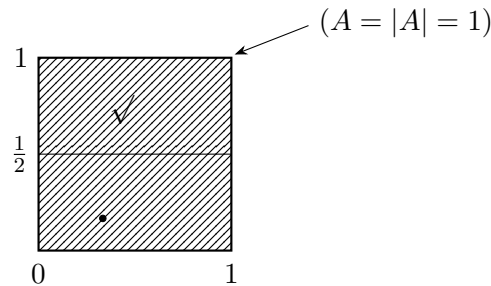


- map is invertible, still chaotic because of ($\lambda_1 = \ln 2$)
- Area is preserved ($A_i = 1, A_f = 1$)

$\Rightarrow$ Jacobian of transformation is 1

- liapunov exponents are $\ln 2, \ln \frac{1}{2}$

$$\text{total liapunov exponent} = \ln 2 + \ln \frac{1}{2} = 0$$



As long as measure is preserved sum of liapunov exponents is zero

All liapunov exponents are zero for integrable hamiltonian syst[?] since it is integrable none of $\lambda(x_0)$ should be +ve, -ve (expand[?] volume space/contracting), sum of these also will be zero, $(0 + 0 + \dots)$

For non-integrable hamilton systems liapunov exponents occur in pairs such that total liapunov exponent is zero. (-ve for each +ve).

Note :-

unlike differential dynamics, in discrete time dynam[?] we can write explicit solution to a map, but stil[?] it will be chaotic.

$$\begin{aligned} x_{n+1} &= 2x_n \bmod 1 \\ x_n &= 2^n x_0 \bmod 1 \quad \longrightarrow \text{explicit fun. of time} \\ &\hookrightarrow \text{still chaotic.} \end{aligned}$$

we can see how we do not see loss of information in Backers map, if we represent $x_n, y_n, x_{n+1}, y_{n+1}$ in binary

$$\begin{aligned} x_n &= 0.a_0 a_1 a_2 a_3 \dots \\ y_n &= 0.b_0 b_1 b_2 b_3 \dots \\ &\downarrow \\ x_{n+1} &= 0.a_1 a_2 a_3 \dots \\ y_{n+1} &= 0.(a_0 b_0) b_1 b_2 b_3 \dots \end{aligned}$$

so we do not loss 'a₀'.

Chaotic but (measure preserving and invertible.)

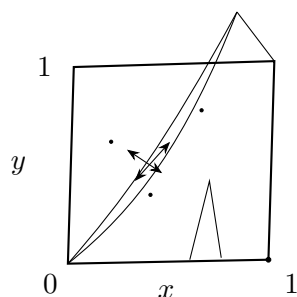
(wow!)

Note :- The general rule to produce chaos in 1-d map is that it has to be non-invertible, for 2 and higher diamen-ncions invertible maps can also produce chaos.

Arnold's cat map :-

$$\begin{aligned}x_{n+1} &= (x_n + y_n) \bmod 1 \\y_{n+1} &= (x_n + 2y_n) \bmod 1 \\ \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix}\end{aligned}$$

determinant of Transformation matrix = 1 (orientation preserving)



The reason it is named as arnold's cat is because he put face of cat in this graph and after few iteration we can't recognize the face of cat.

→ every point is having stretching direction and contracting direction.

***. Gauss (continued fraction) map :-**

$$x_0 = \frac{1}{a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}} \xrightarrow[\text{writing}]{\text{eas of}} \frac{1}{a_0} \oplus \frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} \dots$$

↔ notice '+' sign is written down

If continued fraction terminates x_0 = rational else x_0 is irrational number

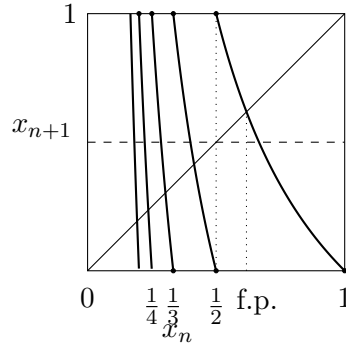
$$x_1 = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}$$

So to get x_1 ; take reciprocal of x_0 , throw away integer part ' a_0 '. So,

$$x_{n+1} = \frac{1}{x_n} - \left[\frac{1}{x_n} \right]$$

↔ integer part of $(1/x_n)$

$$\text{for, } \begin{pmatrix} \frac{1}{2} < x_n < 1 \\ 1 < \frac{1}{x_n} < 2 \end{pmatrix}$$



for any given x_{n+1} there lies infinite pre images.

|slope at each point| > 1

→ unstable at each point.

Q.1) lets take any arbitrary no from 0 to 1. take its reciprocal throw away integer, and repeat the process for long time, what is the probability that final number is less than α . $(0, \alpha)$
 (Ans: $\frac{\ln(1+\alpha)}{\ln 2}$)

Before we calculate let's find fixed point of gauss map.

$$\text{f.p.} \Rightarrow x = \frac{1}{x} - 1$$

$$x^2 + x - 1 = 0 \quad x = \frac{-1 \pm \sqrt{1+4}}{2} \\ = \frac{-1 \pm \sqrt{5}}{2}$$

$$\text{So f.p.} = \frac{\sqrt{5}-1}{2} \quad \text{as} \quad 0 < x < 1$$

$$\rightarrow x + 1 = \frac{1}{x} \\ \rightarrow x = \frac{1}{1+x}$$

continued fraction can be written as

$$x = \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

$$\text{For f.point}_2, \quad x = \frac{1}{x} - 2$$

$$\Rightarrow x^2 + 2x - 1 = 0 \\ x = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \Rightarrow -1 \pm \sqrt{2} \\ \left(x = \sqrt{2} - 1 \right)$$

$$\begin{aligned}
\text{continued fraction} \Rightarrow \quad & \frac{1}{x} = x + 2 \\
\Rightarrow \quad & x = \frac{1}{x + 2} \\
\Rightarrow \quad & x = \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}
\end{aligned}$$

Q.) what short of this no is?

$$x = \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{2 + \dots}}}}$$

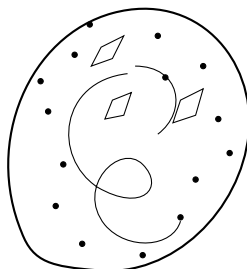
Ans. It would be fixed point of first iterate of map or period 2 cycle.

6 Lecture 20: (classical statistical mechanics)

- statistical mechanics helps in understanding behaviour of systems which involves large no of d.o.f (which are interacting non-linearly with one another).
- It is science of averages. Also it is study at largest/longest length scales and time scales.
- statistical mechanics is partitioned into two parts
 - Equilibrium statistical mechanics.
 - Non-Eqm statistical mechanics.
- Eq^m statistical mechanics arose from question that :- Are there any steady states (eqⁿ states) which the system will finally settle down? If there is How does it behave and so on?
- Thermodynamics is the study of systems at their longest length scales and time-scales.
 - so All fluctuations which happens at very small length scales & time scales are neglected and we look at things at very long time scales and very long length scales.

let us start with hamiltonian of a system. system contains many particles and is isolated from the rest of the universe.

Isolated system (exchange of matter & energy are prohibited).



$$H(q, p)$$

If rules are time-translation invariant then we know that Hamiltonian is constant of motion.
 $\left(\frac{dH}{dt} = 0\right)$

$$H(q, p) = E \quad (\text{total energy of system}).$$

{with uncertainty δE }

phase space becomes $6n$ -dimensional for n -particles. So we need to know $3n$ – constants of motion which are in involution with each other. $\{c_i, c_j\} = 0 \quad i \neq j$.

constants of motion will be

1) H	1
2) total linear momentum (P)	3
3) Total angular momentum (L)	3
4) C.o.M position. ($\vec{R}$)	3
Total	<u>10 COM</u>

To specify state of system we need a point in phase space $(q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n)$. And that point will move on some $6n$ -dimensional hypersurface.

- Due to lack of sufficient nos of c.o.m which are in involution with each other, we know that differential eqⁿ of motion is not integrable and the system of such kind is going to show chaotic behaviour with some positive liapunov exponents. Its likely to wander around completely in strange fashion. such that system is ergodic on energy surface.
- It does not have to be strictly ergodic, there can be great deal of mixing (exponential separation of volume element).

Any way exponential separation $\Rightarrow$ mixing (implies.)

$\Downarrow$

ergodicity

Ergodicity :–

- The volume element of phase space will come closer to every point of phase space in given sufficient time. {defⁿ of ergodicity}.
- The whole phase space is not accessible to the system because we specified its total energy.
- we become ignorant in knowing what happens to system at time ' $t_0 + \Delta t$ '. and assume that every point/volume element of phase space is as equally likely as other points of phase space. We can term this principle as principle of maximum ignorance.

Fundamental Assumption/postulate of eqⁿ statistical mechanics.

- “In a state of thermal eq^m, all accessible micro-states of the system are equally probable”.

“This single statement is enough to derive all equations of stat. mech, whereas thermal eqⁿ turns out to be a special case of it”.

Thermal equilibrium :—

Two systems are said to be in thermal equilibrium if there is no heat flows between them when they are connected to each other.

A system is said to be in thermodynamic eq^m if it is in mechanical, physical and thermal equilibrium.

- state of thermal eq^m also means that long time average of macroscopic variables reach to a steady state and do not change with time. such macroscopic quantities are Energy (internal), Total pressure[?]

“All physical macroscopic quantities are time independent in system of thermal eq^m state.”

Note :—

- Microscopic variables like velocity of a particle in certain d.o.f is not time independent, whereas the average velocity of all particles is time independent which is zero. i.e. system is not moving anywhere.
- The average values can only be independent of time if the probability distribution (over which average is defined) is itself independent of time.

micro state :—

specifying the state of every constituent of the system. The set $(q_1 \cdots q_n, p_1 \cdots p_n)$ is microstate of the system.

or we can say that position in phase space is microstate where as the total (avg) momentum of all particles is macrostate.

By accessible microstate we restrict the velocity a particle can have, i.e.

$$H(q, p) = E_{\text{total}}$$

so a particle of such isolated system can not have $K.E > E$. So some micro states are not accessible anymore.

“All those micro states which are compatible with all restrictions applied on system are termed as accessible micro-state.”

“microstate tells what each and every particle is doing and macrostate is giving some gross (overall) information.”

If microstate energy is the only parameter of microstate — (In quantum stat mech)

“If a isolated system has 10J of total energy then we can have a single particle which has energy close to 10J and rest. of all are close to zero energy. This (microstate) is equally probable than other state; followed from postulate of eq^m stat. mech.”

In phase-space probability density obeys

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

$\rho = \text{const. of motion}$

$$\frac{d\rho}{dt} = \{\rho, H\} + \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial \rho_{eq}}{\partial t} = \{H, \rho_{eq}\}$$

if probability distribution is independent of time

$$\frac{\partial \rho_{eq}}{\partial t} = 0 = \{H, \rho_{eq}\} \Rightarrow \text{implies } \underline{\rho_{eq} = f(H)}.$$

remember ρ_{eq} is not physical observable.

$$\rho_{eq} = \rho_{eq}(H) = \delta(H(q, p) - E)$$

since such isolated system uses microstates, such ensemble is termed as microcanonical ensemble. A stat. ensembles with $\rho_{eq} = \delta(H(q, p) - E)$ in phase space with which corresponds to isolated system in thermal eqⁿ is called microcanonical ensemble

> If we assume we can not have a point in phase space but a volume shell of finite resolution then only we can define probability of occupying certain state (volume element in phase space (shelle)).

$$\Delta q \Delta p \geq \frac{\hbar}{2}$$

for 1 particle volume element

$$(\Delta q_1 \Delta p_1)(\Delta q_2 \Delta p_2)(\Delta q_3 \Delta p_3)$$

$$\sim \hbar^3 \text{ or } h^3$$

for N particle the volume element would be

$$\sim h^{3N}.$$

If volume of accessible phase space is ' M ',

$$\# \text{ of micro states} = \frac{M}{h^{3N}} = \Omega(E) \text{ (finite \& large)}$$

let probability of any microstate = P $\sum_i P_i = 1$

$$P \cdot (\text{no. of microstates}) = 1$$

$$\Rightarrow \left\{ P = \frac{1}{\Omega} = \frac{h^{3N}}{M} \right.$$

Toy Model :-

set of N distinguishable (identifiable) coins. where each coin has $\{H, T\}$ two states.

$$\{2^N \text{ microstates}\} \quad \underbrace{\text{htt} \quad - \quad \text{hhh} \quad - \text{tth} \quad - \quad \text{hhh}}_{\text{'N' letters}} \longrightarrow \text{microstate}$$

(still a state
of system as
a whole)
but describes
what each object is
doing

Let,

Total no. of heads = H

of Tails = T

$$H + T = N$$

Let,

$$H - T = M$$

(N, M) or (N, H) or (H, T) are macrostate of the system.

For given N ,

$$\# \text{ of macrostates} = N + 1.$$

ex for $N = 3$ let (N, H) represents macrostate

we have: $(3, 0), (3, 1), (3, 2), (3, 3)$ 4 macrostates.

no. of microstates are much larger than no. of macrostates.

$$2^N > N + 1$$

$$N = d.o.f.$$

> probability of any possible/specific microstate would be $\left(\frac{1}{2^N}\right)$.

> Probability of attaining a given macrostate :-

we have $0 \leq H \leq N$

$$P(H) = {}^N C_H \cdot \left(\frac{1}{2}\right)^H \left(\frac{1}{2}\right)^{(N-H)}$$

Suppose coin is biased then

$$P(h) = p$$

$$P(t) = 1 - p = q$$

then,

$$P(H) = {}^N C_H p^H q^{N-H}$$

$$\sum_{H=0}^N P(H) = (p + q)^N = 1^N = 1 \quad \text{binomial expansion.}$$

Generating function for the binomial distribution :-

$$\begin{aligned} f(x) &= \sum_{H=0}^N P(H) x^H \\ &= (px + q)^N = \sum_{H=0}^N ({}^N C_H p^H q^{N-H}) x^H \end{aligned}$$

$P(H)$ = coefficient of x^H in expanded generating function

Property of G.F :-

$$f(1) = 1 ; (p + q)^N = 1$$

$$\langle H \rangle = ? \quad (\text{avg. no. of Heads})$$

$$\langle H \rangle = \frac{\sum_{H=0}^N P(H) H}{\sum_{H=0}^N P(H)} = \frac{\sum_{H=0}^N P(H) H}{1}$$

to find $\langle H \rangle = \sum_{H=0}^N P(H) H$ we differentiate G.F

$$f'(x) = \sum_{H=0}^N P(H) \cdot H x^{H-1}$$

$$f'(1) = \sum_{H=0}^N P(H) \cdot H = (N(px + q)^{N-1} \cdot p)_{x=1}$$

so,

$$\langle H \rangle = \sum_{H=0}^N P(H) H = N(p + q)^{N-1} \cdot p = Np$$

$$\boxed{\langle H \rangle = Np}$$

$$\langle H^2 \rangle = ?$$

find

$$f''(1) = (Np^2(N-1)(px + q)^{N-2})_{x=1} = p^2 N(N-1) \quad (1)$$

$$f''(x) \Big|_{x=1} = \left(\sum_{H=0}^N P(H) H(H-1) x^{H-2} \right)_{x=1} = \langle H(H-1) \rangle$$

$$= \langle H^2 - H \rangle$$

$$= \langle H^2 \rangle - \langle H \rangle \quad (2)$$

Equating (1) and (2)

$$p^2 N(N-1) = \langle H^2 \rangle - \langle H \rangle$$

$$\Rightarrow \langle H^2 \rangle = p^2 N^2 - p^2 N + \langle H \rangle$$

$$\Rightarrow \langle H^2 \rangle = p^2 N^2 - p^2 N + Np$$

Also, Now we can find variance,

variance = mean square – square of mean

$$\text{var}(H) = \langle H^2 \rangle - \langle H \rangle^2 \quad (= \langle (H - \langle H \rangle)^2 \rangle)$$

$$= N^2 p^2 - p^2 N + Np - N^2 p^2$$

$$\text{var}(H) = Np(1 - p)$$

$$\sigma^2 = \text{var}(H) = Npq \quad , \text{ as, } (q = 1 - p)$$

Therefore $\Delta H = \text{std. deviation} = \text{uncertainty} = \sqrt{Npq}$

$$\boxed{\Delta H = \sigma = \sqrt{Npq}}$$

Therefore relative fluctuation will be :-

$$\text{fluctuation about mean} = \frac{\Delta H}{\langle H \rangle} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}}$$

$$\text{Relative scatter} = \frac{1}{\sqrt{N}} \sqrt{\frac{q}{p}}$$

(dispersion about mean)

> It says relative fluctuation becomes smaller as no. of coin toss increases / no. of coins increases.

In thermodynamics we deal with $\sim 10^{23}$ d.o.f so relative fluctuations becomes very small. This is the main reason “why thermodynamics works”.

“In statistical mechanics generating function is called as partition function (laplace transformation / green fn.)”

Note :-

out of 100 coin toss $P(50H)$ or $P(50T)$ is most probable. Because of $^{100}C_{50}$ is biggest for all H in $(^{100}C_H)$.

each microstate is equally probable, but each macrostate is made up of many microstates. so attaining 1H and 99T is having lesser no of micro-states then that of 50H, 50T So our example that any one molecule can have nearly all total energy and

Rest of all are nearly at rest; Indeed such micro-states are equally probable but the macrostate has contribution from very less no. of microstates so such macrostate is less likely to exist then other possible macrostates.

7 Lecture 21

N coins, H heads, T tails

$$P(H) = {}^N C_H p^H q^{N-H}$$

p = prob of heads

q = prob of tails

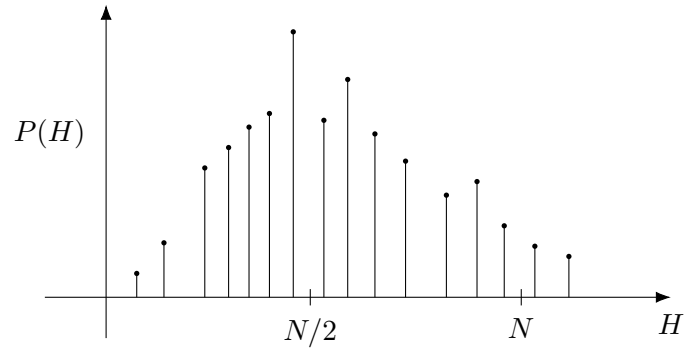
$$f(x) = (px + q)^N$$

$$\langle H \rangle = Np$$

$$\Delta H = \sqrt{Npq}$$

$$\begin{aligned} f(x) &= \sum_{n=0}^N P(n) x^n \\ &= \sum_{n=0}^N {}^N C_n p^n q^{N-n} x^n \\ &= (px + q)^N \end{aligned}$$

What does this distribution looks like.



$$P(H) = \frac{N!}{H!(N-H)!} p^H q^{N-H}$$

we would like to find out what does ${}^N C_H$ do at large no. of 'N'

What is $N!$ (for large N) :-

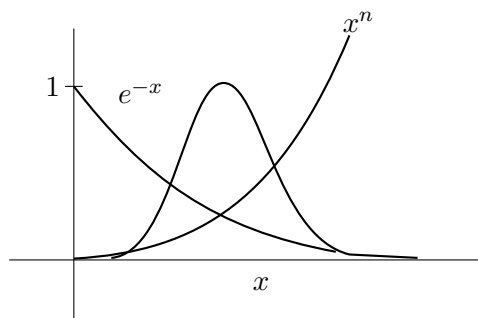
Stirling formula :-

$$\begin{aligned} n! &= n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \\ &= n^n \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \\ &= n^n e^{-n} \sqrt{2\pi n} \cdot \left\{1 + \frac{1}{12n} + \dots\right\} \\ 1! &= 1^1 e^{-1} \sqrt{2\pi} \quad \text{so} \quad \text{is } e \simeq \sqrt{2\pi} ? \end{aligned}$$

defⁿ

$n!$ is defined as

$$\int_0^\infty dx e^{-x} x^n = \Gamma(n+1) \quad (n = 1, 2, 3, \dots)$$



$$\begin{aligned} g(x) &= e^{-x} x^n = e^{-x} e^{\ln x^n} = e^{-x+n \log_e x} = e^{-(x-n \ln x)} \\ g'(x) &= e^{-(x-n \ln x)} \cdot (-1) \left(1 - \frac{n}{x}\right) = 0 \Rightarrow \boxed{g'(n) = 0} \end{aligned}$$

let, $f(x) = x - n \ln x$

$$f(n) = n - n \ln n$$

$$\begin{aligned}
f'(x) &= 1 - \frac{n}{x} \\
f''(x) &= 0 + \frac{n}{x^2} \\
f(x) &= f(n) + f'(n)(x-n) + \frac{f''(n)(x-n)^2}{2!} + \dots \\
f(x) &= (n - n \ln n) + 0 + \frac{(x-n)^2}{2n} + \dots
\end{aligned}$$

$$\begin{aligned}
n! &= \int_0^\infty e^{-x} x^n dx = \int_0^\infty e^{-(x-n \ln x)} \\
\Rightarrow n! &= \int_0^\infty e^{-\left((n-n \ln n) + \frac{(x-n)^2}{2n} + \dots\right)} dx \\
&= \int_0^\infty (e^{-n} \cdot n^n) \cdot e^{-\frac{(x-n)^2}{2n}} \{1 + \dots\} dx
\end{aligned}$$

for very large 'n'

$$\begin{aligned}
n! &= e^{-n} n^n \int_{-\infty}^\infty dx e^{-\frac{(x-n)^2}{2n}} dx \\
&= e^{-n} n^n \sqrt{2\pi n} \cdot \left\{1 + \text{Order}\left(\frac{1}{n}\right)\right\}
\end{aligned}$$

$$\int_{-\infty}^\infty e^{-\frac{t^2}{a}} dt = \sqrt{\pi a}$$

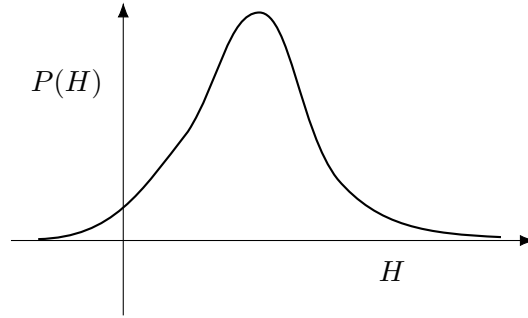
Gaussian integral

$$I = \int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\pi/a}$$

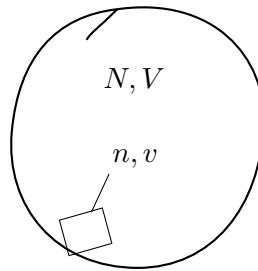
also

$$\begin{aligned}
I &= \int_{-\infty}^\infty e^{-ay^2} dy \\
I^2 &= \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-a(x^2+y^2)} dx dy \quad ; \quad \begin{cases} x^2 + y^2 = r^2 \\ x = r \cos \theta \\ y = r \sin \theta \\ dA = (r d\theta) dr \end{cases} \\
&= \int \int e^{-ar^2} r dr d\theta \\
&= \int_{-\infty}^\infty e^{-ar^2} r dr \int_0^\pi d\theta = \left(\sqrt{\pi/a}\right)^2 \\
&\quad \left(I = \sqrt{\pi/a}\right)
\end{aligned}$$

so for large N



Take ideal gas contained in volume V .



N = total no. of particles.

What is probability that v contains n particles?

$P(n)$ = Probability that ' v ' contains n particle

Probability that a given particle is in volume ' v ' = $\frac{v}{V}$

assume that all the particles are moving independent of each other, therefore the probability that there are ' n ' of them inside ' v ' is

$$= {}^N C_n \left(\frac{v}{V} \right)^n \left(1 - \frac{v}{V} \right)^{N-n}$$

Here ' n ' is random variable whose no. can vary from $(0 \leq n \leq N)$.

$$\text{here } p = \frac{v}{V}, \quad q = 1 - \frac{v}{V}$$

this is the binomial distribution.

lets put $\rho = \frac{N}{V}$

$$\Rightarrow \frac{1}{V} = \frac{\rho}{N}$$

so,

$$P(n) = {}^N C_n \left(\frac{v\rho}{N} \right)^n \left(1 - \frac{v\rho}{N} \right)^{N-n} \quad \left(\begin{array}{l} \text{binomial} \\ \text{distribution} \end{array} \right)$$

What happens to $P(n)$, when $N \rightarrow \infty$ and $V \rightarrow \infty$ but keeping ' ρ ' fixed.

$$\begin{aligned} P(n) &\xrightarrow{N \rightarrow \infty} \frac{N!}{n! (N-n)!} \left(\frac{v\rho}{N} \right)^n \left(1 - \frac{v\rho}{N} \right)^{N-n} \\ &= \frac{N^N e^{-N} \sqrt{2\pi N}}{(n^n e^{-n} \sqrt{2\pi n}) (N-n)^{N-n} e^{-(N-n)} \sqrt{2\pi (N-n)}} \left(\frac{\rho v}{N} \right)^n \left(1 - \frac{\rho v}{N} \right)^{N-n} \end{aligned}$$

$$\left(\begin{array}{l} \text{we can't} \\ \text{use } n! = e^{-n} n^n \sqrt{2\pi n} \\ \text{as 'n' can take 0 too } \infty \end{array} \right) \quad (n!)$$

$$\boxed{\rho v = \bar{n}}$$

$$P(n) \longrightarrow = \frac{e^{-\bar{n}} (\bar{n})^n}{(n!)} \quad \text{where } n = 0, 1, 2, 3 \dots \infty$$

and $\bar{n} = \rho v$ = avg. no of particles in sub volume 'v'.

$$\left(\bar{n} = Np = N \frac{v}{V} \right) = (\bar{n} = \rho v)$$

we have assumed that particles behave like newtonian particle mech. and is distinguishable from one another.

In $\left(P(n) = \frac{e^{-\rho v} (\rho v)^n}{n!} \right)$ formula, (N, V) don't appear anymore, as $(N, V \rightarrow \infty)$ Such that their ratio $\rho = \frac{N}{V}$ is finite (this is called thermodynamic limit of system when $N \rightarrow \infty, V \rightarrow \infty$ such that $\rho = \frac{N}{V}$ is finite). Statistical mech. Reduce to thermodynamics in thermodynamic limit.

Poisson Distribution :-

for random var. $n = 0, 1, 2 \dots \infty$

$$P(n) = \frac{e^{-\lambda} \lambda^n}{n!} \quad (\lambda = \langle n \rangle)$$

λ is avg value of random variable

Generating function

$$f(x) = \sum_{n=0}^{\infty} P(n) x^n = e^{\lambda(x-1)} \quad (\text{How ?})$$

$$f(1) = 1 \quad \text{for} \quad \sum_{n=0}^{\infty} P(n) = 1$$

$$\text{variance} = \langle (n - \langle n \rangle)^2 \rangle = \lambda = \langle n \rangle = \langle n^2 \rangle - \langle n \rangle^2$$

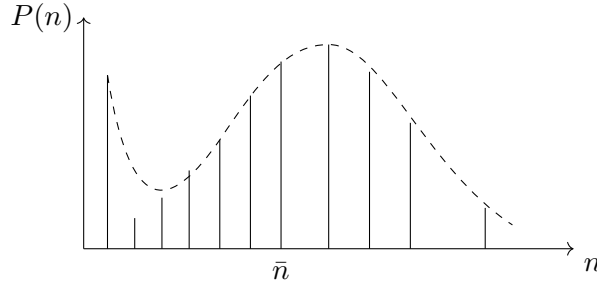
In poisson distribution variance = mean

$$\text{Relative fluctuation} = \frac{\sigma}{\langle n \rangle} = \frac{\Delta n}{\langle n \rangle} = \frac{\sqrt{\langle n \rangle}}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}}$$

for poisson's distribution, $\langle n \rangle = \lambda$, $(\Delta n)^2 = \lambda$. & all higher moments = λ ., so its a single parameter distribution;

In gaussian distribution except for $\langle n \rangle$ & $(\Delta n)^2$ all higher moments vanish.

$\bar{n}$ = avg. no of particles in subvolume 'v'.



if $\bar{n}$ itself become very large then deviation from mean ($n - \bar{n}$) takes gaussian curve. and call that a continuous variable if $\bar{n}$ is very large.

from

$$P(n) = \frac{e^{-\bar{n}} (\bar{n})^n}{n!}$$

using sterling formula again, and replacing variable to

$$x = n - \bar{n}$$

$$P(x) \propto \frac{e^{-\frac{x^2}{2\sigma^2}}}{\sqrt{[?]}} \quad (\text{Gaussian distribution})$$

“ The whole point of above discussion is that if we start with a binomial distribution of burnoulli trials and then from that we take no. of particles / trials to be very large such that probability of succes in trial is vanishingly small. Then Binomial distribution goes over into poisson distribution. Then if mean value of poisson distribution is very large compared to unity, The deviation from the mean ($n - \bar{n}$) is approximatly a contineous variable that has a Gaussian shape”.

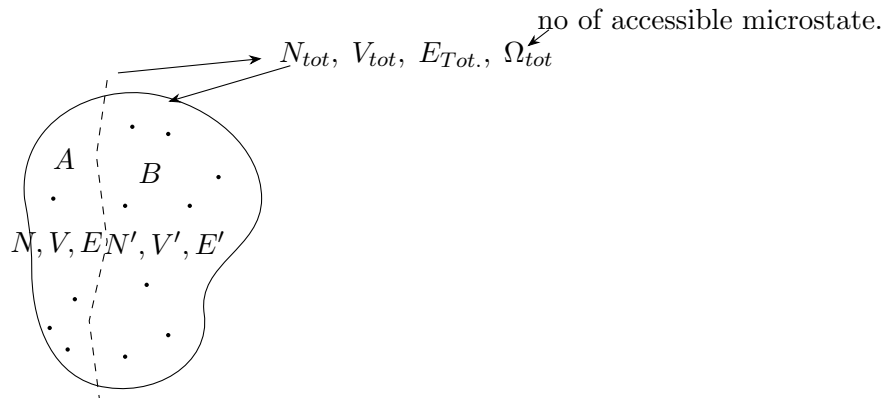
So A probability distribution can shifts over to another probability distribution.

Generating function :- $f(x) = \sum_{n=0}^{\infty} P(n)x^n = P(0)x^0 + P(1)x^1 + P(2)x^2 + \dots P(n)x^n$

$$f(x) = \sum_{n=0}^{\infty} P(n)x^n = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} x^n \Rightarrow e^{-\lambda} \sum \frac{(\lambda x)^n}{n!} = e^{-\lambda} e^{\lambda x}$$

$$\boxed{f(x) = e^{\lambda(x-1)}}$$

lets go back to our problem. of isolated system in thermal equilibrium.



Imagine the system is made of two sub-systems, in thermal eq^m with each other, such that

$$\begin{aligned}N + N' &= N_{Tot} \\V + V' &= V_{Tot} \\E + E' &\simeq E_{tot}\end{aligned}$$

(Isolated, in thermal eq^m)

not necessarily

$(E + E' \neq E_{Tot}$ always if pot. energy plays role)

$E + E' \simeq E_{Tot}$ to a very good approximation.

“as no. of d.o.f which are interacting near partition is $(\frac{1}{10^8})^{th}$ of the total d.o.f”. (assuming short range forces”).

Then at any instant of time, what is the probability that Energy of A is E .; $P(E)$

$P(E)$ = no. of microstate of this entire system such that A has energy E . ($\Omega(E)$) (also (no. of microstates such that B has energy E' .)

$$= \frac{\Omega(E) \Omega'(E')}{\Omega_{tot}(E_{tot})} \quad (\text{normalization})$$

$\Omega(E)$ & $\Omega'(E')$ can have totally different kind of functions.

As A can be a large container of oil and B can be our atmosphere. so A, B have different d.o.f.

8 Lecture 22

Note that Energy of A and B are not fixed. Because there are fluctuations.

E, V, N, N', V', E' . are variable quantities.

what we were trying is to find, is

“Given the postulate of equal apriori probabilities of all the accessible microstates of the total system, what can we say about probability distribution of (for instance say) Energy in system A .”

$P(E) =$

$$\frac{\# \text{ of microstates of tot. system such that } A \text{ has energy } E. (\& B \text{ has } E'.)}{\text{Total no. of microstates of total system such that T.E} = E_{tot}.}$$

If A & B are assumed as independent system (they actually are not) then $\Omega_{tot}(E) = \Omega(E) \cdot \Omega'(E')$

$$= \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{tot}(E_{tot})}.$$

$P(A \& B) = P(A) \cdot P(B)$ only iff A, B are independent events

In above discussion we have assumed that interection Energy is fairly negligible.

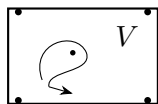
$\Omega(E)$ is actually a fun. of V, N also

Remember always

$$\begin{aligned}\Omega(E) &= \Omega(E, V, N) \\ \Omega'(E') &= \Omega'(E', V', N')\end{aligned}$$

lets see how big these no. are :

Suppose we take a single particle & put that into volume V and the particle is moving around with some energy $\leq E$. lets calculate how many microstates it has.



No. of microstates is the no. of shells in phase-space

$$\text{true in any dimensions} \leftarrow \left(E = \frac{p^2}{2m} \right) \quad |p| \leq \sqrt{2mE}$$

let no. of such microstates be $\phi(E) = \frac{\text{phase space volume}}{\text{min volume}(h^3)}$

$$\phi(E) = \frac{1}{h^3} \int_V d^3q \int_{|p| \leq \sqrt{2mE}} d^3p \quad \rightarrow \text{phase space volume}$$

$d^3p = 4\pi p^2 dp$ volume element

$$= \frac{V}{h^3} \int_0^{\sqrt{2mE}} 4\pi p^2 dp \propto \frac{V}{h^3} E^{3/2} \quad \rightarrow \text{no of dimensions}$$

$(p \propto E^{1/2})$.

$$\begin{aligned}&= \frac{V}{h^3} \frac{4\pi}{3} (p^3) \Big|_0^{\sqrt{2mE}} \\ &= \frac{V}{h^3} \frac{4\pi}{3} \cdot E^{3/2} 2m^{3/2}\end{aligned}$$

relation b/w E & p for a free particle is

$$E^2 = c^2 p^2 + m^2 c^4$$

m = rest mass always

mass does not increase with speed, its just $p = \gamma mv$ instead of mv . $\left(\gamma = \frac{1}{\sqrt{1-v^2/c^2}} \right)$

$$E = (c^2 p^2 + m^2 c^4)^{1/2} = c (p^2 + m^2 c^2)^{1/2}$$

If $p \ll mc$

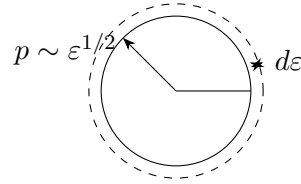
$\{mc = \text{compton momentum}\}$

In 'n' dimentions, $(n-1)$ angles runs from $(0 - \pi$ or $0 - 2\pi)$.

In n -spacial dimen.

$$\begin{aligned}\phi(\varepsilon) &\propto \int_0^{\sqrt{2mE}} p^{n-1} dp \propto \varepsilon^{n/2} \\ E &= mc^2 \left(1 + \frac{p^2}{m^2 c^2} \right)^{1/2} \\ &= mc^2 \left(1 + \frac{p^2}{2m^2 c^2} - \frac{1}{8} \frac{p^4}{m^4 c^4} + \dots \right) \\ &= mc^2 + \frac{p^2}{2m} + \dots = \text{origin of } \frac{p^2}{2m}\end{aligned}$$

No of microstates with energy in $(\varepsilon, \varepsilon + d\varepsilon) \sim \varepsilon^{1/2} d\varepsilon = d\Phi(\varepsilon)$ (given energy shell) $p = \sqrt{2m\varepsilon}$



$$\begin{aligned}
 dV &= 4\pi p^2 dp \\
 &= 4\pi(2m)\varepsilon \frac{(2m)}{2\sqrt{2m\varepsilon}} \\
 dv &\simeq 8\pi\sqrt{2} m^{3/2} \varepsilon^{1/2} d\varepsilon \\
 \Omega(\varepsilon) d\varepsilon &= \frac{\int d^3q \cdot \int d^3p}{h^3} \\
 &= \frac{V \cdot 4\pi p^2 dp}{h^3} \\
 &= \frac{V}{h^3} \cdot 4\pi \cdot 2m\varepsilon \cdot \sqrt{2m} \frac{1}{2\sqrt{\varepsilon}} d\varepsilon \\
 \omega(\varepsilon) &\sim \varepsilon^{1/2} d\varepsilon = g(\varepsilon) d\varepsilon
 \end{aligned}$$

Density of states $(g(\varepsilon) = \varepsilon^{1/2}) = g_i$ (degeneracy at Energy ε).

{no. of microstates per unit energy interval at energy ε }

for n-D : $V \propto \int p^{n-1} dp$ ($\Omega \propto \varepsilon^{n/2}$) for $(n \geq 2)$.

case

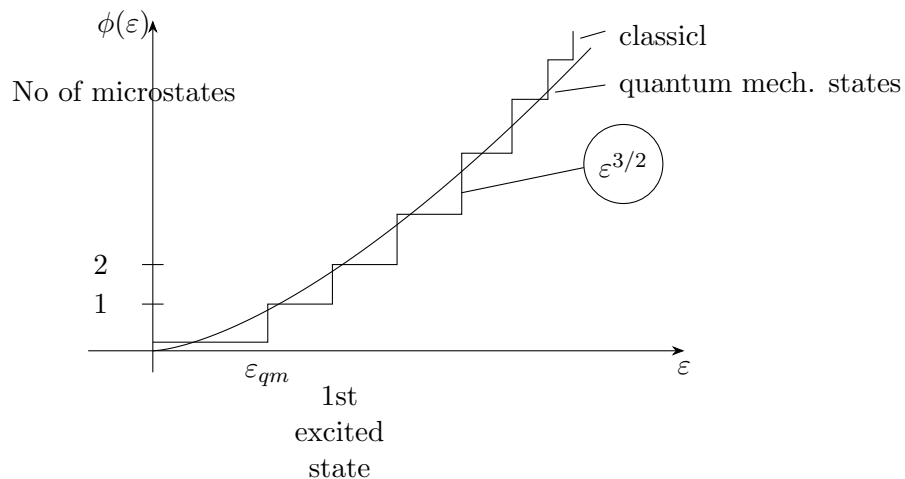
$$\begin{aligned}
 \text{for 2-D container } dv &= 2\pi p dp = 2\pi\sqrt{2m\varepsilon} \frac{2m}{2\sqrt{\varepsilon}} \\
 &= \text{Konstant}
 \end{aligned}$$

density of states becomes constant. in 2-D

case for 1-D

$$\phi(\varepsilon) = \frac{d}{d\varepsilon} \left(\frac{V}{h^3} \varepsilon^{1/2} \right) = \frac{1}{2} \varepsilon^{-1/2} \frac{V}{h^3}$$

Density of states $\uparrow$ as $\varepsilon \downarrow$



As $\varepsilon \uparrow$ no of states increases [3-D case]

(microscopic)

Now, we want to find $\Omega(E)$ so we will like to assume again that particles are free and

$$E = E_1 + E_2 + E_3 + \cdots E_N$$

Sum of energy of individual particles = total energy

For any two particles with '3' states, together they occupy $3^2 = 9$ states.

$$\Omega(E) \propto \varepsilon^{\alpha N} \propto E^{\alpha N}$$

{ if one particle can take $\omega(\varepsilon)$ states = $k\varepsilon^{1/2}$ with energy ε & $\varepsilon + d\varepsilon$

independent N particles will take — $k\varepsilon^{(1/2)N}$ }

if all ε_i are of same order of magnitude ε } approximation

$$E = N\varepsilon$$

So, $\Omega(E) \propto \varepsilon^{\alpha N} \propto \left(\frac{E}{N}\right)^{\alpha N}$.

for very large N , $\Omega(E)$ is astronomically large no.

$$\alpha = \frac{3}{2} \text{ or } \frac{1}{2} \text{ or } \frac{5}{2} \cdots$$

Lecture 22 (time 30:15)

Note

* Most probable macro state i.e. $P(E)$ is maximum = {eq^m state}

So the system which has largest no of microstates, that macrostate is the most probable state.

For Most probable macrostate it implies;

$$\frac{\partial P(E)}{\partial E} = 0 \quad P(E) = \frac{\Omega(E) \Omega'(E')}{\Omega_{Tot}(E_{Tot})}$$

as $P(E)$ is very large, let's take $\log P(E)$ & differentiate.

$$\frac{\partial}{\partial E} (\ln P(E)) = 0$$

Probability

$$dP = \frac{\partial P}{\partial E} dE + \frac{\partial P}{\partial V} dV + \frac{\partial P}{\partial N} dN$$

for eq^m

$$\left(\frac{\partial P(E)}{\partial E} = \frac{\partial P}{\partial V} = \frac{\partial P}{\partial N} = 0 \right)$$

$$\frac{\partial (\ln \Omega(E))}{\partial E} + \frac{\partial}{\partial E} (\ln \Omega'(E')) = 0$$

$$E + E' = E_{tot}$$

$$E' = E_{tot} - E \quad \Rightarrow \quad dE' = -dE$$

there fore

$$\boxed{\frac{\partial \ln(\Omega(E))}{\partial E} = \frac{\partial \ln \Omega'(E')}{\partial E'}}$$

condⁿ for thermal eq^m / most probable macrostate.

A, B two different kind of matter fluid, (gas - liquid).

$$\frac{\partial}{\partial E} (\ln \Omega(E)) = \frac{\partial}{\partial E'} (\ln \Omega'(E')) = [E^{-1}] \rightarrow \text{dimensions}$$

or,

$$\text{some property of } A = \text{some property of } B = \frac{1}{k_B \text{Temp}} \quad \} \underline{\text{Imp.}}$$

defⁿ of temperature

$$(dE = TdS - Pdv + \mu dN$$

$$T = \left. \frac{\partial E}{\partial S} \right|_{V, N}; S = k_B \ln \Omega \Bigg).$$

lets define fun. $\beta(E)$ so for this property of A/B .

$$\boxed{\beta(E) = \frac{\partial \ln \Omega(E)}{\partial E}} = \frac{1}{k_B T(E)}$$

By defⁿ in equilibrium

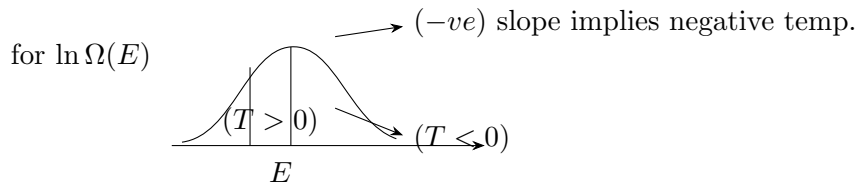
$$\boxed{T_A = T_B}$$

“ Any two parts have same temp in large isolated system under thermal equilibrium”. (large d.o.f).

“ ofcourse each part must be large enough to talk about temp, as temp. is not defined for a single particle”.

If $\Omega(E)$ is not strictly increasing function and if so

$$\frac{\partial}{\partial E} \ln(\Omega(E)) < 0 \Rightarrow \text{negative temperature.}$$



So possibility of $(-ve)$ temp is built in our defⁿ of temp.

$$\text{since } \Omega = \Omega(N, V, E) \quad \text{from } \left(\frac{\partial \ln \Omega}{\partial V} = 0 \right) \& \left(\frac{\partial \ln \Omega}{\partial N} = 0 \right)$$

$$\frac{\partial \ln \Omega}{\partial V} = \frac{\partial \ln \Omega'}{\partial V'} \equiv \left(\frac{P_A}{T_A} = \frac{P_B}{T_B} \right) \Rightarrow (P_A = P_B)$$

$$\frac{\partial \ln \Omega}{\partial N} = \frac{\partial \ln \Omega'}{\partial N'} \Rightarrow \left(\frac{\mu_A}{T_A} = \frac{\mu_B}{T_B} \right) \Rightarrow (\mu_A = \mu_B)$$

(in thermodynamic terms)

$$\begin{aligned} E &= TS - PV + \mu N \\ S &= \frac{E}{T} + \frac{PV}{T} - \frac{\mu N}{T} \\ \left(\frac{\partial S}{\partial N} = \frac{-\mu}{T} \right) ; \left(\frac{\partial S}{\partial V} = \frac{P}{T} \right) \end{aligned}$$

Entropy

$$S(E, V, N) = k_B \ln (\Omega(E, V, N))$$

Thermodynamics in Nutshell :-

1st law of thermodynamics :-

$$dQ = dU + dW$$

dQ = not a perfect differential

$$\boxed{du = dQ - dW}$$

chemical pot.

$$= dU + Pdv - \mu dN \quad \rightarrow \text{Generalized fluxes.}$$

$$dQ = dU - \sum F_i dx_i$$

$\hookrightarrow$ generalized forces

$$E \leftrightarrow U$$

Entropy representation of microcanonical ensemble $S = S(E, V, N)$

$$\boxed{dS = \frac{dE}{T} + \frac{Pdv}{T} - \frac{\mu dN}{T}}$$

tells us

$$\left(\frac{\partial S}{\partial E} \right)_{V,N} = \frac{1}{T}$$

(Energy representation) $\Rightarrow dE = TdS - Pdv + \mu dN \Rightarrow E = E(S, V, N)$

9 Lecture 23

‘E’ is some kind of thermodynamic potential

where, $E = E(S, V, N)$

E = Energy

$$\begin{aligned} T &= \left(\frac{\partial E}{\partial S} \right)_{V,N} \\ P &= - \left(\frac{\partial E}{\partial V} \right)_{S,N} \\ \mu &= \left(\frac{\partial E}{\partial N} \right)_{S,V} \end{aligned}$$

$$\Leftarrow (dE = TdS - Pdv + \mu dN)$$

What is significance of above eqⁿ set ?

we know that $E(S, V, N)$'s' is not a control variable

So lets change that by legendre transformations.

$$E = E(S, V, N)$$

↓

(Internal Energy)

Assumption

" E is homogeneous function of thermodynamic variables $\underline{S, V, N}$ ".

for that we have to assume we are in thermodynamic limit of thermodynamic system i.e. $V \rightarrow \infty, N \rightarrow \infty$ & $\frac{N}{V} = S$ (fixed).

then S, V, N which increase with system size (extensive quantities) which leads to some E .

then we can assume E is homogeneous fun of S, V, N of degree 1.

Euler's theorem

for any homogenous function of degree 'r'

$$f = f(x, y, z \dots)$$

i.e. $f(\lambda x, \lambda y, \lambda z, \dots) = \lambda^r f(x, y, z, \dots)$.

or

$$\left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + \dots \right) = r f$$

Then for $E = E(S, V, N)$

$$S \left. \frac{\partial E}{\partial S} \right|_{V, N} + V \left. \frac{\partial E}{\partial V} \right|_{S, N} + N \left. \frac{\partial E}{\partial N} \right|_{S, V} = E$$

$$S(T) + V(-P) + N(\mu) = E \quad (\text{Euler's relations})$$

So E must be equal to $TS - PV + \mu N$.

or

$$E - TS + PV = \mu N$$

we also know that $G = E - TS + PV$

so $G = \mu N$

$$\text{So } \Rightarrow \mu = \frac{G}{N} = \text{Gibbs free energy per particle}$$

let us find dE

$$dE = (TdS - Pdv + \mu dN) + (SdT - v dP + N d\mu)$$

But laws of thermodynamics tells us that,

$$(dE = TdS - Pdv + \mu dN) \text{ only}$$

So it implies that,

$$SdT - v dP + Nd\mu = 0$$

$$([\text{?}]) \Rightarrow d\mu = \left(\frac{V}{N}\right) dP - \left(\frac{S}{N}\right) dT$$

$$= v dP - s dT$$

volume per particle $\leftarrow$ specific volume per particle

s = entropy per particle $\rightarrow$ specific entropy (per particle)

So, $d\mu = v dP - s dT$ $\rightarrow$ Gibbs–Duhem relation

it implies, $\boxed{\mu = \mu(P, T)}$

So once we know (P, T) we know the chemical potential (μ)

If we assume ‘T’ is some kind of generalised force & ds as some generalised flux

$$\text{then, } dE = (Tds) + \sum_i F_i dX_i$$

$$dE = \sum_i F_i dX_i$$

So homogeneity arguments tells

$$E = \sum_i F_i X_i \Rightarrow \left(\sum_i X_i dF_i = 0\right).$$

$\downarrow$

(Generalised Duhem relation)

This comes from our assumption of E to be a homogeneous function of degree 1.

If E is not homogeneous function.

Legendre transforms

$$E(S, V, N) \longrightarrow H(S, P, N)$$

(Enthalpy)

$$H = E + PV$$

$$dH = dE + Pdv + v dP$$

$$= Tds - Pdv + \mu dN + Pdv + v dP$$

$$dH = Tds + v dP + \mu dN$$

So $H = H(S, P, N)$

$$E(S, V, N) \longrightarrow F(T, V, N)$$

$$F = E - TS$$

$$dF = TdS + \mu dN - Pdv - TdS - SdT$$

$$dF = -SdT - Pdv + \mu dN \quad \Rightarrow F = F(T, V, N)$$

$$E(S, V, N) \longrightarrow G(T, P, N)$$

$$G = E - TS + PV. \quad (\text{Gibbs free energy})$$

$$E(S, V, N) \longrightarrow \Phi_1(S, V, \mu)$$

$$\Phi_1 = E - \mu N$$

$$E(S, V, N) \longrightarrow \Phi_2(S, P, \mu)$$

$$\Phi_2 = E + PV - \mu N$$

$$E(S, V, N) \longrightarrow \Phi_3(T, V, \mu)$$

$$\Phi_3 = E - TS - \mu N$$

But we know $E = TS - PV + \mu N$

$$\Rightarrow E - TS - \mu N = -PV$$

So $\boxed{\Phi_3 = -PV} \rightarrow$ “grand potential”

↓

The reason is that this potential Φ_3 is connected to grand canonical ensemble but others above mentioned potentials are only linked to Canonical ensemble of stat. mech.

In,

$$E = TS - PV + \mu N$$

$(T, P, \mu) \rightarrow$ Intensive variables

$(S, V, N) \rightarrow$ Extensive ”.

<u>Field variables (Intensive)</u>	<u>State variables. (extensive)</u>
T	S
P	V
μ	N
Stress σ_{ij}	Strain $\epsilon_{ij}(= q_i)$
$\vec{E}$	$P \rightarrow$ polarisation per unit volume
$\vec{H}$	M ; magnetic dipole moment per unit volume

$$dF = -s dT - P dv + \mu dN \quad \text{and} \quad ; \quad dG = -s dT + v dP + \mu dN$$

lets look at these 2 relations,

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V, N}$$

So,

$$\left(\frac{\partial F}{\partial T} \right)_{V, N} = \left(\frac{\partial G}{\partial T} \right)_{P, N}$$

Similarly,

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T, N}$$

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T, V}$$

It says these thermodynamic potentials ($U, G, F, H \dots$), their 1st partial derivative with their individual variables on which they depend are other thermodynamic variables.

$$\begin{aligned}
dQ &= dU + dW \\
TdS &= dE + P dv - \mu dN \\
\left(\frac{dQ}{dT}\right)_{V,N} \text{ or } \left(\frac{\partial Q}{\partial T}\right)_{V,N} &= C_V = T \left(\frac{\partial S}{\partial T}\right)_{V,N} = \left(\frac{\partial E}{\partial T}\right)_{V,N} \quad +0-0 \\
C_V &= T \left(\frac{\partial S}{\partial T}\right)_{V,N} \quad \text{but,} \quad S = - \left(\frac{\partial F}{\partial T}\right)_{V,N}
\end{aligned}$$

So

$$C_V = -T \left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N}$$

response functions are 2nd derivative of thermodynamic potentials

But we know C_V can not be negative } lesea teliers principle.

So,

$$_V \left(\frac{\partial^2 F}{\partial T^2} < 0\right).$$

$$C_P = T \left(\frac{\partial S}{\partial T}\right)_{P,N} = -T \left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} \quad \text{as,} \quad \left(S = - \left(\frac{\partial G}{\partial T}\right)_{P,N}\right).$$

also

$$(C_P > C_V) \quad \text{so,} \quad \left|\frac{\partial^2 G}{\partial T^2}\right| > \left|\frac{\partial^2 F}{\partial T^2}\right|.$$

Isothermal compressibility (fixed T,N)

$$K_T = \frac{1}{\text{bulk modulus}} = \frac{-\Delta v}{V \Delta P} = -\frac{1}{v} \left(\frac{\partial v}{\partial P}\right)_{T,N} \quad (\text{as } P \uparrow \quad v \downarrow)$$

Since $\left(V = \left(\frac{\partial G}{\partial P}\right)_{T,N}\right)$

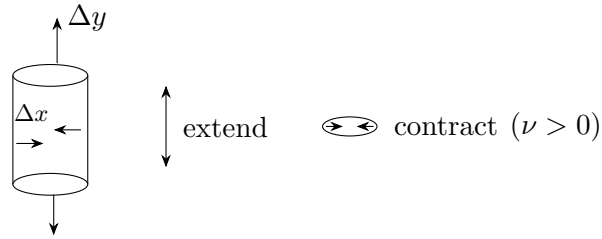
$$\left(\frac{\partial v}{\partial P}\right)_{T,N} = \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}$$

$$K_T = -\frac{1}{v} \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}$$

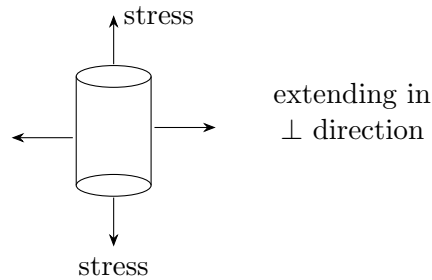
Poisson's ratio (ν) :-

Poisson ratio is measure of Poissons effect, that describes the expansion or contraction of a material in direction perpendicular to the direction of loading

$$\left(\nu = \frac{\Delta x}{\Delta y} = \frac{\text{contraction strain}}{\text{extension strain}}\right)$$



It's possible that ($\nu < 0$) such that upon extending in one direction object/material also expand in $\perp$ dir.



It turns out $\boxed{-1 < \nu \leq \frac{1}{2}}$

For ideal gas

$$\begin{aligned}
 PV &= nRT \\
 P dv + v dP &= nR dT \\
 v dP &= -P dv \\
 \frac{dv}{v} &= -\frac{dP}{P} \\
 \Rightarrow K_T &= -\frac{1}{v} \frac{\partial v}{\partial P} = -\frac{1}{V} \cdot \left(\frac{-v}{P} \right) = \frac{1}{P}
 \end{aligned}$$

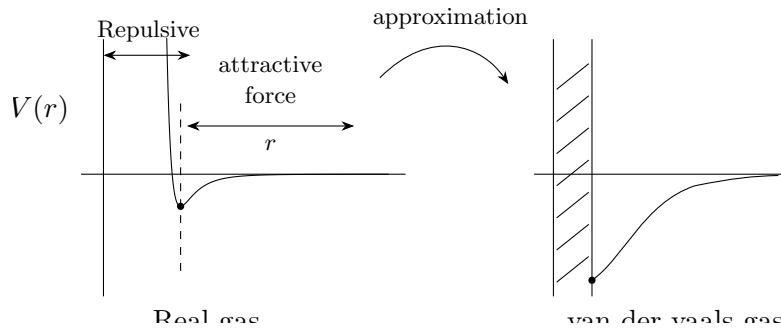
If K_T is high pressure is low.

for adiabatic process

$$\begin{aligned}
 P v^\gamma &= \text{const} \\
 dP v^\gamma + P \gamma P^{\gamma-1} dV &= 0 \\
 v^{\gamma-1} (v dP + \gamma P dv) &= 0 \\
 \Rightarrow K_T &= -\frac{1}{v} \frac{\partial v}{\partial P} = +\frac{1}{v} \cdot \frac{v}{\gamma P} = \frac{1}{\gamma P} \\
 \left(K_{S,N} &= \frac{1}{\gamma P} \right).
 \end{aligned}$$

What happens if we have van der vaals gas :-

$$P = \frac{nRT}{V - bn} - \frac{an^2}{V^2}$$



{This repulsion arose from pauli exclusion principle

$$V(r) = V_0 \left[\left(\frac{a}{r} \right)^{12} - \left(\frac{a}{r} \right)^6 \right] \quad \text{"6-12" potential} \quad \left(\begin{array}{l} \text{lennard-Jonnes} \\ \text{potential} \end{array} \right)$$

$\left(\frac{a}{r} \right)^6 \rightarrow$ attractive force potential

$\left(\frac{a}{r} \right)^{12} \rightarrow$ need quantum mech effect. (repulsive term).

Attractive force term arise from dipole moment attraction.



Random fluctuations can cause dipole moment.

interaction energy of any of dioples goes like

$$(U) \sim \vec{p} \cdot \vec{E}$$

greater the field greater the seperation

So

$$\vec{p} = \alpha \vec{E}$$

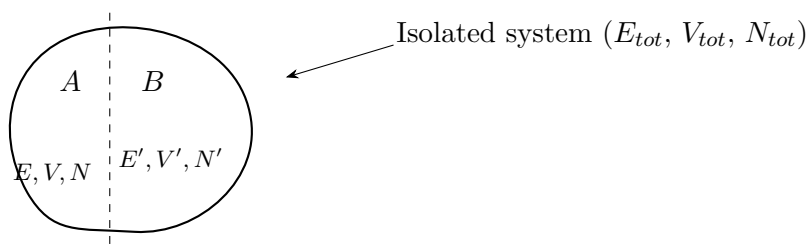
$$U \sim \alpha |E|^2 \quad \vec{E} \text{ of dipole} = \frac{2kp}{r^3}$$

$$U \propto E^2 \propto \frac{1}{r^6}$$

This formula holds true only for molecules with temperory dipole moment.

10 Lecture 24

Let us try to understand the distribution of energy in a subsystem of a (large isolated system.



condⁿ of eqⁿ implies,

$$\frac{\partial \ln \Omega}{\partial E} = \frac{\partial \ln \Omega'}{\partial E'} = \frac{1}{k_B T}$$

$$(T_A = T_B).$$

Probability that subsystem A has energy E

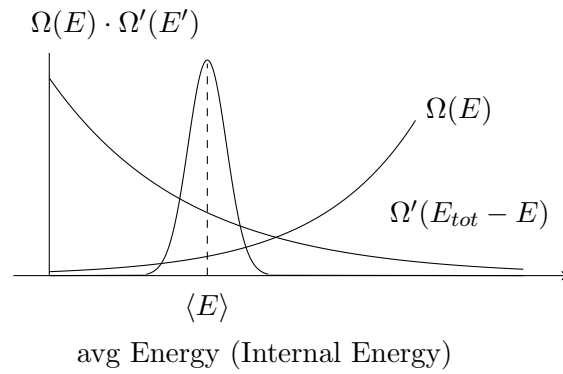
$$P(E) = \frac{\text{No of microstates of total system such that A has energy 'E'}}{\text{Total no of accessible microstates of system with energy } E_{tot}}.$$

$$P(E) = \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{tot}(E_{tot})}$$

where $(0 \leq E \leq E_{tot})$.

We know that

$$\Omega'(E') = \Omega'(E_{tot} - E) \propto (E_{tot} - E)^N$$



$$\ln P(E) = \ln \Omega(E) + \ln \Omega'(E_{tot} - E) - \ln \Omega_{tot}(E_{tot})$$

let us expand $\ln P(E)$ about some point $\bar{E}$ or $\langle E \rangle$.

$$\ln P(E) \approx \ln P(\bar{E}) + (E - \bar{E}) \left. \frac{\partial \ln P(E)}{\partial E} \right|_{(E=\bar{E})}^{=0} + \frac{(E - \bar{E})^2}{2!} \left. \frac{\partial^2 \ln P(E)}{\partial E^2} \right|_{E=\bar{E}}$$

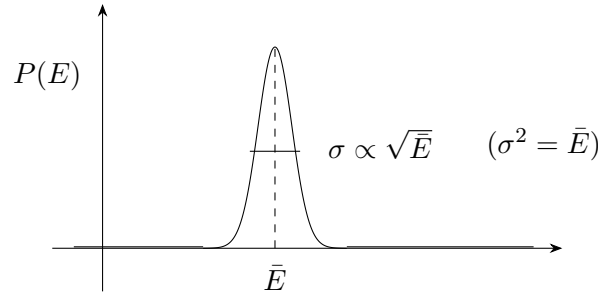
Since $P(E)$ is max at $E = \bar{E}$, $\left(\frac{\partial \ln P(E)}{\partial E} \right)_{\bar{E}} = 0$

$$\ln P(E) = \ln P(\bar{E}) + \frac{(E - \bar{E})^2}{2!} \left. \frac{\partial^2 \ln P(E)}{\partial E^2} \right|_{\bar{E}}$$

Since slope of slope at $\bar{E}$ is negative.

$$P(E) = P(\bar{E}) \cdot e^{-K \frac{(E - \bar{E})^2}{2!}}$$

$$P(E) = (\text{Normalisation const}) \cdot e^{-\alpha \frac{(E - \bar{E})^2}{2\bar{E}}} \quad \text{'}\alpha\text{' to make dimensionless.}$$



So relative fluctuations. goes like $\frac{\sqrt{\bar{E}}}{\bar{E}} = \frac{1}{\sqrt{\bar{E}}}$

$$\left(\bar{E} = N \frac{f}{2} k_B T \right)$$

relative fluctuations about mean $\propto \frac{1}{\sqrt{N}}$

So for ($N \approx 10^{24}$) fluctuations in eq^m about mean value in avg Energy $\propto 10^{-12}$. which is negligible. That is why in thermodynamics we only talk about $\langle E \rangle$ or $\langle E \rangle$ rather than $\langle E_{tot} \rangle$.

$$\begin{aligned} \ln P(\varepsilon) &= \ln \Omega'(E') - \ln \Omega_{tot}(E_{tot}) \\ &= \ln \Omega'(E_{tot} - E) - \ln \Omega_{tot}(E_{tot}) \end{aligned}$$

Since ($E_{tot} \gg \varepsilon$) let us expand $\ln \Omega'(E')$ about (E_{tot}).

$$\ln P(\varepsilon) = \left(\ln \Omega'(E_{tot}) - \varepsilon \left. \frac{\partial \ln \Omega'(E')}{\partial E'} \right|_{E'=E_{tot}} + \dots \right) - \ln \Omega_{tot}(E_{tot})$$

$$P(\varepsilon) = (\text{const}) e^{-\beta \varepsilon} \quad \left| \quad \beta = \frac{\partial \ln \Omega'(E')}{\partial E'} = \text{inverse temp} = \frac{1}{k_B T} \text{ of heat bath} \right.$$

to find (const) we need to know $\Omega'(E_{tot})$ (info about heat bath). but we don't need that too, we can normalize $P(\varepsilon)$ and find normalisation const.

$$P(\varepsilon) = \frac{e^{-\beta \varepsilon}}{\sum_{\varepsilon} e^{-\beta \varepsilon}}$$

So Normalisation const. is fixed by properties of subsystem alone and we eliminate whatever happens outside"

for a moment, let us assume (A) has discrete energy levels.

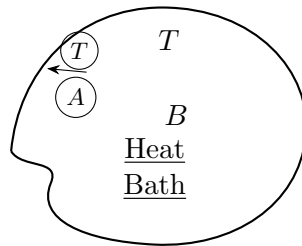
$$\left. \begin{array}{l} \text{degenerate} \\ \text{states} \end{array} \right\} \begin{array}{l} \text{---} \text{---} \text{---} \text{---} \varepsilon_3 \\ \text{---} \text{---} \varepsilon_2 \\ \\ \text{---} \varepsilon_1 \\ \text{---} \varepsilon_0 \end{array}$$

$$P(\varepsilon_i) = \frac{e^{-\beta\varepsilon_i}}{\sum_j e^{-\beta\varepsilon_j}} \quad \rightarrow \text{summed over all states not levels.}$$

The only information about heat bath to be known is its temp. unlike (case of microcanonical ensamble where $E = \bar{E}$ at eq^m) The total energy of system in thermal eqⁿ with heat bath is not fixed. It is fluctuating with in diff energy with $P(\varepsilon_i) = e^{-\beta\varepsilon_i} / \sum_i e^{-\beta\varepsilon_j}$ but the avg. value of $\langle \varepsilon_i \rangle$ is independent of time, because $P(\varepsilon_i)$ is independent of time

The factor $\left(\sum_{\text{states } j} e^{-\beta\varepsilon_j} \right)$ is called ‘Z’ the canonical partition function.

and our system

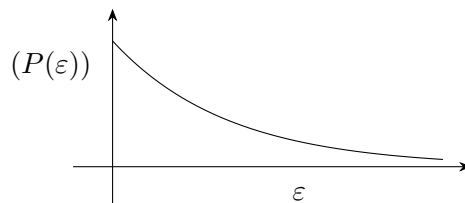


in Thermal-eq^m at temp T is called “canonical ensamble”.

$$\left(Z = Z(T, V, N) = \sum_{\text{states } j} e^{-\beta\varepsilon_j} = \sum_{\text{levels } i} g_i e^{-\beta\varepsilon_i} \right)$$

The subsystem “A” in thermal eq^m of this kind is not in fixed energy state. It has got probability distributions for certain energy values. For a quantum system “A” is not even in pure energy eigen state. It will be in superposition of all energy eigen states. {mind blowing fact.

$$(P(\varepsilon) \propto e^{-\beta\varepsilon_i}).$$



distribution of energy goes like exponential decay rather than steep gaussian distribution.

β = property of bath.

In terms of energy levels,

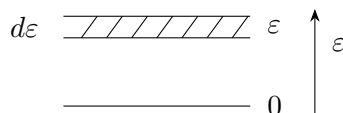
$$Z = \sum_{\text{levels } j} (g_j) e^{-\beta\varepsilon_j}$$

$\rightarrow$ degeneracy of level ε_j .

for continuous energy levels,

$$P(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon}.$$

now instead of counting degeneracy (g_j) we ask



how many states with energy ε & $\varepsilon + d\varepsilon$

$$= g(\varepsilon) d\varepsilon$$

$\hookrightarrow$ density of states

$g(\varepsilon)$ = no of states in $d\varepsilon$ range per unit energy ε

$$Z(T, V, N) \equiv Z = \int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon \quad (\text{general canonical partition fun.})$$

So,

$$p(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon}.$$

So Z is Laplace transform of $g(\varepsilon)$, in which transform variable is β , instead of 's'. (Laplace transform or Moment generating functions for continuous variables)



$$\left. \begin{aligned} g(\varepsilon) &\longrightarrow \text{system properties} \quad (\text{mech. aspect classical/quantum}). \\ e^{-\beta\varepsilon} &\longrightarrow \text{statistical aspect.} \end{aligned} \right\}$$

$$e^{-\beta\varepsilon} g(\varepsilon) \longrightarrow \underline{\text{stat. mech.}}$$

(Q) What is average energy?

$$\begin{aligned} \langle E \rangle &= \sum_{(\text{states } j)} P(E_j) E_j \\ &= \frac{\sum_{(\text{states } i)} E_i e^{-\beta E_i}}{Z(T, V, N)} \\ &= \frac{\sum_{(\text{states } i)} E_i e^{-\beta E_i}}{\left(\sum_{(\text{states } j)} e^{-\beta E_j} \right)} = \frac{1}{Z} \left(\sum_{(\text{states } i)} E_i e^{-\beta E_i} \right). \end{aligned}$$

$$\boxed{\langle E \rangle = -\frac{\partial(\ln Z)}{\partial \beta}} \quad \left(\frac{-1}{Z} \cdot \frac{\partial Z}{\partial \beta} = +\frac{1}{Z} \sum E_i e^{-\beta E_i} \right)$$

$$= -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

So, ($C_v \geq 0$) $\therefore$

In thermodynamics we only deal with $\langle E \rangle$. So we can't calculate (C_v) or $\underline{C_p}$. So C_p, C_v are input in Thermo-dynamics.

11 Lecture 25: Connection with thermodynamics

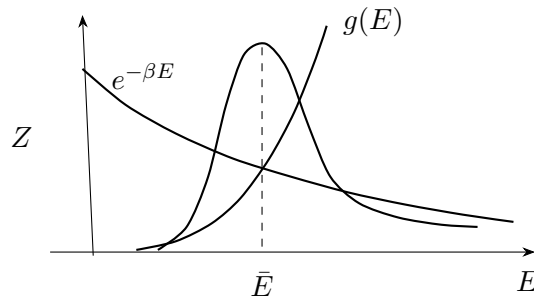
Recall that the canonical partition function of small subsystem in heat bath of temp. $T = \frac{1}{k_B \beta}$

$$Z = \sum_{\text{states } j} e^{-\beta E_j} \longrightarrow \int_0^\infty dE e^{-\beta E} g(E)$$

$$g(E) \propto E^{1/2}$$

$$g(E) \propto E^{\frac{N}{2}} \text{ or } \propto E^N \quad \text{for } \underline{N = 10^{24}}.$$

Lets plot (Z vs E)



$$Z = \int_0^\infty dE e^{-\beta E} g(E) = e^{-\beta \bar{E}} \Omega(\bar{E})$$

(?) maybe $\sim$ (by me) $(\Omega(E) \approx \Omega(\bar{E}) = \int_0^\infty g(E) dE)$

$\Omega(\bar{E})$ = total no of microstates

So,

$$\ln Z = \ln \Omega(\bar{E}) - \beta \bar{E}$$

$$-k_B T \ln Z = -k_B T \ln \Omega(\bar{E}) + \bar{E}$$

But we know that entropy (S) is defined as

$$\boxed{S = k_B \ln \Omega}$$

$$\begin{aligned}
-k_B T \ln Z &= -TS + \bar{E} \quad (\bar{E} = \langle E \rangle) = U \text{ (Internal energy)} \\
&= F/A \text{ (Hemholtz free energy).} \\
&\Rightarrow \boxed{Z = e^{-\beta F}}
\end{aligned}$$

So, the whole idea was to write

$$Z = \sum_{\text{states } j} e^{-\beta E_j} \quad \text{in terms of} \quad e^{-\beta E_{\text{eff}}}$$

$\hookrightarrow$ effective energy

$$(Z = e^{-\beta F}). \quad (F = \text{Helmholtz free energy}).$$

It needs coherent punch from bath molecules to provide maximum energy to subsystem 'A'. So it is less likely to get high energy from a heat bath; that is the reason the energy distribution goes like

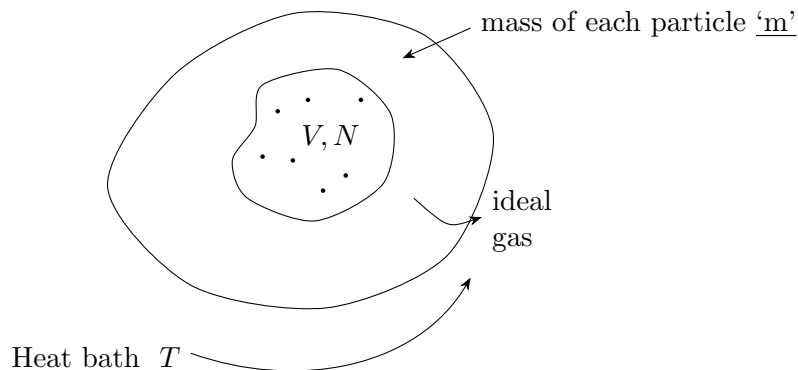
$$P(E_i) = \frac{e^{-\beta E_i}}{\sum_{\text{states } j} g_j e^{-\beta E_j}}$$

Let us find Pressure:

We know that

$$\begin{aligned}
F &= U - TS \\
dF &= dU - TdS - SdT \\
&= TdS - PdV + \mu dN - TdS - SdT \\
dF &= -SdT - PdV + \mu dN \\
P &= -\left(\frac{\partial F}{\partial V}\right)_{T,N}
\end{aligned}$$

(Partition function for classical ideal gas):



Assume, All particles are independent of each other and there is no interaction b/w the particles.

$$Z = \sum_{\substack{\text{states} \\ \text{of} \\ \text{gas}}} e^{-\beta E} = \sum e^{-\beta(\varepsilon(1)+\varepsilon(2)+\varepsilon(3)+\dots)}$$

where $\varepsilon(i)$ is energy of i th particle of whole system.

If all particles are independent of each other, then partition function factorize into –

$$Z = \sum e^{-(\beta\varepsilon + \beta\varepsilon + \beta\varepsilon + \dots)} = \sum_{\text{states of gas}} \left(e^{-\beta\varepsilon} \cdot e^{-\beta\varepsilon} \cdot e^{-\beta\varepsilon} \dots \right)$$

$$Z = \left(\sum_{\text{states of any particle}} e^{-\beta\varepsilon} \right)^N$$

→ Only true if particles are distinguish-able.

$$Z = \left(\int_0^\infty e^{-\beta\varepsilon} \underbrace{g(\varepsilon)d\varepsilon}_{\Omega(E)} \right)^N = \left(\frac{1}{h^3} \left(\int_V d^3r \int d^3p e^{-\frac{\beta p^2}{2m}} \right) \right)^N$$

$$Z = \left(\int_0^\infty e^{-\beta E} \Omega(E) dE \right)^N$$

$$= \left(\frac{V}{h^3} \left(\int e^{-\frac{\beta p_x^2}{2m}} dp_x \int e^{-\frac{\beta p_y^2}{2m}} dp_y \int e^{-\frac{\beta p_z^2}{2m}} dp_z \right) \right)^N$$

$$= \left(\frac{V}{h^3} \left((2\pi m k_B T)^{1/2} \cdot (2\pi m k_B T)^{1/2} (2\pi m k_B T)^{1/2} \right) \right)^N$$

$$Z = \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)^N$$

$$\Omega(\varepsilon) = \frac{1}{h^3} \int_V d^3r \int_{-\infty}^\infty d^3p \quad \left(\int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \right)$$

$$Z = e^{-\beta F}$$

$$\ln Z = -\beta F \Rightarrow \left(F = \frac{-1}{\beta} \ln Z \right) = -k_B T N \ln \left(\frac{V (2\pi m k_B T)^{3/2}}{h^3} \right)$$

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T,N} = \frac{1}{\beta} \cdot \frac{1}{Z} \cdot \frac{\partial Z}{\partial V} \Big|_{T,N}$$

$$= \frac{1}{\beta Z} \cdot \frac{1}{h^3}$$

$$= N k_B T \cdot \frac{1}{V \left(\frac{(2\pi m k_B T)^{3/2}}{h^3} \right)} \cdot \frac{(2\pi m k_B T)^{3/2}}{h^3}$$

$$P = \frac{N k_B T}{V} \Rightarrow \boxed{PV = N k_B T}$$

When we will do Q-M ; $E = pc$ for photons in black body cavity at temp T . then

$$\bar{E} \neq \frac{3}{2} k_B T \quad (\text{even in 3-D}).$$

$\hookrightarrow E = p^1 c$ So

$$\underline{\bar{E} \propto 3k_B T^4 f(\cdot)}$$

$\hookrightarrow$ Bose statistics.

So,

$$PV = Nk_B T \quad ; \quad U = \frac{3}{2} Nk_B T$$

$$U = \frac{3}{2} \cdot PV$$

$$\Rightarrow \boxed{PV = \frac{2U}{3}}$$

$\rightarrow$ avg energy density.

$$\text{or, } \left(P = \frac{2u}{3} \right) \quad \left(u = \frac{U}{V} = \text{avg Energy per unit volume} \right).$$

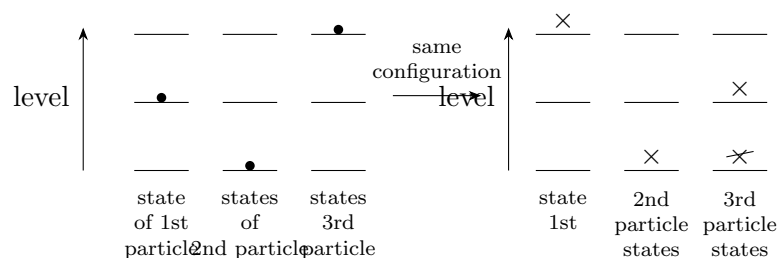
This relation is even true for quantum gases. (quantum ideal gas).

(Q) Is it correct that

$$Z = \sum e^{-\beta E} = \left(\sum e^{-\beta \epsilon} \right)^N ?$$

summed over all states of gas summed over states of any particle

No, its not true (always) here we have overcounted the number of states, since the particles are indistinguishable. (physical fact).



since particles are indistinguishable, we have overcounted the number of states.

In quantum statistics we will correct this mistake.

Instead of asking in which level particle 1 is.. and 2 is.. we ask how many particles are in ground state, 1st excited state. and so on....

So in nutshell in quantum statistics we do row wise counting instead of column wise counting performed in classical statistics. From here idea of occupation no. arose in quantum statistics.

$$dF = -SdT - PdV + \mu dN$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V,N} \quad ; \quad e^{-\beta F} = Z$$

$$\left(F = \frac{-1}{\beta} \ln Z \right)$$

and,

$$Z = \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)^N$$

$$\Rightarrow F = \frac{-N}{\beta} \ln \left(\frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} \right) = -Nk_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)$$

So,

$$\begin{aligned} S &= - \left(\frac{\partial F}{\partial T} \right)_{V,N} = +Nk_B \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \\ &\quad + \frac{Nk_B T}{\frac{V}{h^3} (2\pi m k_B T)^{3/2}} \cdot \frac{V}{h^3} \cdot \frac{3}{2} (2\pi m k_B T)^{1/2} (2\pi m k_B) \\ &= -Nk_B \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) + \frac{3}{2} Nk_B \end{aligned}$$

at $T = 0$ $\ln \left(\frac{V}{h^3} (0) \right) \rightarrow \infty$, so ideal gas is no more a good assumption for our real gas, there are interactions

and physical real gas condenses to liquid & solid which also can't be explained by assuming gas to be ideal.

Also this formula of entropy is more flawed and it is due to incorrect partition function arose from overcounting of states (also known as Gibbs paradox).

Let us find

$$\begin{aligned} \mu &= + \frac{\partial F}{\partial N} \Big|_{T,V} \quad ; \quad F = -Nk_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \\ \mu &= -k_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \end{aligned}$$

this formula is also not sensible

Even the formula for Free energy $F = -Nk_B T \ln \left(\frac{V}{h^3} ()^{3/2} \right)$ as, we expect free energy to be extensive quantity.

The 'V' in $\ln \left(\frac{V}{h^3} ()^{3/2} \right)$ is trouble here.

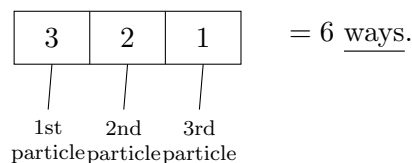
($F \propto$ any one extensive variable

$$F = Nf(\text{intensive variable}).)$$

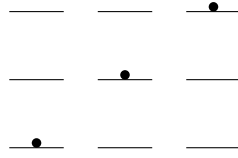
Correction to partition function :-

There are ($n!$ no of ways) particles can be arranged

ex 3 particles.



So A rough way to correct Z is to divide by ($n!$) but that is only true if no two particles attains same (state)

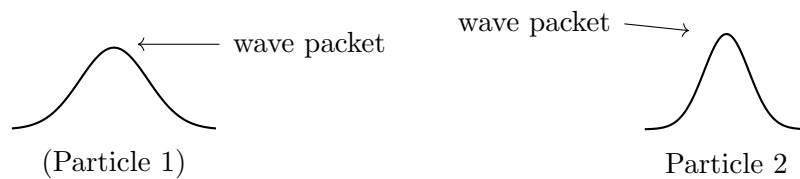


But that is only very likely if no. of states available

Since particles are quantum object, we have to move to quantum statistics for more precise understanding.

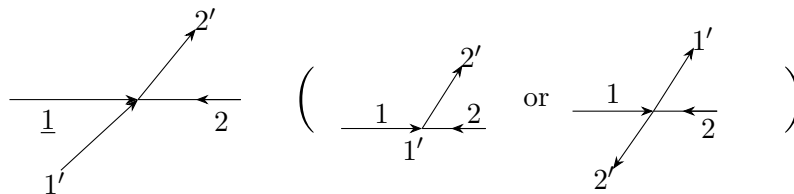
(Criterion for classical statistics to be valid) :-

In Q.M. particles are indistinguishable and x & p of particle can't be precisely defined instantaneously.



We can say that particle 1 is at this position $(x_1 \pm \Delta x_1)$ & particle 2 is at some other position $(x_2 \pm \Delta x_2)$, iff the avg distance b/w them is much larger than spread of wave packets. (i.e. non-interacting identical particles can be distinguishable).

When particles 1 & 2 interact, we can't identify them.



(particles are exchanged).

So, for classical statistics to work

$$\text{mean interparticle separation} \gg \lambda_{\text{debroglie}}$$

we still have to divide by $(N!)$ to count states correctly.

If we have N particles in volume V then volume available for each particle is $\left(\frac{V}{N}\right)$

So linear dimension available is $\left(\frac{V}{N}\right)^{1/3}$

$$\left(\frac{V}{N}\right)^{1/3} \gg \frac{h}{p_{rms}} \quad \left(\lambda_{deB} = \frac{h}{p}\right).$$

$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{3}{2} k_B T \quad p_{rms} = \sqrt{3mk_B T}$$

for one particle

$$\sqrt{\langle p^2 \rangle} = p_{rms} \Leftarrow = (3mk_B T)^{1/2} \text{ or } (mk_B T)^{1/2}$$

$$\left(\frac{V}{N}\right)^{1/3} \gg \frac{h}{(mk_B T)^{1/2}}$$

$$\left(\frac{V}{N}\right) \gg \frac{h^3}{(mk_B T)^{3/2}} \quad \text{or} \quad \left(\frac{1}{n}\right) \gg \frac{h^3}{(mk_B T)^{3/2}}$$

$$\boxed{1 \gg \frac{nh^3}{(mk_B T)^{3/2}}} \quad \frac{nh^3}{(mk_B T)^{3/2}} = \text{degeneracy factor}$$

So

$$\boxed{\frac{nh^3}{(mk_B T)^{3/2}} \ll 1}$$

is criteria for statistical to be valid. (n = number density of particles)

eg) Nitrogen $m \approx 10^{-26}$ kg

$$h = 10^{-34}$$

$$n = 10^{24} \text{ per cubic meter}$$

$$T = 300 \text{ K}, \quad k_B = 1.38 \times 10^{-23}$$

$$\frac{10^{24} \cdot 10^{-34 \times 3}}{(10^{-26} \cdot 10^{-23} \cdot 300)^{3/2}} = \frac{10^{-102+24}}{(10^{-51})^{3/2}}$$

$$= \frac{10^{-78}}{10^{-75}} \approx 10^{-3} \ll 1.$$

So classical statistics works.

If ($n \uparrow$) this approximation fails. (like in case of liquids/neutron stars. the degeneracy factor is of order of 10^6).

Also at low temp ($T \rightarrow 0$) this limit/condition fails and we use quantum statistics.

(Lec -26 – is of probability distributions so, It will be written in mathametical notebook of V. balki sir).

12 Lecture 27

(After) (0 : 00)

(Phase transitions)

Thermodynamics is a kind of limiting case of statistical mech. where fluctuations are neglected and so on; And these fluctuations can be taken into account if we use stat. mech. and it provides correction to thermodynamic cases, goes beyond thermostatics & calculate (C_P, C_V) which are input parameters in thermodynamics.

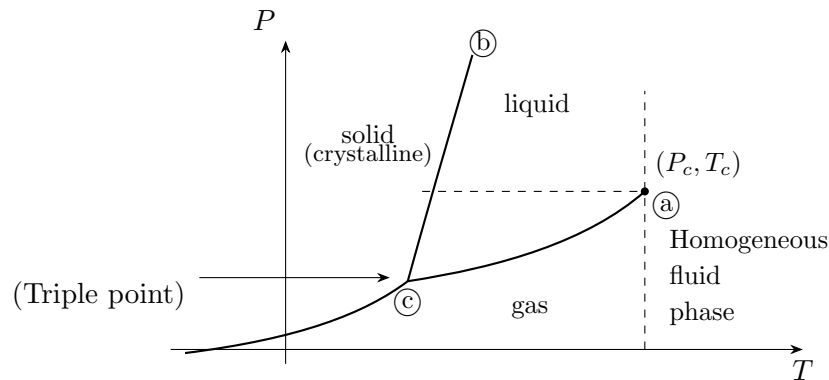
There is a place where thermodynamics fails (completely and no longer valid); it is hard problem of phase transitions of matter. In principle stat. mech. (eq^m stat. mech.) should tell us everything about system at thermal eqⁿ, including processes at phase transitions.

Thermodynamics fails at critical point (P_c, V_c, T_c).

We will use vander vaals gas to demonstrate real gases.

We have thermodynamic variables (P, V, T) out of which we only use any two of them, as third is found of using equation of state (which is different for different phases of matter – solid, liquid, gas).

Phase diagram



- Ⓐ → liquid-gas coexisting curve or (Boiling curve)
- Ⓑ → solid-liquid coexisting curve or melting curve
- Ⓒ → solid-gas coexistent curve or sublimation (curve).

every point on this phase plane is assumed to at thermal eq^m state of the system.

(phase – A homogeneous thermodynamic eq^m system)

(Ice melting of glaciers is measured using rise of water level but Ice can directly convert to vapour phase, so this is a challanging problem).

(Note :- It looks like the melting curve goes on with same slope, but it does not happen, As pressure increases solid have more densely packed atoms, at very high pressure (FCC) Face centered cubic crystals formes.

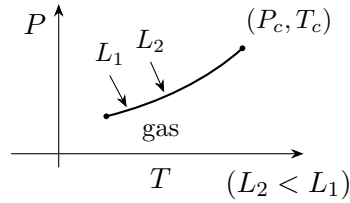
(Gibbs phase rule) :- For one kind of molecular species at most we can have only three coexisting phases. For more than one molecular species we can have more than 3 kind of phase coexisting at common point.

Critical Point (V_c, T_c, P_c) :-

The boiling curve/liquid-gas coexisting curve stops at some point on phase plane called critical point where it becomes impossible to distinguish liquid phase from gas phase.

At this point the conventional thermodynamics fails. on right of (P_c, T_c) or for $(T > T_c)$ we can't tell diff between liquid & phase. (Homogeneous fluid phase).

To convert form liquid to gas phase, we supply latent heat.



this transition is discontinuous as density of system changes discontinuously

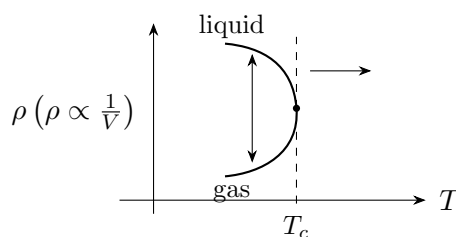
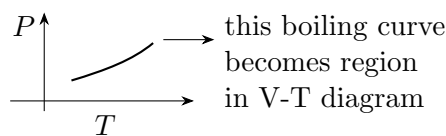
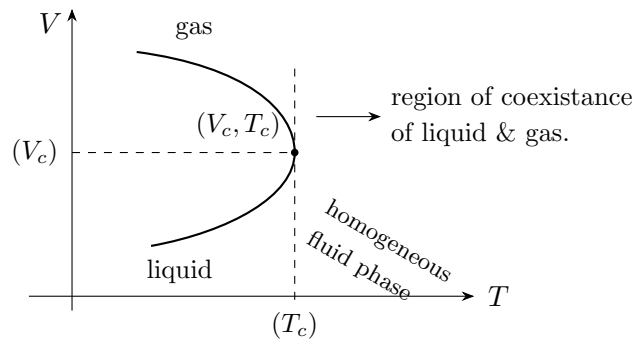
As we go up this curve (i.e. increase pressure), latent heat becomes less and less, & latent heat vanishes at critical point (i.e. we require no heat to convert from liquid to gas phase).

Also the surface tension also vanishes at critical point i.e. the meniscus (the curved upper surface of liquid) also disappears. “we have mush at critical point.”

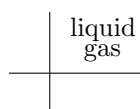
Everything that distinguish liquid from gas disappears, even the density start matching.

Lets draw melting curve in $(V - T)$ plane

volume of gas is very high

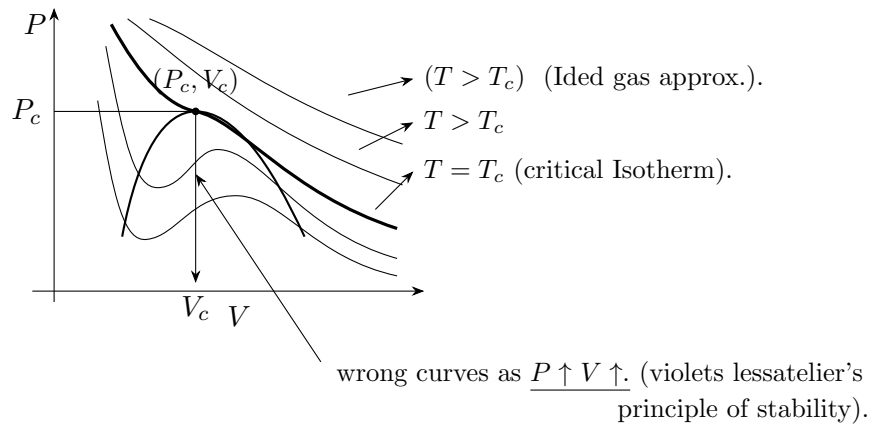


discontinuous phase transition jump in density (ρ).



Critical point is 2nd order or continuous phase transition, and coexisting curve is series of discontinuous / 1st order phase transitions.

P-V diagram :- (only for boiling curve).



$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2} \quad \text{or} \quad \left(\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT \right)$$

To find critical point (Inflection point).

$$\left. \frac{\partial P}{\partial V} \right|_{(T_c, P_c, V_c)} = 0$$

also

$$\left. \frac{\partial^2 P}{\partial V^2} \right|_{(P_c, V_c, T_c)} = 0$$

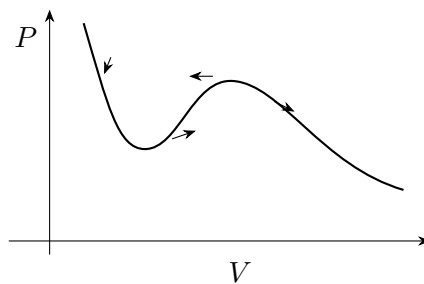
and from eqⁿ of state, we can find (P_c, V_c, T_c) .

There is no universal eqⁿ of state as eqⁿ of state depends upon interactions and interactions are hard to model universally.

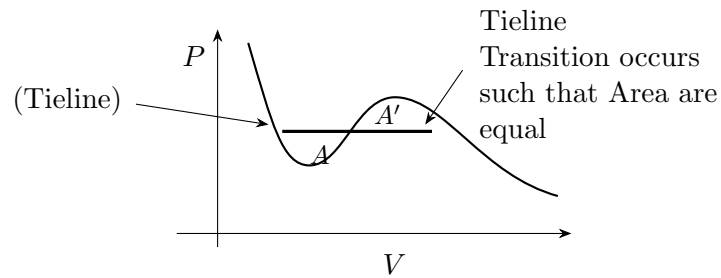
The portion where $\frac{\partial P}{\partial V}$ is positive makes compressibility

$$K_B = -\frac{1}{V} \frac{\partial V}{\partial P}$$

negative and violates thermodynamic stability. So we have to correct this portion.

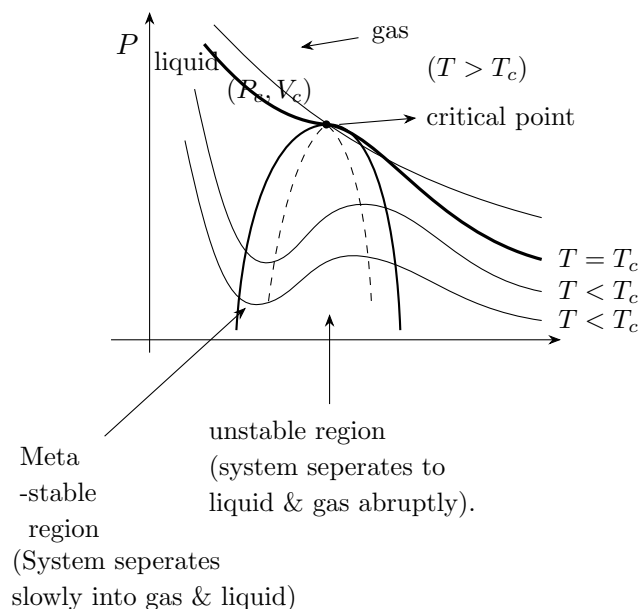


hysteresis is the problem here. So to avoid, it is observed that, flip



So some portions of allowed curve also gets omitted in observation. So vaander vaals gas eqⁿ of state is not accurate, we not only have to remove unstable region but slightly bigger region also.

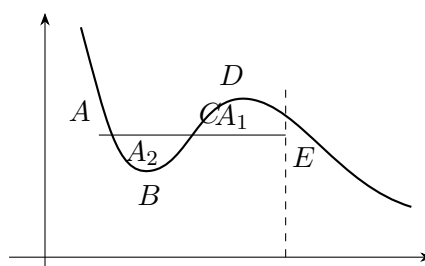
So envelope goes like this –



Any system on unstable region will instantaneously separate into some liquid & remain to gas depending upon lever rule.

Note :- In case of binary alloys (liquid alloys), the metastable portion can be made extremely slow, It may take hours, yeares to happen.

Maxwells equal Area rule :-



(maxwell's tieline construction)

(Is not quite correct though as It goes through ($\textcircled{\text{D}}$ $\rightarrow$ $\textcircled{\text{B}}$) i.e. from thermodynamic unstable region.)

let us consider $G(T, P, N) = U - TS + PV$

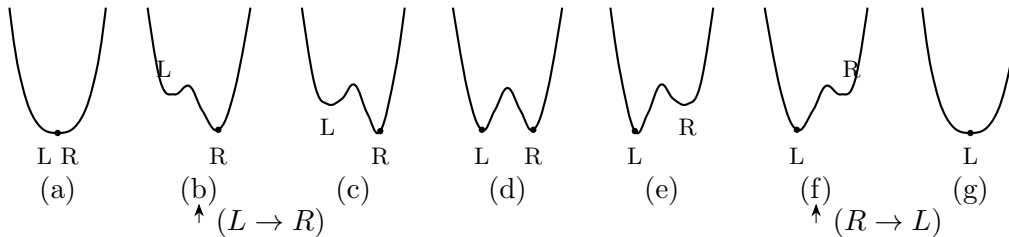
$$\begin{aligned} dG &= dU - TdS - SdT + PdV + VdP \\ &= -SdT + VdP + \mu dN \end{aligned}$$

for fixed (N, P, T) ($dG = 0$)

On an isotherm, $dG = VdP$ (as $dT = 0$).

$$\begin{aligned} G_E - G_A = 0 &= \int_A^E dG = \int_A^B VdP + \int_B^C VdP + \int_C^D VdP + \int_D^E VdP \\ &\quad \left(G = \frac{\mu}{N} \text{ should be same in gas phase \& liquid phase} \right) \\ &= \underbrace{\left(\int_A^B VdP - \int_C^B VdP \right)}_{(A_1)} + \underbrace{\left(- \int_D^C VdP + \int_D^E VdP \right)}_{(-A_2)} = 0 \\ &= \boxed{(A_1 = A_2) = A} \end{aligned}$$

Mechanical example of hysteresis (Double well potentials).



If we start at (R) i.e. right well, Transition occurs to 'L' at (f).

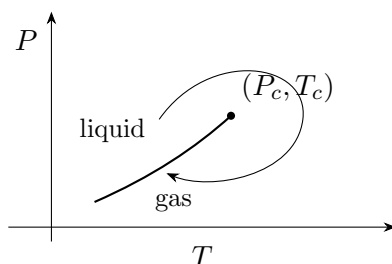
If we start at 'L' i.e. left wall (g) then transitions occur at (b). This is called hysteresis in mechanical case.

There is a branch of mathematics called catastrophe theory where people talk about change in generic potentials. (BTW) we can't have such minimum in thermal eqⁿ.

13 Lecture 28

Near critical point, many systems behave (independent of their interactions/details) identically.

We can convert from liquid to gas or gas to liquid without encountering a sharp transitions. (This is only possible because boiling curve ends at critical point).



we can't have continuous transitions from liquid to solid or vice versa. (Because this curve (melting curve) does not end. It does not end because solid has crystalline symmetry

It is symmetric under group of specific transformations like space rotations, translations. On the other hand liquid is homogeneous and isotropic, therefore liquid has much greater degree of symmetry. Solid has long range order.

order and symmetry are kind of opposites. More ordered is phase less symmetric it is; in the sense that the set of transformations under which it remains unchanged is smaller and smaller.

The most disordered state is in fact most symmetric because it looks exactly the same in every direction.

on the other hand,

liquid and gas phases are both symmetric (in every dir.).

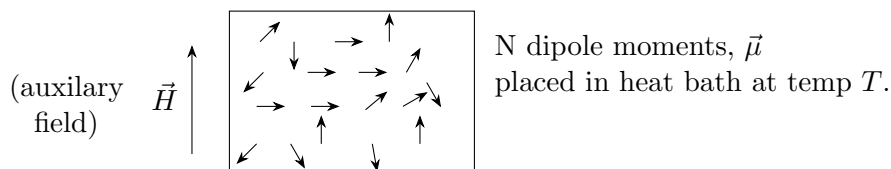
So $\exists$ a critical point

we can understand behaviour at critical point via fluid-magnet analogy.

– Model of paramagnetism

Assumption – system consists of elementary magnetic dipole moments (atomic (Spin + Angular)). oriented randomly in presence of ext. $\vec{H}_{ext}$ $\left(H = \frac{\vec{B}}{\mu_0} \right)$.

magnetic dipole moment will try to align in direction of $\vec{H}_{ext}$. So system will have some kind of symmetry.



P.E of a dipole $\vec{\mu}$ in field $\vec{H}$

$$U = -\vec{\mu} \cdot \vec{H}$$

$$U_{min} = -\mu H \quad (\theta = 0)$$

$$U_{max} = +\mu H \quad (\theta = \pi).$$

lets look at very simple model, such that

$$U = +\mu H \text{ or } -\mu H \quad (\text{one-dimension case})$$

(magnetic dipole moment of e^- can only take two values).

$$\begin{array}{ccc} E_1 & \text{-----} & (\mu H) \quad \uparrow \vec{\mu} \quad \downarrow \vec{H} \\ E_0 & \text{-----} & (-\mu H) \quad \uparrow \vec{\mu} \quad \uparrow \vec{H} \end{array}$$

lets find magnetisation (or better average Magnetisation). in $(1 - D)$

$$\vec{M} = N \text{ (Avg dipolemoments of each dipole moment).}$$

($\hookrightarrow N\langle\mu\rangle$ is only true iff all $\vec{\mu}$ are independent assuming no-dipole-dipole interactions).

(since we only want to see $\vec{M}$ along dir. of $\vec{H}$. lets drop vector)

$$\begin{aligned} M &= N \left(\frac{\mu P(\mu) + (-\mu) P(-\mu)}{P(\mu) + P(-\mu)} \right) \\ &= \frac{N (\mu e^{\beta\mu H} - \mu e^{-\beta\mu H})}{e^{\beta\mu H} + e^{-\beta\mu H}} \quad \left(\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \right). \end{aligned}$$

$$\text{Av. magnetisation } (M) = N\mu \tanh(\beta\mu H)$$

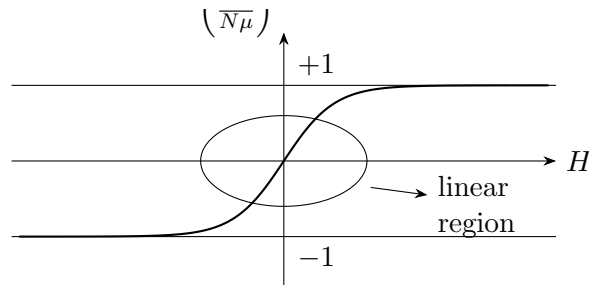
So, $\boxed{M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right)}$ is magnetic eqⁿ of state.

fluid (P, V, T) of [?] (H, M) system

$$\begin{array}{cc} \uparrow & \uparrow \\ \text{field} & \text{responce} \\ \text{(Intensive)} & \text{extensive} \end{array} \qquad \begin{array}{cc} \uparrow & \uparrow \\ \text{field} & \text{responce} \\ \text{(Intensive)} & \text{(extensive)} \end{array}$$

$\left(\frac{M}{N\mu}\right)$ vs H :-

$\left(\frac{M}{N\mu}\right)$ is magnetisation per atom per dipole moment.



our old defⁿ of susceptibility (χ)

$$\begin{aligned} \vec{M} &\propto \vec{H} \\ \vec{M} &= \chi \vec{H} \\ \frac{M}{H} &= \text{const} \end{aligned}$$

is not quite right as, this graph is non linear in most range of $\vec{H}$.

Isothermal susceptibility

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_{T,N} \Big|_{H=0}$$

or

(χ_T = slope at origin)

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \Big|_{H=0}$$

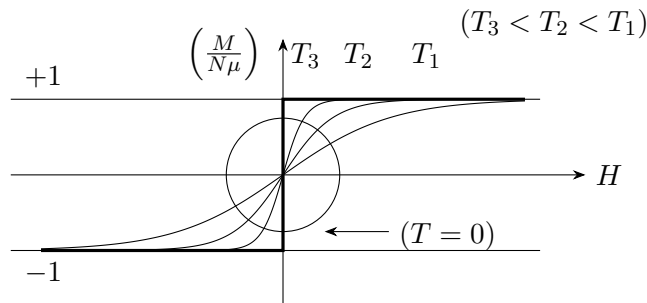
$\tanh \theta \approx \theta$ for very small θ .

$\tanh ax \approx ax$

$$= N\mu \left(\frac{\mu H}{k_B T} \right)$$

$$\left(\chi_T = \frac{N\mu^2 H}{k_B T} \right) \rightarrow \text{Curie's law (pierre curie)} \quad \left(\chi_T \propto \frac{1}{T} \right)$$

So slope at origin increases with decrease in temp.



at $T = 0$ (absolute zero)

$$\chi \rightarrow \infty$$

or slope $\rightarrow \infty$

We have neglected interaction between magnetic dipoles.

At high temperature, we can neglect dipole-dipole interaction but at very low temp, dipole-dipole interactions become large as compared to random fluctuation driven by heat bath.

All phase transitions of this kind is dependent on minimization of free energy

$$F = U - TS$$

$$U = \langle E \rangle \text{ or } \bar{E} \text{ or } \langle \mathcal{H} \rangle \quad (\text{Hamiltonian})$$

It so happens that at sufficiently low temperatures the ordering tendency of internal energy (U) overcomes effect of entropy.

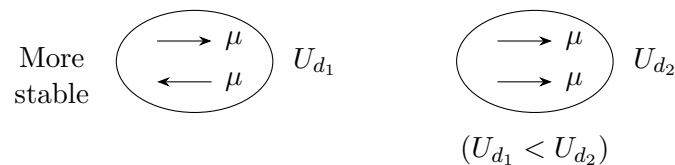
So Free energy (F) is mostly governed by U as ' T ' is very small and ' S ' (entropy) also becomes small at low temperatures ($S = k_B \ln \Omega(E)$) ($\Omega(E)$ decreases with $\underline{T}$).

So at low temp, ‘ F ’ is governed by U and it is ordered phase, and at high temp ‘ F ’ is governed by ‘ TS ’ which is very large compared to U , and this is disordered phase.

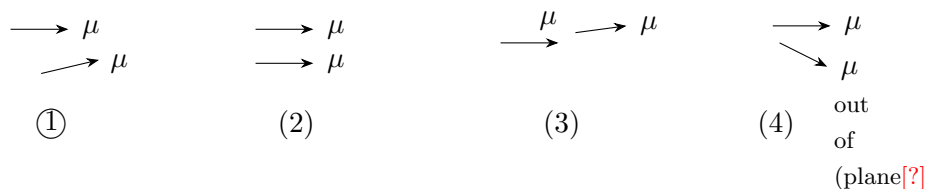
- “This is the main reason low temp is ordered state and high temp is disordered state.”

If dipole–dipole interaction is taken in account our plot or relation $\left(M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right) \right)$ fails.

For two dipoles, if U_d represents interaction energy



dipole–dipole interaction is not isotropic, below configuration does not have same energy.



If we rely on dipole–dipole interactions alone, we won’t have bar magnet at all. Because one ($\uparrow \mu$) will flip another in reverse direction ($\downarrow \mu$), so that the net dipole moment cancels out & magnetisation becomes zero.

$$(\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \cdots).$$

net ($M = 0$).

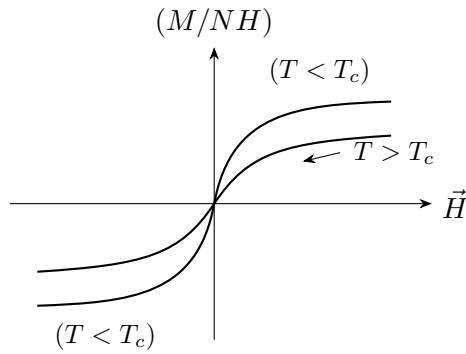
But we do see permanent magnets, so the phenomena of permanent magnetism is more subtle than this and is not connected to dipole–dipole interactions, It is connected to a quantum mechanical effect called the exchange Interaction which actually favours ($\uparrow \uparrow$) over ($\uparrow \downarrow$). and it in fact overcomes dipole–dipole interactions. ($\uparrow \downarrow$).

Exchange Interaction is very strong short range interaction It decays exponentially. (where as dipole–dipole inter-action is long range $\left(\frac{2K(\vec{p} \cdot \hat{r})}{r^3} \right)$ & weak).

There are magnets where dipole–dipole interaction becomes more important/dominant and it leads to Anti-ferromagnetism : ($\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$).

If we take exchange interactions in account we see that the curve b/w $\left(\frac{M}{NH} \right)$ vs $\vec{H}$, attains infinite slope at finite temp. ($T > 0K$) instead of ($T = 0K$).

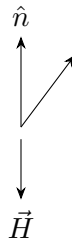
Below curie temp (T_c) the curve attains infinite slope at origin.



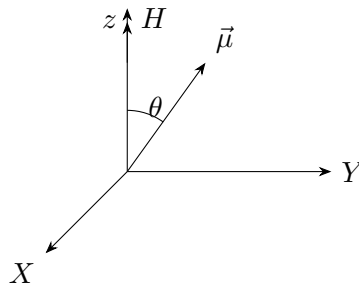
So this is discontinuous phase transitions at (T_c) curie temp.

This phase transition is from paramagnetic phase to ferromagnetic phase (M is very high even at $H = 0$).

For general case $(E = -\mu H \cos \theta)$ (3-D case)



lets compute the magnetisation.



– Also assuming no dipole-dipole interaction.

$$M = N \langle \mu \rangle$$

(our magnetisation)

$$\langle \mu \rangle = \frac{\int \mu \cos \theta P(\theta) d\Omega}{\int P(\theta) d\Omega} = \frac{\int \mu \cos \theta e^{+\beta \mu H \cos \theta} d\Omega}{\int e^{+\beta \mu H \cos \theta} d\Omega}$$

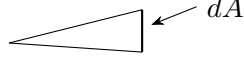
($d\Omega$ = solid angle)

Here we have ignored K.E, Rotational energy of these dipole moments, we have only used magnetic energy.

or

$$\int d\Omega = \int_{-1}^1 d(\cos \theta) \int_0^{2\pi} d\varphi$$

$$\int d\Omega = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi$$



$$\begin{aligned} d\Omega &= \frac{dA}{r^2} \\ &= \frac{(r d\theta)(r \sin \theta d\varphi)}{r^2} \end{aligned}$$

$$d\Omega = \sin \theta d\theta d\varphi$$

$$\int d\Omega = \iint \sin \theta d\theta d\varphi$$

$$M = N \langle \mu \rangle$$

$$= N\mu \frac{\int \cos \theta e^{\beta\mu H \cos \theta} \sin \theta d\theta \int_0^{2\pi} d\varphi}{\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi}$$

$$(\cos \theta = t, \quad -\sin \theta d\theta = dt)$$

$$= \frac{N\mu (2\pi) (-) \int_1^{-1} t e^{\beta\mu H t} dt}{2\pi \int_0^\pi e^{\beta\mu H \cos \theta} \sin \theta d\theta}$$

$$= \frac{2\pi N\mu \int_{-1}^1 t e^{\beta\mu H t} dt}{2\pi \int_{-1}^1 e^{\beta\mu H t} dt}$$

$$= \frac{N\mu \left(\frac{t e^{\beta\mu H t}}{(\beta\mu H)} - \frac{e^{\beta\mu H t}}{\beta^2 \mu^2 H^2} \right)_{-1}^1}{\left(\frac{1}{\beta\mu H} \right) (e^{\beta\mu H})_{-1}^1}$$

$$= \frac{N\mu^2 H \beta \left(e^{\beta\mu H} \left(\frac{1}{\beta\mu H} - \frac{1}{\beta^2 \mu^2 H^2} \right) - e^{-\beta\mu H} \left(\left(\frac{-1}{\beta\mu H} \right) - \frac{1}{\beta^2 \mu^2 H^2} \right) \right)}{(e^{\beta\mu H} - e^{-\beta\mu H})}$$

$$= N\mu \left[\frac{e^{\beta\mu H} + e^{-\beta\mu H}}{e^{\beta\mu H} - e^{-\beta\mu H}} - \frac{1}{\beta\mu H} \right]$$

$$M = N\mu \left[\coth(\beta\mu H) - \frac{1}{\beta\mu H} \right]$$

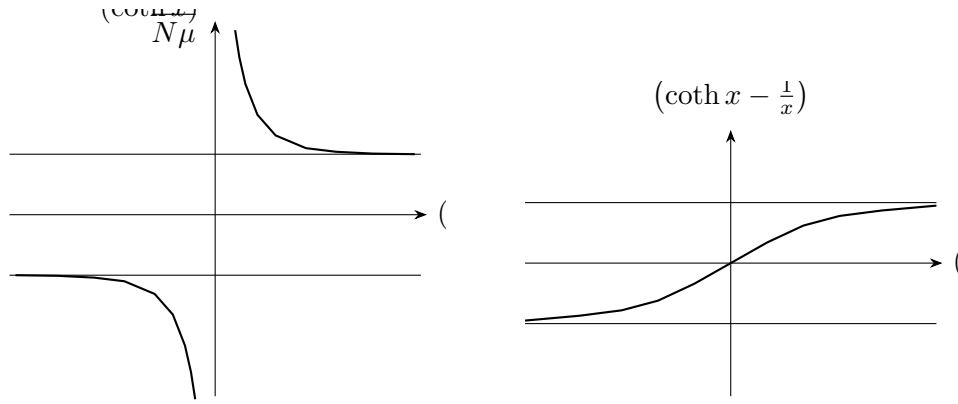
Previously when we only assumed ($E = \pm\mu H$)

$$M = N\mu \tanh x \quad x = \beta\mu H$$

↓ Now when we used all possible orientations

$$M = N\mu \left(\coth x - \frac{1}{x} \right) \quad = \text{Langevin functions.}$$

Let us plot $\left(\frac{M}{N\mu} \right)$ vs $\left(\coth x - \frac{1}{x} \right)$ vs (x)



$$\begin{aligned} \coth x &= \frac{\cosh x}{\sinh x} = \frac{\left(1 + \frac{x^2}{2} + \dots\right)}{x \left(1 + \frac{x^2}{6} + \dots\right)} \\ &= \left(\frac{1}{x}\right) \cdot \left(1 + \frac{x^2}{2} + \dots\right) \left(1 - \frac{x^2}{6} + \dots\right) \\ &= \frac{1}{x} \left(1 + x^2 \left(\frac{1}{2} - \frac{1}{6}\right) + \dots\right) \\ &= \frac{1}{x} \left(1 + \frac{2x^2}{6}\right) \\ \coth x &\approx \frac{1}{x} + \frac{x}{3} \end{aligned}$$

So $\left(\coth x - \frac{1}{x} \right) \approx \left(\frac{x}{3} \right)$

So,

$$\frac{M}{N\mu} = \left(\coth \frac{\mu H}{k_B T} - \frac{k_B T}{\mu H} \right)$$

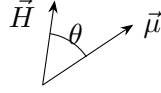
So, near origin ($H = 0$)

$$\left. \frac{M}{N\mu} \right|_{H \rightarrow 0} \approx \frac{\mu H}{3k_B T}$$

So $\left(\chi_T = \frac{N\mu^2}{3k_B T}\right)$ (χ_T = Isothermal susceptibility)

→ No of dimensions is '3' in our case

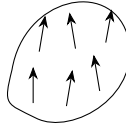
(For 2-D) :- μ is allowed to have any dirn. in a plane



$$M = N\langle\mu\rangle$$

$$M = N \frac{\int_0^{2\pi} \mu \cos \theta e^{+\beta\mu H \cos \theta} d\theta}{\int_0^{2\pi} e^{\beta\mu H \cos \theta} d\theta}$$

$$\frac{M}{N\mu} = \frac{\int_0^{2\pi} \cos \theta e^{\beta\mu H \cos \theta} d\theta}{\int_0^{2\pi} e^{\beta\mu H \cos \theta} d\theta} \quad \leftarrow \text{(Bessels function } J_0)$$



$$\frac{M}{N\mu} = \frac{\int \cos \theta \left(1 + \frac{\beta\mu H \cos \theta}{1} + \frac{(\beta\mu H \cos \theta)^2}{2!} + \dots\right) d\theta}{\int_0^{2\pi} \left(1 + \beta\mu H \cos \theta + \frac{(\beta\mu H \cos \theta)^2}{2!} + \dots\right) d\theta}$$

We are only interested in finding (χ_T) i.e. slope at origin. So, let's expand $(e^{\beta\mu H \cos \theta})$ to first order.

$$\begin{aligned} \frac{M}{N\mu} &= \frac{0 + \beta\mu H \int_0^{2\pi} \cos^2 \theta d\theta}{2\pi + \beta\mu H \left(\underbrace{\int_0^\pi \cos \theta d\theta}_{\rightarrow 0} \right)} = \frac{\beta\mu H \pi}{2\pi} \\ &= \frac{\beta\mu H}{2} \end{aligned}$$

So slope

$$\left. \frac{\partial M}{\partial H} \right|_{H \rightarrow 0} = \frac{N\beta\mu^2}{2} = \frac{\mu^2 N}{(2k_B T)} \quad \rightarrow \text{dimension is 2.}$$

$$\chi_T = \frac{N\mu^2}{2k_B T}$$

what we saw that $(\chi_T \propto \frac{1}{T})$ but it is not valid at small temp ($T \rightarrow 0$ or $T = 0$). So we need to fix it.

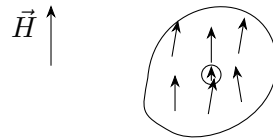
Just like to go from ideal gas to real gas (use interactions) we used the van der Waals gas eqn, similarly here we use Weiss molecular field theory.

Weiss molecular field theory

The idea is that in the ferromagnetic medium each dipole moment experience not only the ext. field $\vec{H}$ but also the internal mag. field due to other dipole moments.

$$H \xrightarrow{\text{replace}} H_{\text{eff}} = H + \lambda \frac{M}{N}$$

↓ to make intensive quantity to add to $\underline{H}$.



$$H_{\text{int}} \propto M \qquad H_{\text{int}} = \lambda M$$

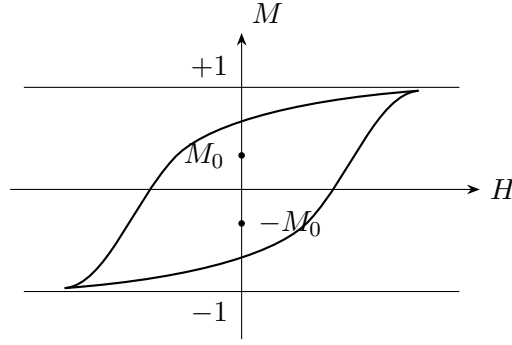
$$M = N\mu \tanh\left(\frac{\mu(H + \frac{\lambda M}{N})}{k_B T}\right) \quad (\text{for 1-D}).$$

This is transcendental equation, we can only solve it by Numerical methods.

Let us see its behaviour near $\underline{H = 0}$.

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_B T}\right) \quad \left(\begin{array}{l} M_0 = \text{magnetisation} \\ \text{per unit} \\ \text{dipole moments} \end{array} \right)$$

- 1) $M_0 = 0$ is always a solution (Paramagnetic solution). ($H = 0, M = 0$)
- 2) But ferromagnet have hysteresis.



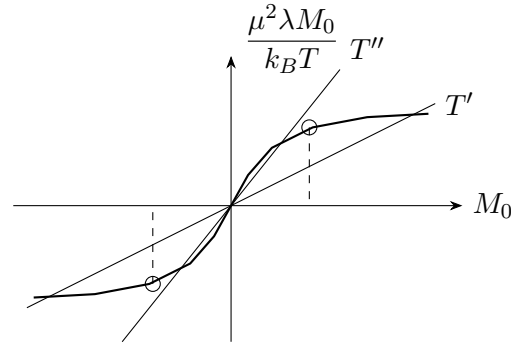
Near $M \approx 0$

$$M_0 = N\mu \left(\frac{\mu\lambda M_0}{Nk_B T} \right)$$

$$\approx \frac{N\mu^2\lambda M_0}{Nk_B T} \approx \frac{\mu^2\lambda M_0}{k_B T}$$

Lets plot $\left(\frac{\mu^2\lambda M_0}{k_B T} \right)$ vs (M_0)

(at high temp $T' > T_0$, slope is less), (for smaller temp T'' slope is even higher).



Intersection happens if slope > 1

$$\frac{\mu^2\lambda}{k_B T} > 1 \quad \text{or} \quad \boxed{T < \frac{\mu^2\lambda}{k_B}}$$

So $\left(T = \frac{\mu^2\lambda}{k_B} \right)$ is critical point.

The negative solution for M_0 is unstable; so we only have one positive root of M_0 . Which leads to ferromagnetism.

So below a certain critical temp, the cooperative tendency of all these magnetic moments to align in same dir. will dominate over disrupting tendency of entropy (S) and we have order.

$$F = U - TS \quad (U \gg TS)$$

at low temp (T) $F \approx U$ (ordered phase) (ferromagnetic)

But above (T_c) disorder wins and interaction is not strong enough to maintain order.

$$F = U - TS$$

$$(TS \gg U)$$

This is how phase transition occurs b/w paramagnet and ferromagnet. Now we will see how it is similar to case of fluids.

14 Lecture 29

We can also have ferroelectric transitions of permanent electric dipole moments, which are accompanied by structural phase transitions (crystalline phase transitions which would lead to shapes of unit cells which have permanent electric dipole moments.

The behaviour near the critical region of eqn of state

$$M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$$

would be very similar to that of fluid near its

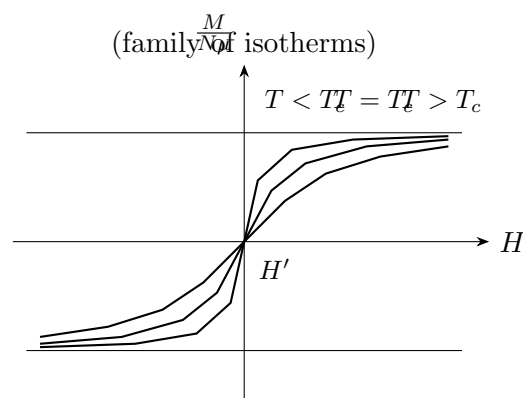
critical region. We would like to find out the non zero magnetisation (M) for $H = 0$.

$$M_0 = N\mu \tanh \left(\frac{\mu \lambda M_0}{N k_B T} \right)$$

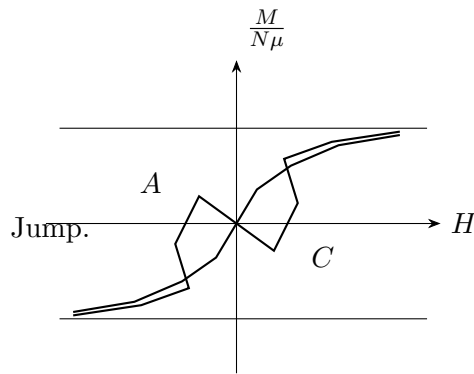
(Digression) also our model $M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$ does not include hysteresis phenomena.

Hysteresis actually arose from presence of multiple domains in magnetic material. These domains align in right direction in presence of strong $\vec{H}$. For a single domain our model works perfectly as a single domain can't show hysteresis.

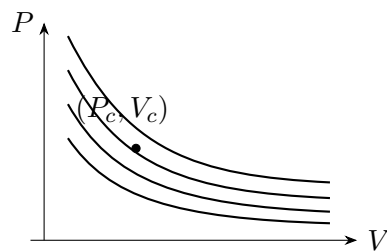
Family of Isotherms :-



$$\left(\text{Critical Isotherm} \quad \frac{\partial M}{\partial H} = 0 \ \& \ \frac{\partial^2 M}{\partial H^2} = 0 \right)$$



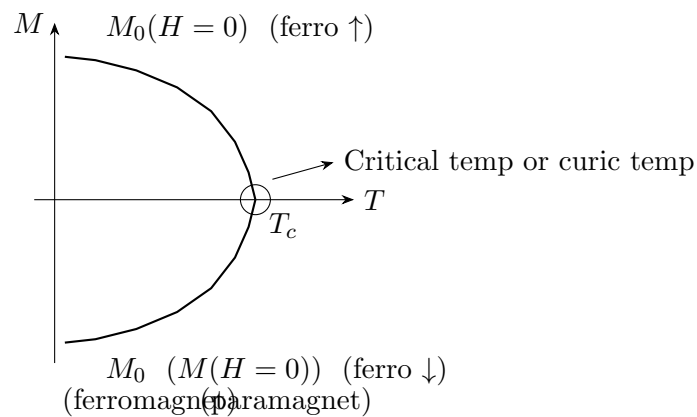
curve $A \rightarrow C$ is unphysical as ($H \uparrow$ in $-ve$ direction & $M \uparrow$ in $+ve$ direction).



← Isotherms in case of fluid.

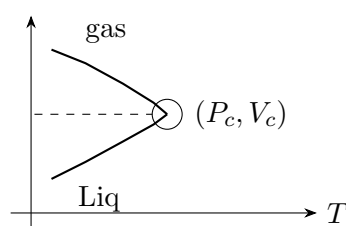
$$\rightarrow \left(\text{Critical Isotherm } \frac{\partial P}{\partial V} \bigg|_{(P_c, V_c)} = 0, \text{ \& } \frac{\partial^2 P}{\partial V^2} \bigg|_{(P_c, V_c)} = 0 \right)$$

let us plot (M vs T)



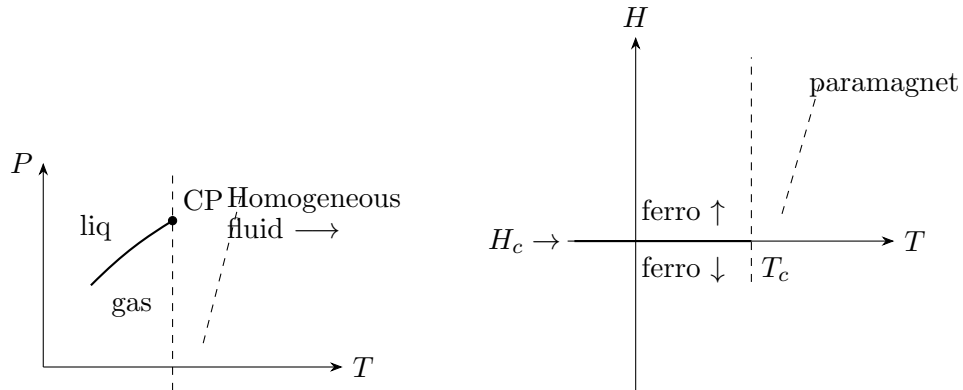
In case of (magnet)

In case of fluid



→ discontinuous transitions

→ parabolic region

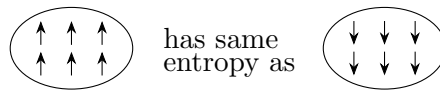


$$\frac{dP}{dT} = \frac{L}{T\Delta V} = \frac{T\Delta S}{T\Delta V} = \frac{\Delta S}{\Delta V} = \frac{S_{gas} - S_{liq}}{v_{gas} - v_{liq}}$$

↓ specific volume

$$\frac{dH}{dT} = \frac{\Delta S}{\Delta M}$$

In case of fluid as well as f. magnet, we have one line ending at critical pt. [?] from ferro ↑ to ferro ↓ [?] with ΔS [?]



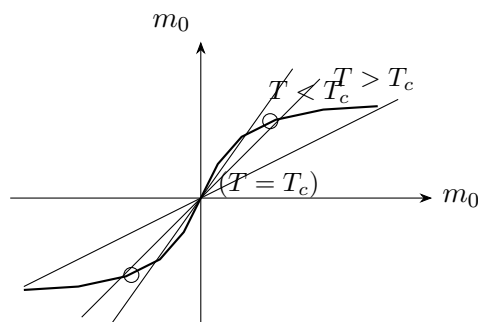
$$\left(\frac{dH}{dT} = \frac{\Delta S}{\Delta M} = \frac{0}{\text{nonzero}} = 0 \right) \quad \checkmark$$

let's find

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_B T}\right)$$

let $\left(\frac{M_0}{N\mu} = m_0\right)$

$$m_0 = \tanh\left(\frac{\mu^2\lambda m_0}{k_B T}\right)$$



T_c is given by when slope is 45°

$$\left(\frac{\mu^2 \lambda}{k_B T} = 1 \right)$$

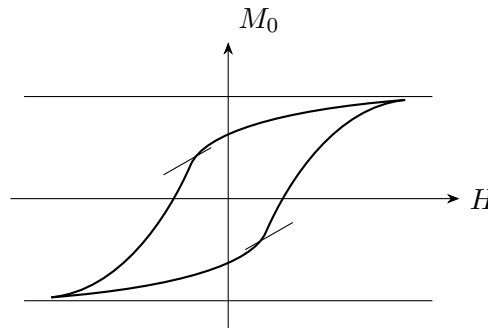
$$\left(T_c = \frac{\mu^2 \lambda}{k_B} \right).$$

$(M_0 = 0)$ root is not stable for $(T < T_c)$ as magnet becomes ferromagnet instead of paramagnet.
We can show it using $F = U - TS$ $(T < T_c)$ $(F \approx U) \rightarrow$ ordered phase

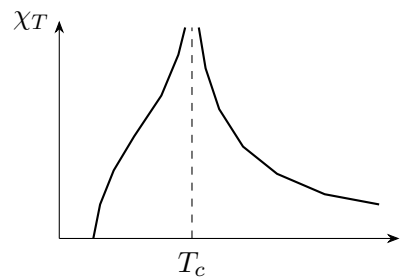
Susceptibility at $(H = 0)$.

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \Big|_{H=0}$$

even though (M) is discontinuous we can define χ_T as slope are continuous.



Lets plot $(\chi_T$ vs T)



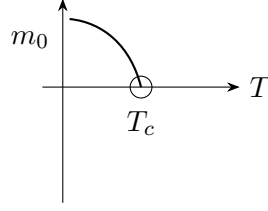
$$m_0 = \tanh \left(\frac{\mu^2 \lambda m_0}{k_B T} \right) = \tanh \left(\frac{T_c m_0}{T} \right)$$

at $T = T_c$ $m_0 = \tanh m_0 \Rightarrow (m_0 = 0)$ is only solution (paramagnet).

but below $(T < T_c)$ we have other solutions.

let us expand $\tanh x$, near $[?]$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\left(1 + x + \frac{x^2}{2!} + \dots \right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right)}{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right)}$$



$$= \frac{2 \left(x + \frac{x^3}{3!} \right)}{2 \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right)} = \left(x + \frac{x^3}{3!} + \dots \right) \left(1 - \frac{x^2}{2!} - \dots \right)$$

$$= \left(x - \frac{x^3}{2!} + \frac{x^3}{3!} - \frac{x^5}{3! 2!} + \dots \right)$$

$$\tanh x \approx x + x^3 \left(\frac{1}{6} - \frac{1}{2} \right) = x - \frac{x^3}{3} + \dots$$

$$m_0 = \tanh \left(\frac{T_c}{T} m_0 \right)$$

$$m_0 \approx \frac{T_c}{T} m_0 - \frac{1}{3} \left(\frac{T_c}{T} m_0 \right)^3$$

for m_0 other than zero,

$$m_0 \left(1 - \frac{T_c}{T} \right) \approx -\frac{1}{3} \left(\frac{T_c}{T} m_0 \right)^3$$

$$1 - \frac{T_c}{T} \approx -\frac{1}{3} \left(\frac{T_c}{T} \right)^3 m_0^2$$

$$\left(\frac{T_c - T}{T} \right) \approx \frac{1}{3} \left(\frac{T_c}{T} \right)^3 m_0^2$$

($T < T_c$) if T is very near to T_c ($\frac{T_c}{T} \approx 1$)

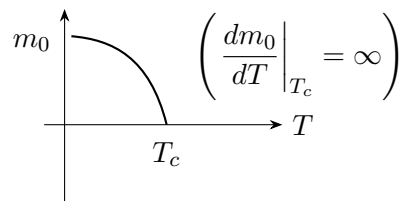
$$\Rightarrow \boxed{m_0 \sim \pm \sqrt{T_c - T}}$$

$m_0 \approx \pm (T_c - T)^{1/2} \rightarrow$ mean field critical exponent (β)

here ($\beta \neq \frac{1}{k_B T}$) (Just a symbol)

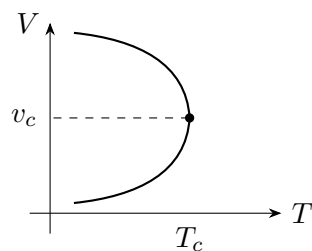
$$\left. \frac{dm_0}{dT} \right|_{T=T_c} = \frac{(-1)}{\pm 2\sqrt{T_c - T}} \rightarrow \infty.$$

So behaviour of m_0 near T_c is



Exactly same relation arose in case of fluid.

$$\left((v - v_c) \sim \pm \sqrt{T_c - T} \right)$$



(H.W) show that $(v - v_c) \sim \pm \sqrt{T_c - T}$.

we can see that (in both cases critical exponent is $(\frac{1}{2})$) which arose because (both models are examples of mean field theory models. i.e. In van der vaals case we assume $P \neq \frac{RT}{v-b}$ but $P = \frac{RT}{v-b} - \frac{a}{v^2}$ because of attraction of each molecules. In same way

here we assume every magnetic moment another magnetic moment and internal $\vec{H}$ is proportional to magnetisation itself, we have exactly the same philosophy and that is what led to critical exponent.

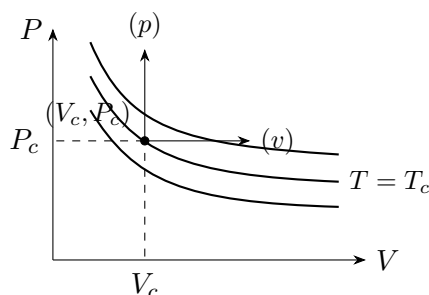
(Q) Does experiments agree with critical exponent to be $(1/2)$ or how accurately we can get closer to T_c and controll temp. to see it.

Ans $\sim$ In most cases we essentially see exponent closer to $1/2$ but, but, if we do careful experiments near T_c we do not see half at all. we see $(1/3$ or $0.3)$ for it we have to get very-very close to T_c to see it.

Temp is one of hardest thing to controll, best we can controll is of (milli-degree) of accuracy.

Frequency is most accurately measurable quantity, we can measure frequency in one part in 10^{15} . There are astronomical objects/clocks (pulsars) whose rate of change of time period can be computed to one part in 10^{19} .

(Q) How does critical Isotherm behave? (near (V_c, P_c)).



shifting origin to see behaviour near (V_c, P_c)

$$\left(\frac{\partial P}{\partial V} \right)_{T_c} = \left(\frac{\partial^2 P}{\partial V^2} \right)_{T_c} = 0$$

$$p = \frac{P - P_c}{P_c} \quad v = \frac{V - V_c}{V_c}$$

The curve near (V_c, P_c) is like

$$(p \sim |v|^\delta) \longrightarrow \text{mean field critical exponent } (\delta)$$

or $(|v| \sim p^{1/3})$

lets look at critical isotherm for ferro magnetic transitions

$$M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$$

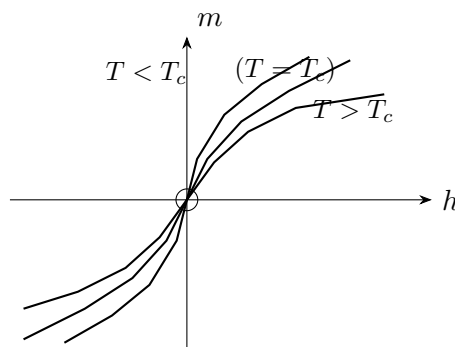
$$\text{let } m = \frac{M}{N\mu}$$

$$\begin{aligned} m &= \tanh \left(\frac{\mu}{k_B T} (H + \lambda m \mu) \right) \\ &= \tanh \left(\frac{\lambda \mu^2}{k_B T} \left(\frac{H}{\lambda \mu} + m \right) \right) \\ &= \tanh \left(\frac{T_c}{T} (h + m) \right) \end{aligned}$$

$$\text{let } \left(\frac{H}{\lambda \mu} = h \right)$$

for critical isotherm $(T = T_c)$

$$m = \tanh(h + m)$$



$\longrightarrow$ critical isotherm

at $(h = 0), (m = 0)$ m is zero. (for $T > T_c$)

$$m = \tanh(h + m) \approx (h + m) - \frac{(h + m)^3}{3}$$

$$\Rightarrow (m + h)^3 \approx h$$

h is much smaller (near origin) so $m^3 \sim h$

$$\boxed{m \sim h^{1/3}} \quad (h \text{ is field variable})$$

Just like $(|v| \sim p^{1/3})$ (p is field variable)

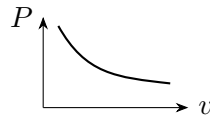
(m, v) are response variables.

Similarly, χ_T and compressibility behave exactly same near critical point.

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \bigg|_{H=0} ;$$

$$k_T = \frac{-1}{v} \frac{\partial v}{\partial P}$$

$$\frac{\partial P}{\partial v} = 0 \text{ at critical point}$$



$\left(\begin{array}{l} \text{curie} \\ \text{weiss law} \end{array} \right) \Leftarrow$

$$\chi_T = \frac{1}{|T - T_c|}$$

$$(\chi_T|_{T=T_c} = \infty) \longleftrightarrow (k_T = \infty)$$

$$\chi_T = \frac{1}{|T - T_c|^\gamma} \longrightarrow \gamma \text{ critical point}$$

$(\gamma \neq C_P/C_V)$ just a symbol.

Experimental values of $(\gamma \sim 1.3)$.

All these phase transitions can be explained by Landau's theory ((1937). (Idea of broken symmetry leads to such phase transitions of all kinds of systems).

15 Module 13, Lecture 34: The rotation group and all that

Outline :-

- (1) Rotations of the coordinate axis
- (2) orthogonality of rotation matrices.
- (3) proper and im-proper rotations
- (4) Generators of infinitesimal rotations in 3-D
- (5) Lie algebra of generators
- (6) Rotation generators in 3-D transforms like a vector.
- (7) The general rotation matrix in 3-dimensions.

The rotation group :-

We could just as well consider rotations in N-Dimensions so frequently we will talk about N-Dimensional euclidean space or occasionally we will talk about rotation in 3-D space (euclidean space).

It turns out that there are major differences between rotations in even-dimensions or rotation in odd dimensions.

The notation used here will be of cartesian tensors, the index notation with summation convention. so indices $i = 1, 2, 3 \dots N$ in N-Dimensional case or $i = 1, 2, 3$ in 3-D case.

So let us look at simplest case (i.e. 3-dimensional rotation). coordinate of a point is represented as X_i $i = 1, 2, 3$ and we know that under rotation of coordinate axis, when origin is fixed, the distances do not change.

and the rotation is a linear transformation, in the sense that

$$X_i \longrightarrow X'_i = R_{ij}X_j$$

$$\downarrow$$

- a) homogeneous transformation as origin does not change.
- b) linear transformation as each new coordinate is linear combination of old coordinates.

$$\text{If } \vec{r} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \text{then} \quad R = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix}$$

$$\vec{r}' = R \vec{r}$$

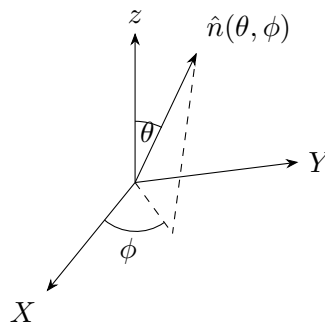
ex, for rotation along z axis (in x-y plane) about angle ' φ '.

$$R = \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

But more general way of writing a rotation is to parametrise this rotation in some suitable form.

$$R(\hat{n}, \varphi) \longrightarrow \text{Rotation matrix for axis of rotation } \hat{n}(\theta, \phi)$$

by amount of rotation ' φ '



Note :- “Rotation about an axis is artifact of 3-D (rotation about an axis is peculiar to 2-D) The reason is we could have rotated in 2-D (plane)/(euclidean plane) i.e the rotation occurs about origin (0,0) and there is no third dimension at all (axis)

If we rotate in 4-D for $\vec{r} = (X_1 X_2 X_3 X_4)$, The rotation can be done in X_1 - X_2 plane leaving X_3 - X_4 subspace unchanged. So we do not have any unique axis about which such rotation can be performed.”

The more general way of rotation is to say rotation in a plane so for 2-D one plane exist and 2 axis

for 3-D – 3 mutually orthogonal planes exist, and 3 axis

for 4-D – 6 mutually ” planes exist, and 4 axis.

So, for three dimensions only, no. of mutually orthogonal planes is equal to no. of axis.

$$\left(\text{No of } \perp \text{ planes} \Rightarrow \frac{n(n-1)}{2} = n \quad \text{only for } n = 3 \right)$$

technically

So If we say rotation about $\hat{z}$ axis, we mean (rotation in x-y plane.

$$\left\{ \begin{array}{l} \text{for 4-D we have 6 mutually orthogonal planes.} \\ i =, \quad \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \quad (N_{C_2}) \text{ planes} \\ \left. \begin{array}{l} 12 \quad 13 \quad 14 \\ 23 \quad 24 \\ 34 \end{array} \right\} \text{ set of } \perp \text{ planes. in 4-D.} \end{array} \right\}$$

(Q) Can we associate an axis with rotation always ?

for even dimensional case $n = 2, 4, 6 \dots$

we can not associate unique axis wito define rotation.

So we can not always associate an unique axis of rotation for N-D (even) euclidean (euclidean) space. | “But we can always associate an unique axis for odd dimensional euclidean space.”

we could have rotation about an axis in 4-D space, but all rotations in 4-D need not have unique axis associated with them. (i.e. we can rotate about x_1 axis leavng x_1 unchanged but there are other rotations where we can not associate a unique axis).

Lets come back to rotation in 3-D. what more we can say about such rotations?

In general these rotations form a group. because of these

(1) combination of two rotations $R_1 R_2 = R_3$ is again a rotation.

(2) Every rotation has an inverse $RR^{-1} = I = R^{-1}R$

$$\left(\begin{array}{l} R^{-1} \text{ is inverse rotation such that in succession we} \\ \text{obtain } (RR^{-1}) = I \text{ (i.e. No rotation at all).} \end{array} \right)$$

(?) (So its clear that there is correspondence between set of abstract rotations in N-dimensional space and set of matrices.) How?.

In N-Dimensional space matrices are $(n \times n)$ dimension. and are orthogonal with $\det(M) = +1$.

R is an orthogonal matrix.

also set of $(n \times n)$ orthogonal matrix forms group.

hence R is an element of $O(n)$

$\hookrightarrow$ group of $(n \times n)$ Orthogonal matrices.

orthogonality is easily established.

under a rotation :

$$X_i X_i = X'_i X'_i \quad - (1)$$

$\left(\begin{array}{c} \text{distance} \\ \text{b/w two points} \end{array} \right) \longrightarrow$

$$\begin{aligned} X'_i &= R_{ij} X_j, & x'_i &= R_{ik} X_k \\ X'_i X'_i &= R_{ij} R_{ik} X_j X_k \\ &= R_{ji}^T R_{ik} X_j X_k = (R^T R)_{jk} X_j X_k \end{aligned} \quad - (2)$$

on the other hand

$$X_i X_i = X'_i X'_i = X_j X_j = \delta_{jk} X_j X_k \quad - (3)$$

equating equation (3) and (2)

$$\begin{aligned} (R^T R)_{jk} X_j X_k &= \delta_{jk} X_j X_k \\ \Rightarrow (R^T R)_{jk} &= \delta_{jk} \\ \Rightarrow R^T R &= I \quad (R^T = R^{-1}) \end{aligned}$$

So a matrix whose inverse is its transpose is called orthogonal matrix. So R is orthogonal matrix.

(we arrived at this result from our condition that distance between any two points should be unchanged under rotation)

Note :- for finite dimensional matrices left inverse = right inverse if they exists. $R^T R = R R^T = I$.

$$\begin{aligned} \Rightarrow |R^T R| &= 1 \\ \text{as } |R^T| &= |R|, \text{ so } |R|^2 = 1 \\ \Rightarrow |R| &= \pm 1 \end{aligned}$$

If, $\det(R) = +1$, then ' R ' represents proper rotations. because $|I|$ itself is $+1$.

all Rotation matrix with $\det(R) = +1$ can be constructed by moving contineously from the Identity. So we start with orignal coordinate system, and we want to go to proper rotation R , we can do so by sucession of infinitesimal transformations, each of which has determinant $+1$.

So this property of connected rotation is what defines proper rotation.

On the other hand there are matrices whose determinant is (-1) ($|R| = -1$) and they are discontineous transformations.

There is no way of reaching R such that $|R| = -1$ by contineously moving from identity by making succession of infinitesimal transformations. There has to be discrete transformation somewhere. It would still be orthogonal but it can not have $|R| = +1$

ex

$$\left. \begin{matrix} x'_1 = -x_1 \\ x'_2 = x_2 \\ x'_3 = x_3 \end{matrix} \right\} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \left\{ \begin{matrix} \text{Reflection} \\ \text{transformation} \end{matrix} \right.$$

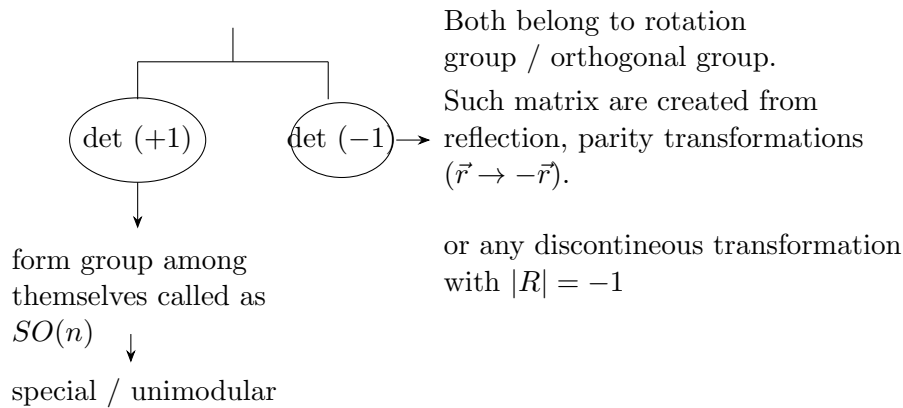
$$x' = RX$$

here 'R' represents improper rotation.

Once we have performed improper rotation (R_1) any proper rotation (R_2) made. Can not make proper transformation.

$$\begin{aligned} R_1 R_2 &= R_3 \\ |R_1 R_2| &= |R_3| \\ |R_1| |R_2| &= -1 \times 1 = -1 \Rightarrow |R_3| = -1 \Rightarrow R_3 \text{ is improper Rotation.} \end{aligned}$$

So this set $O(n)$ has two disjoint pieces.



If $f, g \in SO(n)$

- 1) $f \cdot g = h \in SO(n)$
- 2) $f \cdot f^{-1} = I$ and $I \in SO(n)$

So $|R| = +1$ forms group among themselves called $SO(n)$.

where as set of elements with $(\det R = -1)$ does not form group as

$$\text{if } f, g \in \{|R| = -1\}$$

- 1) $f \cdot g \notin A$
- 2) $ff^{-1} = I$ but $I \notin A$ as $|I| = +1$.

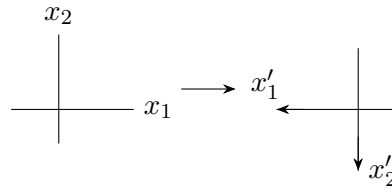
$$A = \{|R| = -1\} \quad \downarrow \text{set of all matrix } |R| = -1 \text{ and are orthogonal}$$

So 'A' does not form group.

"Therefore the parameter space of $O(n)$ is disconnected, there is a piece which is proper (that forms a subgroup) and there is a piece which has $\{|R| = -1\}$. which are

$$(\text{proper transformation}) \times (\text{improper transformation}) = (\text{improper transf.})''$$

we have to be careful as, in even – dimensional space parity transformation is not improper transformation. but is proper rotation.



$$\left. \begin{matrix} x'_1 = -x_1 \\ x'_2 = -x_2 \end{matrix} \right\} \Rightarrow x' = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$|R| = +1$$

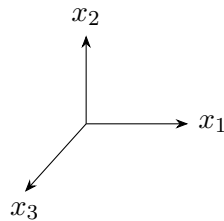
$\hookrightarrow$ proper transformation

but reflection is improper transformation.

$$|R(\hat{n}, \varphi) = ? \rangle$$

$$R(\hat{e}_3, \varphi) = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\downarrow$ represents rotation about $\hat{z}$ axis.



let us see what infinitesimal rotation does to first order.

$$R(\hat{e}_3, \delta\varphi) = \begin{pmatrix} \cos \delta\varphi & \sin \delta\varphi & 0 \\ -\sin \delta\varphi & \cos \delta\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\downarrow$ first order

$$\cos \delta\varphi = \underline{1}$$

$$= \begin{pmatrix} 1 & \delta\varphi & 0 \\ -\delta\varphi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R(\hat{e}_3, \delta\varphi) = I_{3 \times 3} + \delta\varphi \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\downarrow}$$

plays role of generators of these rotations.

Repeated action by these generators (matrices) will lead to the rotation R .

In Q.M it turn out that these generators of rotation correspond to physical quantities, namely the components of angular momentum. and in Q.M we like to represent these generators as hermitian operators. so let us write

$$\begin{aligned} R(\hat{e}_3, \delta\varphi) &= I_{3 \times 3} + i(\delta\varphi) \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= I + iJ_3(\delta\varphi) \end{aligned}$$

where $J_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is hermitian.

i.e. $(J_3^\dagger = J)$.

we need to make sure that repeated action of these generators indeed makes a finite rotation.

This means that for a finite rotation amount ' φ ' we can reach to it in ' n ' infiniteasimal rotations by ' $\delta\varphi$ ' such that

$$\lim_{n \rightarrow \infty} n(\delta\varphi) = \varphi$$

This implies

$$R(\hat{e}_3, \varphi) = (I + iJ_3(\delta\varphi)) (I + iJ_3(\delta\varphi)) (I + iJ_3(\delta\varphi)) \cdots N \text{ times}$$

$$= (I + iJ_3(\delta\varphi))^n$$

$$\delta\varphi \rightarrow \frac{\varphi}{n}$$

$$R(\hat{e}_3, \varphi) = \lim_{n \rightarrow \infty} \left(I + \frac{iJ_3\varphi}{n} \right)^n$$

$$\text{if } (iJ_3\varphi = X) \quad I \ll 1$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \\ &= e^x \end{aligned}$$

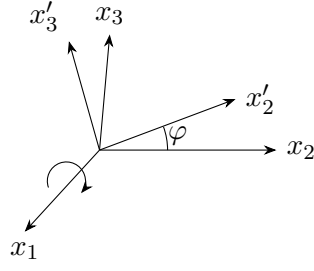
$$\begin{aligned} R(\hat{e}_3, \varphi) &= e^{iJ_3\varphi} \left(I + \frac{iJ_3\varphi}{1} + \frac{(iJ_3\varphi)^2}{2!} + \frac{(iJ_3\varphi)^3}{3!} + \cdots \right) \\ &= \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

It turns out that the matrices like J_3 obey lie algebra (obey certain algebraic relation) and forms group too which is called as Lie group.

what is $R(\hat{e}_1, \varphi) = ?$

$$R(\hat{e}_1, \delta\varphi) = I + iJ_1(\delta\varphi) \quad (\text{Infinitesimal rotation})$$

(what is $J_1 = ?$)



$$x'_1 = x_1$$

$$x'_2 = x_2 \cos \varphi + x_3 \sin \varphi$$

$$x'_3 = -x_2 \sin \varphi + x_3 \cos \varphi$$

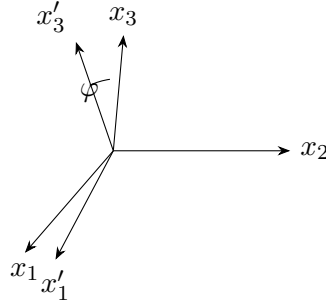
$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{pmatrix}$$

$$R(\hat{e}_1, \delta\varphi) = I + i(\delta\varphi) \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$\hookrightarrow \boxed{J_1}$$

Similarly,

$$R(\hat{e}_2, \delta\varphi) = I + i(\delta\varphi)J_2$$



$$x'_1 = x_1 \cos \varphi - x_3 \sin \varphi$$

$$x'_2 = x_2$$

$$x'_3 = x_1 \sin \varphi + x_3 \cos \varphi$$

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \cos \varphi & 0 & -\sin \varphi \\ 0 & 1 & 0 \\ \sin \varphi & 0 & \cos \varphi \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$R(\hat{e}_2, \delta\varphi) = \begin{pmatrix} 1 & 0 & -\delta\varphi \\ 0 & 1 & 0 \\ \delta\varphi & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= I + i\delta\varphi \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

$$J_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}$$

If the no of mutually orthogonal planes is 3, then there are three generators, corresponding to infinitesimal rotations in each of three planes.

Interesting thing is, these generators (J_1, J_2, J_3) they form lie algebra i.e.

- 1) They form LVS among themselves such that
- 2) commutator of any two of them belong to same LVS.

$$[J_j, J_k] = i\epsilon_{jkl}J_\ell$$

(Infinitesimal transformation) generators

where,

$$\begin{cases} \epsilon_{jkl} = \epsilon_{klj} = \epsilon_{ljk} = 1 \\ j, k, \ell = 1, 2, 3 \\ \epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1 \\ \epsilon_{213} = \epsilon_{321} = \epsilon_{132} = -1 \\ \epsilon_{112} = \epsilon_{212} = \epsilon_{322} = \epsilon_{122} = 0 \\ \text{(if any two are same)} \end{cases}$$

$$\left\{ \begin{array}{l} \epsilon_{jkl} \text{ has 27 components, 21 are zero, and} \\ 3 \text{ of them are } +1 \text{ and 3 of them are } -1. \end{array} \right\}$$

Lie algebra of $SO(3)$ is represented as “ $so(3)$ ”.

General Rotation about ($\hat{n}$) axis :-

$$R(\hat{n}, \varphi) = e^{i(J_1 n_1 + J_2 n_2 + J_3 n_3)\varphi}$$

where,

$J_1 n_1 + J_2 n_2 + J_3 n_3$ = Infinitesimal generator in $\hat{n}$ direction

φ = amount of rotation

The difficulty is in fact that $J_i J_j$ do not commute with each other. so we have difficulty in exponentiation of $(J_1 n_1 + J_2 n_2 + J_3 n_3)$.

$$\text{for, } [A, B] \neq 0 \quad e^{A+B} \neq e^A \cdot e^B$$

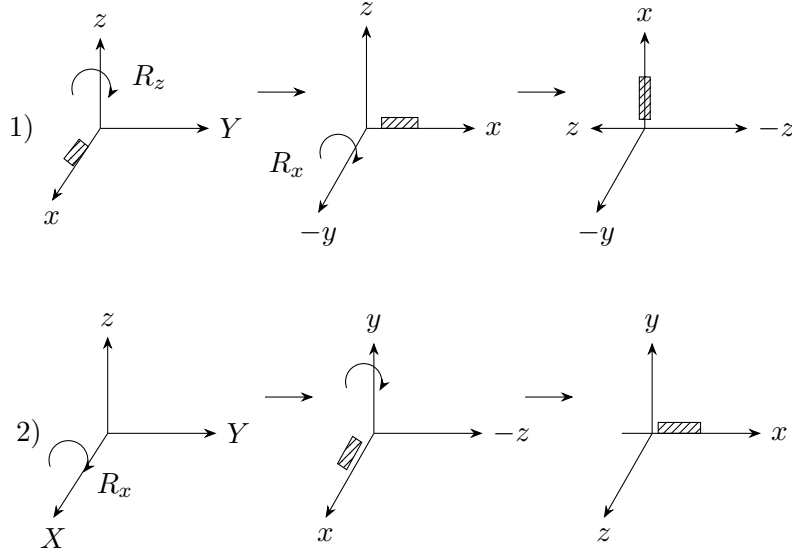
$$\begin{aligned} & \downarrow \\ & \left(I + \frac{(A+B)}{1!} + \frac{(A+B)^2}{2!} + \dots \right) \\ \Rightarrow & \left(I + \frac{(A+B)}{1} + \frac{A^2}{2!} + \frac{B^2}{2!} + \frac{AB}{2!} + \frac{BA}{2!} \right) + \dots \\ & \neq e^A \cdot e^B \end{aligned}$$

“and this problem that $[J_j J_k] \neq 0$ is at root of all the phenomena we see in 3-D euclidean space.”

Because it essentially says rotation about two different axis do not commute with each other.

i.e.

$$R_1(\hat{e}_1, \varphi_1) R_2(\hat{e}_2, \varphi_1) \neq R_2(\hat{e}_2 \varphi) R_1(\hat{e}_1 \varphi)$$



we can see both rotations do not end up having different state. (rotation).

On the other hand if we are on a plane, If we rotate by angle φ and subsequently by angle φ' . The result is rotation by angle $(\varphi + \varphi')$.

we could have added them in either order.

So Rotation group in 2-D which is group of 2×2 orthogonal matrices, $(SO(2))$ is called abelian group.

$$R(\varphi)R(\varphi') = R(\varphi + \varphi') = R(\varphi')R(\varphi)$$

But that is not true in 3-D euclidean space because

$$[J_j J_k] = i\epsilon_{jkl} J_l \neq 0.$$

Because of this $e^{i(J_1 n_1 + J_2 n_2 + J_3 n_3)\varphi}$ is not trivial to find out.

In N-Dimensions, this problem is much worse, as we have n_{C_2} such generators and general transformation can be non-trivial to find out.

(These generators $(J_1 J_2 J_3)$ itself transform like vectors under rotation

“vector in 3-D is set of 3 elements which transform the exactly same way coordinates/axis transform under rotation”.

$$\vec{a} = (a_1 \ a_2 \ a_3)$$

under rotation $\vec{a} \longrightarrow \vec{a}'$

$$\text{if } x'_i = R_{ij} X_j$$

$$\text{and if } a'_i = R_{ij} a_j$$

then we call $\vec{a}$ “vector”.

The physical law is invariant when it is expressed in terms of quantities whose transformation law is prescribed to us.

So the laws are form invariant provided they are expressed in terms of covariant objects (objects whose transformation law is already encoded in the object itself).

It turns out that $J = (J_1, J_2, J_3)$ where $\vec{J}$ transforms exactly as coordinates themselves under rotation.

16 Module 13, Lecture 35

Outline :-

- (1) The finite rotation formula for a vector
- (2) The general form of elements of $U(2)$ and $SU(2)$
- (3) Relation between the groups $SO(3)$ and $SU(2)$.

$$R(\hat{n}, \psi) = e^{i(J_1 n_1 + J_2 n_2 + J_3 n_3)\psi} = e^{i(\vec{J} \cdot \hat{n})\psi} \quad (n_1^2 + n_2^2 + n_3^2 = 1)$$

let $M = i(\vec{J} \cdot \hat{n})$

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad J_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad J_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$M = i(J_1 n_1 + J_2 n_2 + J_3 n_3) = i \begin{pmatrix} 0 & -n_3 i & i n_2 \\ i n_3 & 0 & -i n_1 \\ -i n_2 & i n_1 & 0 \end{pmatrix}$$

$$M = \begin{pmatrix} 0 & n_3 & -n_2 \\ -n_3 & 0 & n_1 \\ n_2 & -n_1 & 0 \end{pmatrix}$$

or $M_{ij} = \epsilon_{ijk} n_k$ (M_{ij} = matrix element)

$$\begin{aligned} M_{ij}^2 &= M_{il} M_{lj} \\ &= (\epsilon_{ilk} n_k)(\epsilon_{ljm} n_m) \\ &= -\epsilon_{lik} \epsilon_{ljm} n_k n_m \\ &= -(\delta_{ij} \delta_{km} - \delta_{im} \delta_{kj}) n_k n_m \\ &= n_i n_j - \delta_{ij} \end{aligned}$$

$$(\epsilon_{ilk} = -\epsilon_{lik})$$

$$\begin{aligned} (M^3)_{ij} &= (M^2)_{ik} M_{kj} \\ &= -M_{ij} \quad (\text{or}) \quad (M^3 = -M) \end{aligned}$$

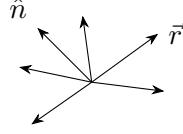
$$e^{iM\psi} = aI + bM + cM^2$$

$$R_{ij}(\hat{n}, \psi) = \boxed{(\cos \psi) \delta_{ij} + (1 - \cos \psi) n_i n_j + (\sin \psi) \epsilon_{ijk} n_k}$$

$$x_i \rightarrow x'_i = R_{ij} x_j = (\cos \psi) x_i + (1 - \cos \psi) (\vec{r} \cdot \hat{n}) n_i + (\sin \psi) \epsilon_{ijk} x_j n_k$$

$$\vec{r} \xrightarrow{R(\hat{n}, \psi)} \vec{r}' = \underline{(\cos \psi) \vec{r} + (1 - \cos \psi) (\vec{r} \cdot \hat{n}) \hat{n} + (\sin \psi) (\vec{r} \times \hat{n})}$$

(Finite rotation formula)



we can see that first term $(\cos \psi) \vec{r}$ is along the vector itself, second term has component along axis of rotation ($\hat{n}$) and third component is perpendicular to plane of $\vec{r}$ and $\hat{n}$.

If $\vec{r} \cdot \hat{n} = |\vec{r}|$ ($\vec{r}$ is along $\hat{n}$) then there is no rotation at all. for vector $\vec{r}$.

$$\begin{aligned} \vec{r}' &= (\cos \psi) \vec{r} + (1 - \cos \psi)(\vec{r} \cdot \hat{n}) \hat{n} \\ &= (\cos \psi) \vec{r} + (1 - \cos \psi) |\vec{r}| \hat{n} \quad \left(\begin{array}{l} \hat{n} = \hat{r} \\ |\vec{r}| \hat{n} = \vec{r} \end{array} \right) \\ &= (\cos \psi + 1 - \cos \psi) \vec{r} = \vec{r} \end{aligned}$$

so any set of 3 no. which satisfy finite rotation formula is a vector.

but 'J' which transforms like vector does not obey finite rotation formula; as this is not the rule of transformation for operators.

$$\vec{r} \xrightarrow{R(\hat{n}, \psi)} \vec{r}'$$

$$R(\hat{n}, \psi) = (\cos \psi) \delta_{ij} + (1 - \cos \psi) n_i n_j + (\sin \psi) \epsilon_{ijk} n_k$$

where,

$$R(\hat{n}, \psi) = e^{i(\vec{J} \cdot \hat{n})\psi} \quad \text{and} \quad [J_k, J_l] = i \epsilon_{klm} J_m$$

($\hat{J}$ need not be 3×3 matrix. $\hat{J}$ can have any dimensionality depends on object it acts upon.)

we know that J_1, J_2, J_3 involves σ_i (pauli) matrix in them. and it turns out that if $J_j = \frac{1}{2} \sigma_j$ where J_i = angular momentum operators. for $\text{spin}(\frac{1}{2})$.

$$\begin{aligned} [J_i J_j] &= \frac{1}{4} [\sigma_i, \sigma_j] = \frac{1}{4} 2i \epsilon_{ijk} \sigma_k \\ &= i \epsilon_{ijk} \left(\frac{\sigma_k}{2} \right) \end{aligned}$$

$$\boxed{[J_i J_j] = i \epsilon_{ijk} J_k} \quad \text{①}$$

These operators J_j satisfies exactly same lie algebra as generators of rotation group $[J_k J_l] = i \epsilon_{klm} J_m$.

(The fact is rotation in 3-D space is characterized by lie algebra of infinitesimal generators (J_1, J_2, J_3).)

any set of matrices which satisfies eq. ① represents rotation group in 3-D. (any set of 3 objects of any dimensionality which satisfies lie algebra, is represents rotation in 3-D). For instance we have ket vector $|\psi\rangle$, describing state of particle/system and if we change coordinate axis by rotation about $\hat{n}$ axis by angle ψ . Then $|\psi\rangle$ changes to $|\psi'\rangle$ in changed frame.

$$|\psi'\rangle = U(R) |\psi\rangle$$

$\hookrightarrow$ unitary transformation which depends on $R(\hat{n}, \psi)$.

But $U(R)$ turns out to be

$$U(R) = e^{i(\vec{J} \cdot \hat{n})\psi} \quad \} \text{ Rotation operator in hilbert space.}$$

as in general $|\Psi\rangle$ is infinite dimensional vector, so $\vec{J}$ has to be $\infty \times \infty$ dimensional matrices.

Here rotation in coordinate system (in 3D) has induced a transformation in hilbert space given by the operator U . (where ' U ' turns out to be infinite dimensional matrix).

ex.

Since $[J_i, J_j] = i \epsilon_{ijk} J_k$ where $(J_j = \frac{1}{2} \sigma_j)$ ($\vec{J}_j$ is 2×2 matrix)

Therefore $\exists$ (2×2) matrix representation of the group $SO(3)$.

It is called $SU(2) \hookrightarrow$ unitary matrices.

$$\boxed{U(\hat{n}, \psi) = e^{i(\vec{\sigma} \cdot \hat{n})\frac{\psi}{2}}} \quad \} \text{ only acts upon } 2 \times 2 \text{ matrix or } (2 \times 1) \text{ matrix}$$

$$U(\vec{r} \cdot \sigma) U^{-1} = (\vec{r}' \cdot \sigma)$$

(as the matrices $(\frac{\sigma_i}{2} = J_i)$ satisfies lie algebra, U represents rotation in 3-D space.) But, we need 3 piece of information to specify vector in 3-D so it has to be act upon 2×2 matrix rather than (2×1) matrix.

$$U^\dagger = e^{-i(\sigma^\dagger \cdot \hat{n})\frac{\psi}{2}} \quad (\sigma^\dagger = \sigma)$$

as, $UU^\dagger = U^\dagger U = I$; $\Rightarrow U^\dagger = U^{-1} \rightarrow$ ' U is unitary matrix. (2×2) .

But,

(Q) But how we implement (2×2) rotation matrix on 3-D space?

For any 3-D vector $\vec{a} = (a_1, a_2, a_3)$

we can construct $\vec{a} \cdot \vec{\sigma} = a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3$

$$(\vec{a} \cdot \vec{\sigma}) = \begin{pmatrix} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{pmatrix}$$

where coordinate (x_1, x_2, x_3) are represented as: $\begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}$

just as

$$\begin{aligned} \vec{r} &\xrightarrow{R(\hat{n}, \psi)} \vec{r}' \\ (\vec{r} \cdot \sigma) &\xrightarrow{U} (\vec{r}' \cdot \vec{\sigma}) \end{aligned}$$

$$U(\vec{r} \cdot \sigma) U^{-1} = (\vec{r}' \cdot \sigma)$$

which is:-

$$e^{i(\sigma \cdot \hat{n})\psi/2} (\vec{r} \cdot \vec{\sigma}) e^{-i(\sigma \cdot \hat{n})\psi/2} = (\vec{r}' \cdot \vec{\sigma}) \quad \text{--- ②}$$

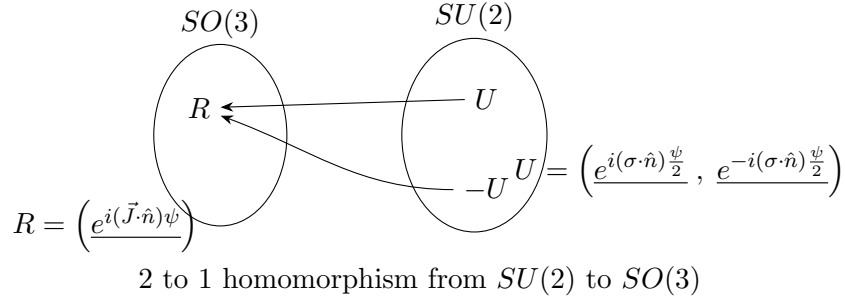
exponential of (2×2) matrix. $e^{i(\sigma \cdot \hat{n})\frac{\psi}{2}}$:-

$$\boxed{e^{i(\vec{\sigma} \cdot \hat{n})\psi/2} = I(\cos(\psi/2)) + i(\vec{\sigma} \cdot \hat{n}) \sin \frac{\psi}{2}} \quad \text{--- ③}$$

$$\left\{ \text{In general. } e^{i(\vec{\sigma} \cdot \vec{a})\theta} = I(\cos \theta) + \frac{i(\vec{\sigma} \cdot \vec{a})}{|\vec{a}|} \sin \theta \right\} \quad \text{--- ④}$$

If we put back eq ③ in eq ② we get exactly the finite rotation formula.

So there is a connection between every 3×3 orthogonal matrix with ($\det R = +1$) of $SO(3)$ grp has 2×2 matrix of $SU(2)$ group.



If we change $U \rightarrow (-U)$

$$\begin{aligned} U(\vec{r} \cdot \sigma)U^{-1} &= (\vec{r}' \cdot \sigma) \\ (-U)(\vec{r} \cdot \sigma)(-U^{-1}) &= (\vec{r}' \cdot \sigma) \end{aligned}$$

So for every one element of $SO(3) \ni 2$ such unitary matrix of $SU(2)$ group.

Every element of $SO(3)$ group is parameterised by three parameters (θ, ϕ, ψ) or $(\hat{n}(\theta, \phi), \psi)$. That should be same in $SU(2)$.

What are no. of independent parameters in $SU(2)$.

$$\text{General } (2 \times 2) \text{ matrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

M has to be unitary.

$$M^\dagger = \begin{pmatrix} \alpha^* & \gamma^* \\ \beta^* & \delta^* \end{pmatrix}$$

$$MM^\dagger = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha^* & \gamma^* \\ \beta^* & \delta^* \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & \alpha\gamma^* + \beta\delta^* \\ \gamma\alpha^* + \delta\beta^* & |\gamma|^2 + |\delta|^2 \end{pmatrix} = I$$

(M to be unitary matrix)

$$MM^\dagger = I$$

$$\Rightarrow \begin{aligned} |\alpha|^2 + |\beta|^2 &= 1 & \alpha\gamma^* + \beta\delta^* &= 0 & \gamma^* &= \frac{-\beta\delta^*}{\alpha} \\ |\gamma|^2 + |\delta|^2 &= 1 & \gamma\alpha^* + \delta\beta^* &= 0 & \Rightarrow \delta &= \frac{-\gamma\alpha^*}{\beta^*} \end{aligned}$$

also $|M| = +1$

$$\Rightarrow \alpha\delta - \beta\gamma = e^{i\theta} \quad (\text{How?}).$$

$\rightarrow |e^{i\theta}| = 1 \quad \{e^{i\theta} \text{ can be any complex } \underline{\text{no}}.$

$$\alpha \left(\frac{-\gamma\alpha^*}{\beta^*} \right) - \beta \left(\frac{-\beta^*\delta}{\alpha^*} \right)$$

$$\frac{-|\alpha|^2\gamma}{\beta^*} - |\beta|^2\gamma = -(|\alpha|^2 + |\beta|^2)\frac{\gamma}{\beta^*} = -\frac{\gamma}{\beta^*}$$

$$\delta = \frac{-\alpha^*\gamma}{\beta^*} = \alpha^*e^{i\theta} \quad \gamma = -\beta^*e^{i\theta}$$

The general (2×2) unitary matrix must be of the form,

$$\begin{pmatrix} \alpha & \beta \\ -\beta^*e^{i\theta} & \alpha^*e^{i\theta} \end{pmatrix}, \quad \text{with } |\alpha|^2 + |\beta|^2 = 1$$

No of real parameters $(\alpha_1 + i\alpha_2), (\beta_1 + i\beta_2) = 4$ parameters but $|\alpha|^2 + |\beta|^2 = 1$ makes it 3 parameters, we also have ' θ ' as 4th parameter again.

If, $|M| = +1$ (then $\theta = 0$) $\Rightarrow$ we have 3 parameters in $SU(2)$ where as original $U(2)$ has four parameters.

Hence in $SU(2)$ group, the most general matrix is

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} \quad \begin{array}{l} \text{with } |\alpha|^2 + |\beta|^2 = 1 \\ \text{or } \alpha_1^2 + \alpha_2^2 + \beta_1^2 + \beta_2^2 = 1 \end{array}$$

So the claim is now that we have two such unitary matrices for a element of $SO(3)$.

17 Module 13, Lecture 36

outline :-

- (1) 2 to 1 homomorphism b/w $SU(2)$ and $SO(3)$
- (2) The parameter space of $SU(2)$ and $SO(3)$
- (3) Double connectivity of $SO(3)$
- (4) The universal covering group of a lie group.
- (5) The groups $SO(n)$ and $spin(n)$
- (6) Tensor and spinnor representations.
- (7) parameter space of $U(n)$ and $SU(n)$
- (8) A bit about fundamental group (or first homotopy group) of a space

Note :- $SU(2)$ is simply connected and connected group.

we have already seen that parameter space of $SU(2)$ elements/group is four-dimensional.

$$\alpha_1^2 + \alpha_2^2 + \beta_1^2 + \beta_2^2 = 1$$

which represents hypersphere / 4-D sphere.

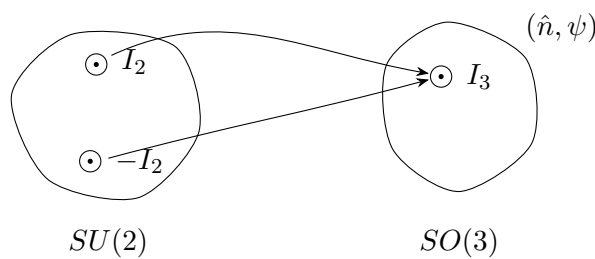
Just as we denote a 3-D sphere by S^2 we denote 4-D sphere by S^3 ($S \rightarrow$ stands for surface).

$S^2 \rightarrow$ surface of balloon

3-D surface — S^3 .

So S^3 is connected (i.e. every other point can be reached from any point without leaving the space). S^3 is also simply connected as any closed loop can be shrunk to a point. (absence of holes in S^3 surface/space).

* 2 to 1 homomorphism between $SU(2)$ and $SO(3)$.



(connected space. $SO(3)$ has $|R| = +1$; proper rotation)

$\{I_2, -I_2\}$ is kernel of $SU(2)$. It is set of elements which are mapped on to $I_{3 \times 3}$ in $SO(3)$.

$\{I_2, -I_2\}$ also form a group by themselves. (group of order 2)

$$I I = I, \quad I \cdot (-I) = -I, \quad (-I)(+I) = -I, \quad (-I)(-I) = I.$$

So they form a cyclic group. (Z_2).

$SO(3)$ is isomorphic (one to one connection) Not to $SU(2)$ but is isomorphic upto a sign ($\pm$).

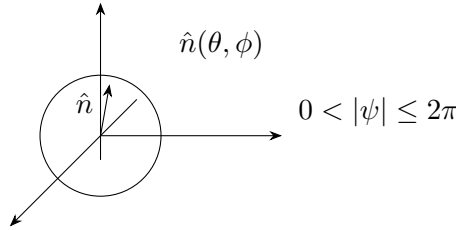
$$SO(3) \simeq SU(2)/Z_2 \rightarrow \text{order } \textcircled{2}$$

$\hookrightarrow$ cyclic group of order (2) $\{I, -I\}$ $\swarrow$ quotiented with cyclic group of order 2

neighborhood of I_2 is same as neighbourhood of I_3 but the global topology is different.

parameter space of $SO(3)$:-

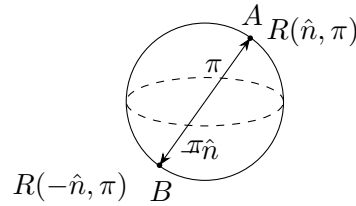
elements of $SO(3)$ are parameterized by $(\hat{n}, \psi)$ where $\hat{n}(\theta, \phi)$ is axis of rotation about $\hat{n}(\theta, \phi)$. This space can be realized by solid sphere of radius 2π . where $\hat{n}$ of point in sphere gives (θ, ϕ) and length of radius/vector provides amount of rotation.



But It is fact of 3-Dimension that Rotation about $\hat{n}$ by π is same as rotation about $(-\hat{n})$ by π

$$R(\hat{n}, \pi) = R(-\hat{n}, \pi)$$

So radius of π is sufficient.



points A and B must be same

So we can not represent this parameter space in 3-D. we need higher dimension to represent/realize the object.

There is also a 2-D object that can not be embeded in 3-D space like klein bottle.

Similerly we can not model this in 3-D space.

The space/parameter space of $SO(3)$ is connected but what kind of connectivity it have?

How to find connectivity of any such objects; weather they are simply connected, doubly connected ...?

what we do is to draw, & map loops on this space and if these loops can be shrunk to a point (contineously without leaving the space) then its simply connected. If these loops can not shrunk to point contineously then it means space is not simply connected and the connectivity depends on how many classes of such paths we can find.

“It will turn out that $SO(3)$ is doubly connected There are exactly two classes of closed paths which are inequivalent to each other”.

Example :-

lets look at simple example, by considering group of rotations in a plane (one generator).

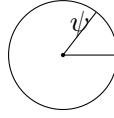
$$SO(2)$$

$$R = \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \quad (0 \leq \psi < 2\pi)$$

$$|R| = 1.$$

→ parameter space is one-dimensional (S^1) $\hookrightarrow$ parameter space of $SO(2) = S^1$

So all the rotations in $SO(2)$ are parameterized by points on circle (S^1).

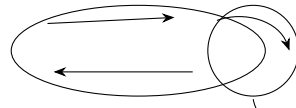


So, parameter space of $SO(2)$ is S^1 .

we can see that parameter space of $SO(2)$ i.e. S^1 is connected. (we can go from any point to another point in S^1 without leaving S^1 (continuously))

Is this parameter space (S^1) simply connected ? (No)

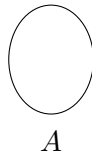
let us map figure A to circle (S^1).



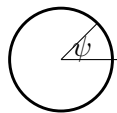
A

This loop can be shrunk to a point

But there is another way of mapping figure/object A on surface S^1 .



A



So $SO(2)$ is not simply connected

as,

This loop can not be shrunk to a point; without cutting it.

we can map it twice (2 times) $\bigcirc$ on the S^1 , and There is no way to unwind too.

we can also map it in opposite direction (-1 times) (-2 times) .. so on.

It means all closed paths, i.e. maps of loop on S^1 , can be put into classes and each classes is specified by the integer (corresponding to no. of times it winds around).

$$n = (\infty, -6, -5, -4, \dots, 0, 1, 2, \dots, \infty).$$

These classes are in correspondence with one to one integers but the integers form group under addition.

“which means, the classes of maps (closed loops on S^1), they themselves form a group called fundamental group of space”.

Note :- In general –

If we look at any space V and ask what are all distinct classes of maps of S^1 on to V which is denoted by $\pi_1(V)$ and it is called fundamental group of that space.

$$\pi_1(V) = \text{fundamental \underline{group} of that space}$$

or, (first homotopy group of space (V))

So what is group composition law of $\pi_1(V)$ –

- (1) If we wind around 6 times and then we wind loop further 7 times, total is something which winds around 13 times.

If this first homotopy group $\pi_1(V)$ has only one element (Identity element) then space is said to be simply connected.

If it contains more than one element then it is non-trivial case. (doubly connected, triply connected – infinitely connected).

So, $V = S^1$, i.e. $(SO(2))$

- (1) what is first homotopy group of $S^1 \Rightarrow \pi_1(S^1) = \text{group of integers}$

$$\pi_1(SO(2)) \sim \pi_1(S^1) = \underline{\mathbb{Z}}$$

($\hookrightarrow$ contains ∞ -elements ; so it is infinitely connected.)

- (2) first homotopy group of $S^2 = \pi_1(S^2)$

as any loop on surface S^2 can be shrunk to a point 

So, $\pi_1(S^2) = 0$ (group with one element).

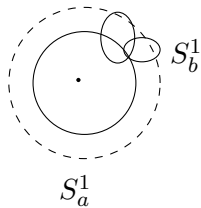
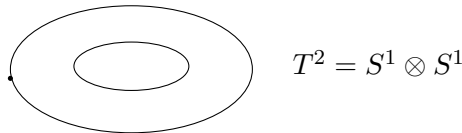
- (3) Similarly $\pi_1(S^n) = 0$ for $(n \geq 2)$ or $(n > 1)$

So $\pi_1(S^3) = 0 \Rightarrow \pi_1(SU(2)) = 0$

S^3 i.e. $SU(2)$ parameter space of $SU(2)$ is simply connected

$\Rightarrow SU(2)$ is simply connected. as $\pi_1(SU(2)) = \pi_1(S^3) = 0 \downarrow$ has only one element

- (4) $\pi_1(T^2)$ or $\pi_1(S^1 \otimes S^1)$



To wind path on torus we need to know how many times we wind around S_a^1 and S_b^1 so we need two integers.

So,

$$\pi_1(T^2) = \mathbb{Z} \otimes \mathbb{Z}$$

Includes ∞ -no of elements so Its connectivity is infinite.

(as there are infinite classes of closed paths that can not be reduced to each other).

Instead of a rotation matrix in 2-D.

$$R = \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \text{ ————— element of } SO(2)$$

let us associate with this rotation^(R) a complex no $e^{i\psi}$ ($e^{2\pi i} + 1 = 0$).

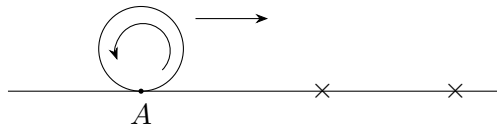
Then it is clear that there is one to one correspondance between 2×2 orthogonal matrices $SO(2)$ and points on unit circle. but points on unit circle also form a group, group of 1×1 matrices, and unitary matrices.

$$\begin{array}{ccc} \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \simeq e^{i\psi} & \text{(one to one correspondance)} & \\ \underline{SO(2)} & & \underline{U(1)} \end{array}$$

as we have seen $\pi_1(S^1)$ is infinitely connected.

we can do the following :-

- (1) Take our circle S^1 and roll it out on real number line.



point 'A' will mapped on Real no line infinite times.

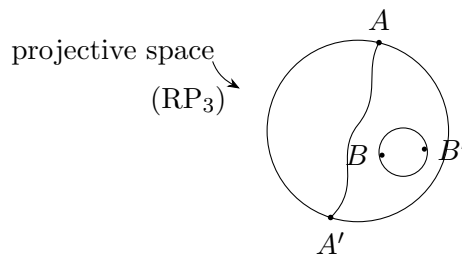
So Real number line provide cover for S^1 .

$$U(1) = SO(2) \simeq R^1/Z \quad \text{just as} \quad SO(3) \simeq SU(2)/Z_2$$

→ R^1 is called universal covering group of $SO(2)$, because it is simply connected on infinite no. real line.

→ Every lie group has got a universal covering group, which is simply connected; (whose parameter space is simply connected).

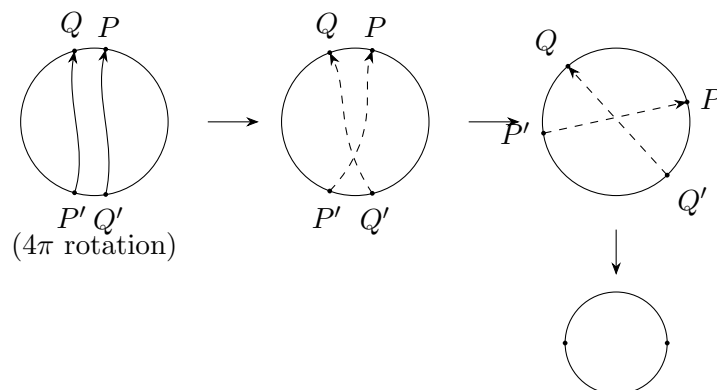
For parameter space of $SO(3)$:-



- (1) $A'A$ is closed path but can not shrung to a point on 2π rotation ($\pi + \pi$).
- (2) path BB' in sphere S^2 is closed path and can be shrunk to a point.

Note

→ $A'A$ can also be shrunk to a point by following trick



Therefore there are only two classes of closed paths which can be shrunk to point. (1) ($\leq 2\pi$) rotation (2) 4π rotation

↑ (general loops) (like BB')

These are objects in this space (3-D) which will not come back to their original value from 2π rotation but guaranteed to come back at 4π rotation. (spinors). of coordinate system/axes

$$\pi_1(SO(3)) = \pi_1(RP_3) = \pi_1(SU(2)/Z_2) = Z_2$$

So, $SO(3)$ is doubly connected. (parameter space of)

Q) what is connectivity of space $SO(n)$ for ($n > 3$).

$$SO(n) \simeq spin(n)/Z_2 \rightarrow \text{doubly connected}$$

$$Spin(3) = SU(2) \quad (\text{only for } n = 3)$$

$$Spin(6) = SU(4) \quad spin(n) \text{ is simply connected.}$$

(parameter space of rotations in higher dimensions $n \geq 3$ is also doubly connected)

No of parameters in $SU(n) = n^2 - 1$

No of _____ in $U(n) = n^2$

($SO(n)$ has nC_2 parameters. and analogue of $SU(2)$ for $SO(n)$ is $spin(n)/Z_2$ & it so happens that $spin(3) = SU(2)$ for $n > 3$)

Note :-

It turns out that

$$\pi_1(SU(n)) = 0 \quad (\text{simply connected})$$

i.e. parameter space of $SU(n)$ is simply connected. for any n .

universal covering group for $SO(3)$ is $SU(2)$.

Idea of universal covering group is extremely useful, because it ignores the complications in global topology of group space and focuses on simply-connected portions.

parameter space of $SU(n)$: If it is a $(n \times n)$ matrix it should have n^2 elements but with complex no entries it has $2n^2$ elements $U^\dagger U = I$ gives n^2 conditions, which reduces parameters to n^2

and $|U| = +1$ also decreases no of parameters to $\underline{n^2 - 1}$. ($U(n)$ has n^2 elements/parameter but $SU(n) = n^2 - 1$ elements/parameter)

local property of group/parameter space is decided by generators. (of infinitesimal rotation)

let us write general generator for $SO(n)$:-

$$J_{jk} (= -J_{kj})$$

$$[J_{jk}, J_{lm}] = \delta_{jl}J_{km} + \delta_{km}J_{jl} - \delta_{jm}J_{kl} - \delta_{kl}J_{jm}$$

For (3-D), $J_{12} \rightarrow J_3, J_{23} \rightarrow J_1, J_{31} \rightarrow J_2$

$\downarrow$ (rotation in 1-2 plane)

(Q) What is $SU(1)$?

Ans. $U(1)$ has 1 element/parameter

whereas $SU(1)$, $|U| = +1$, requires 0 parameters.

so its constant 1×1 matrix with $|M| = 1$

$SU(1)$ has one element only = $[1]$

Note : In general

$$U(n) = U(1) \otimes SU(n)$$

$$\pi_1(U(n)) = \pi_1(U(1)) \otimes \underbrace{\pi_1(SU(n))}_{=0}$$

$$= Z$$

$U(n)$ is infinitely connected.

18 Group Theory: $SO(3)$ and $SU(2)$ (from Quantum Mechanics, Lec 25)

Connection between $SO(3)$ and $SU(2)$:-

We know that in 3-Dimensional space, I would like to generate rotations (linear transformation of the coordinates, which leaves a point (origin) unchanged. Its linear and homogeneous. also $|M_T| = 1$. (M_T = transformation matrix). These three conditions suffice to define rotations and they form a group, this group is called $SO(3)$.

where 'S' stands for $|M_T| = +1$ (special)

'O' stands for orthogonal (because the transformations has to be orthogonal to ensure that distance b/w two points remain unchanged, $|x_1 - x_2| = |x'_1 - x'_2|$).

'3' : comes from 3-dimensions.

The set of 3×3 matrices, which are orthogonal, with determinant +1, they form a group. These matrices form a representation of the abstract group of rotations.

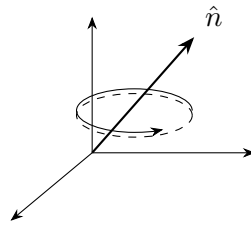
Rotations are operators in their abstract, but they are explicitly represented by set of orthogonal, (3×3) matrices with $\det M_T = +1$.

Parameter space of the set of rotations

what is parameter space of set of rotations ?

In other words what are values of angles that specifies the possible rotations. Now these can be specified in many ways, as you know you can go to one coordinate system to a rotated one by specifying three Euler angles, but you can specify those Euler angles in many different ways, they all turn out to be equivalent with each other but there is nothing unique about it.

The most convenient way of specifying these rotations is to say that the rotations occur about some axis ($\hat{n}$) in space with respect to some fixed coordinate system through a certain amount of rotation. (angle).



$R(\hat{n}, \psi)$ where $\hat{n}$ is specified by (θ, ϕ) , so that

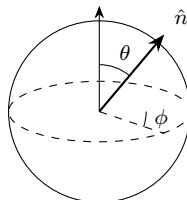
$$\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

$$0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi, \quad 0 \leq \psi \leq 2\pi$$

<u>equation</u>	<u>Locus</u>
$x = \text{const}$	plane
$R = \text{const w.r.t}$	sphere
$\theta = \text{const.}$	half cone
$\phi = \text{const}$	half plane

Now we can specify Rotation using three variables (θ, ϕ, ψ) and we can put these sets (θ, ϕ, ψ) as point in certain space.

Lets model (θ, ϕ, ψ) on sphere (solid sphere).

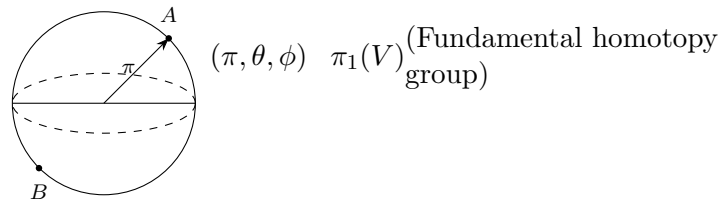


If we cleverly choose $R = 2\pi$ then length of arrow can represent amount of rotation (ψ).

points in such solid sphere coorespondes to only one rotation. (But here point of origin & all points on sphere should be same, so this representation is not good).

But it is a fact of 3 Dimencion that rotating an object through π about $\hat{n}$ is same as rotating about $(-\hat{n})$ through π .

so $R = \pi$ is enough.



But there is further complication, 'A' and 'B' are mathematically identical points; in this parameter space because both corresponds to rotation by π about $\hat{n}$ axis or its opposite and they physically cooresponds to the same rotation.

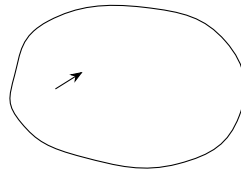
So the parameter space of $SO(3)$ is complicated. It is a solid sphere, its radius is π . Such a space can not be represented in 3-Ecludien dimensions. But we can look at all of its mathematical property given this property.

Connectivity of the parameter space

This space is connected. A connected space is one where you can go from any point in this space to any other point in space continuously, without leaving the space. so this space is certainty connected.

But it is not simply connected, a simply connected space is one where any path, any contineous close path in the space can be contineously deformed or shrunk to a point without leaving the space.

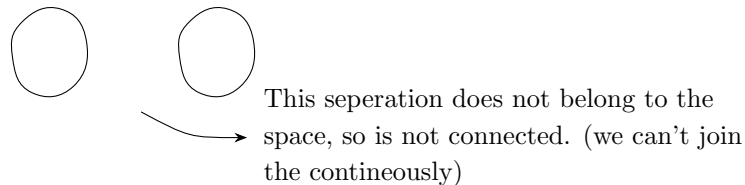
connected and simply connected :



any closed path can be shrunk continuously to a point without leaving this paper. so this space is connected as well as simply connected.

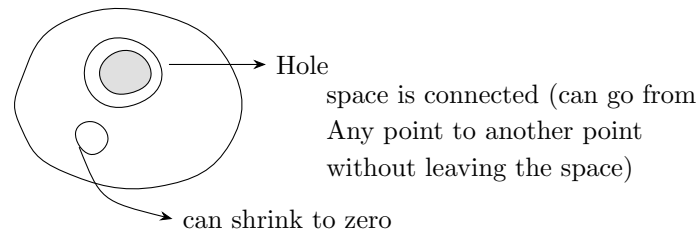
Not connected, Not simply connected

so a space for example has pieces seperated.



connected but not simply connected :-

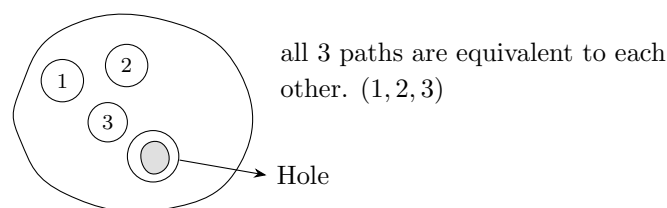
For example :-



path does not shrink to zero. so space is not simply connected.

(Q) If it is Not simply connected, what kind of connectivity it have ?

(1) Paths are equivalent to each other if you can deform one to the other, continuously.



Path ④ is not equivalent to path (1, 2, 3) as path ④ can't be shrunk to a point.

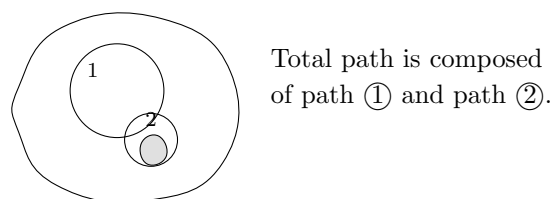
(Q) what are classes of paths that can all be deform to each other.

All paths that can be deformed to each other forms a class known as equivalence class. (They are all equivalent to each other).

Fundamental homotopy group $\pi_1(V)$

(Q) what are different equivalence classes of paths that you have in a space.

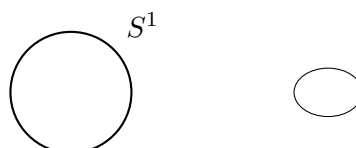
It turns out by obvious rule of composition of class, these equivalence class forms elements of a group. This group is called fundamental group of space.



and together they form a element of a group (3rd path in a group), $\pi(V')$.

example :-

Suppose the space is S^1 (1-D circle). what are equivalence classes of paths ?

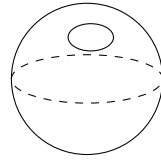


all these paths can be shrunk to a point, while this complete circular path can no longer be shrunk to a point.

Also winding of such path can also not be able to shrink to a point, so

$$\pi_1(S^1) = \mathbb{Z} \quad (\text{Integers})$$

what is $\pi_1(S^2)$:-



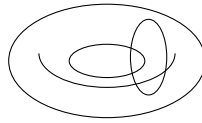
all paths can be shrunk to a point without leaving the space (2-D space).

“You can not lasso basket ball, rubber will slip off and shrink to ($\sim$ zero) again”.

$$\pi_1(S^2) = \{1\} \quad \text{or} \quad \{e\} = 0 \quad (\text{only one element}).$$

Similarly, $\pi_1(S^3) = 0$.

$\pi_1(T^2)$:-

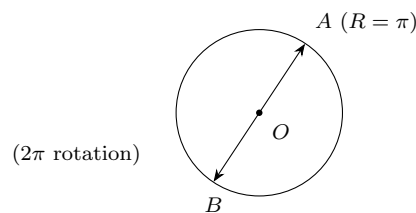


so that

$$T^2 \text{ (torus)} = S^1 \otimes S^1$$

$$\pi_1(T^2) = \mathbb{Z} \otimes \mathbb{Z}$$

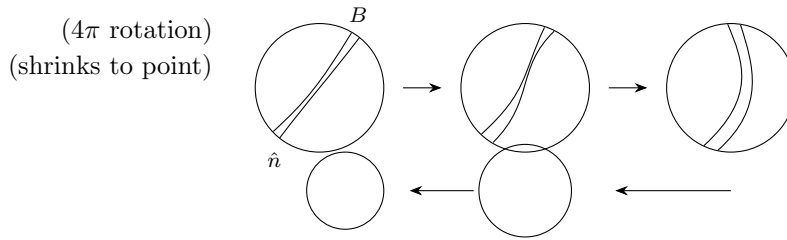
Points on sphere :- (Rotation group points)



since points A and B are identical this represents a closed path in that space (Hard to imagine, lol).

and it is completely distinct from other class of closed paths (which can be shrunk to a point). Because this path (B - O - A closed path) can not be shrunk to a point.

But I can rotate it again (2π angle ($\pi + \pi$)) and shrink to zero by following trick.



so there actually exists 2 class of objects in this parameter space, those that come to themselves after a rotation of 2π and other objects which come to themselves after rotation of 4π .

Tensor and spinor representations

This is the origin of $\frac{1}{2}$ integer valued representations of rotation group.

$$\begin{aligned} j &= 0, 1, 2, 3 \dots & (\text{tensor representations}) \\ j &= \frac{1}{2}, \frac{3}{2}, \frac{5}{2} \dots & (\text{spinor representations}) \end{aligned}$$

so we know that in normal tensor representations

Tensor of rank '0' is a scalar.

Tensor of rank '1' is a vector. $\dots$ and so on.

The spinors sort of speak interpolate between these.

spinors change sign on 2π rotation, so they need one more 2π rotation to get same orientation of object.

$\pi_1(SO(3))$ is doubly connected $= \mathbb{Z}_2$. (just two elements in group).

Lec 25 : Representation of $SO(3)$ in $SU(2)$

let us complete the correspondence that was supposed to bring out between group of 3×3 rotations and certain 2×2 matrixes ($SU(2)$).

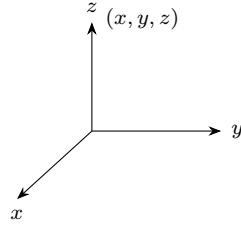
We recognize that rotations in 3×3 space R is specified by unit vector $\hat{n}(\theta, \phi)$ and ψ (amount of rotation) or $R(\hat{n}, \psi)$ led to parameter space (specified by $\hat{n}, \psi$) which was not simply connected and was doubly connected and there are two classes of closed paths in this space which could not be reduced to each other (two inequivalent classes of closed paths).

This was the main reason for the fact that we had single and double valued representations of the rotation group because we also saw that the second class of closed paths could be shrunk to a point (if we do such two rotations (4π total) instead of 2π).

All those representations of the rotation group which transform such that when you went to a rotation of 2π , you return to the original state, are called tensor representations and others (rotation of 4π) were called spinor representations.

Representation of 3×3 Rotation group $SO(3)$ in 2×2 matrix $SU(2)$

We normally represent position by three parameters in 3-D.



so that $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

instead of writing $\vec{r}$ in column matrix we can express another representation of position

$$\vec{r} \cdot \vec{\sigma} = x \sigma_x + y \sigma_y + z \sigma_z = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

So if we have $\vec{r} \cdot \vec{\sigma}$ we can work back and find x, y, z as, (also, any 2×2 matrix can be written in terms of unit matrix and pauli matrixes σ_i)

So once $\vec{r} \cdot \vec{\sigma}$ is given we can work back and find vector $\vec{r}$ itself.

So there is a correspondence between vectors $\vec{r}$ and matrices $\vec{r} \cdot \vec{\sigma}$.

like wise we used Rotation matrix ($g \in SO(3)$) (3×3 matrix)

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = g \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \vec{r}' = g \vec{r}$$

in exactly same way we have 2×2 matrix (U) corresponding to element ' g ' such that $U(\vec{r} \cdot \vec{\sigma})U^{-1} = \vec{r}' \cdot \vec{\sigma}$.

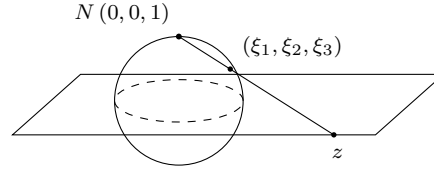
Therefore this ' U ' is representative of rotation matrix ' g '.

(Q) what kind of relation is there between ' g ' and ' U ' ?

- ' U ' should also be parametrised by (θ, ϕ, ψ) or $(\hat{n}, \psi)$
- It should be a 2×2 matrix.
- These matrices will turn out to be unitary with $|U| = +1 \therefore U \in SU(2)$.
- The elements of $SO(3)$ were real but elements of $SU(2)$ can also be complex numbers.
- The advantage of using $SU(2)$ over $SO(3)$ is that group $SU(2)$ is simply connected. (its like a sphere).

Lec 25 (continued) : Stereographic projection and rotations

we can use stereographic projection to map surface of unit sphere to a plane (complex plane).



point on sphere (ξ_1, ξ_2, ξ_3) . For every point on sphere we have a point on plane and vice versa.

example :- The North pole is mapped on to ' ∞ '. The south pole is mapped on to origin. Equator $(\xi_1^2 + \xi_2^2 = 1)$ remains mapped to $(|z| = 1)$.

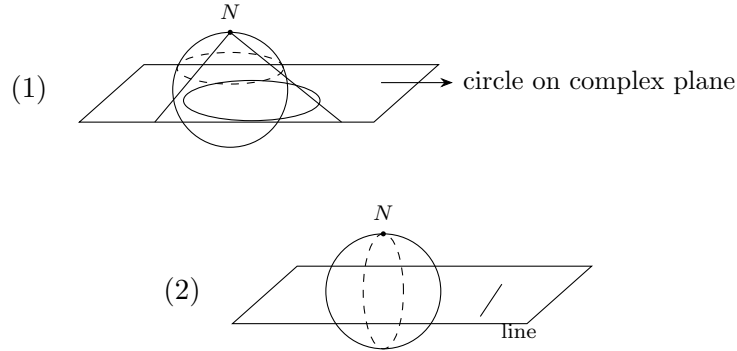
$$\xi_1 = \frac{2x}{|z|^2 + 1}, \quad \xi_2 = \frac{2y}{|z|^2 + 1}, \quad \xi_3 = \frac{|z|^2 - 1}{|z|^2 + 1}$$

$$(\xi_1^2 + \xi_2^2 + \xi_3^2 = 1) \quad \text{unit sphere}$$

So we map this Riemann sphere on this complex plane using stereographic projection, which are; $x = \frac{\xi_1}{1 - \xi_3}$, $y = \frac{\xi_2}{1 - \xi_3}$.

This projection has many interesting property; example : It preserves/maps circles on spheres to either circles or straight lines.

Any latitude is mapped on the circle with cocentric origin, any longitude is mapped on the straight line passing through the origin. (does it pass through origin ?)



Rotation in 3-D space and the Riemann sphere

Now Rotation in 3-D space would corresponds to rotating Riemann sphere and what does it do on the plane ? It induces of course a transformation on plane as well. It is easy to check (prove yourself) that if we rotate in physical 3-D space (rotate in x - y plane about $\hat{z}$ axis) then the rotation matrix is (in simple form).

if, $\hat{n} = \hat{e}_z$, $\psi = \gamma$

$$g \in SO(3) = \begin{pmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If we put that back to Riemann sphere and ask what that transformation corresponds to (the rotation in ξ_1 - ξ_2 plane about ξ_3 axis) Then it is easy to see that

$$U = \pm \begin{pmatrix} e^{i\gamma/2} & 0 \\ 0 & e^{-i\gamma/2} \end{pmatrix}$$

So the Transformation is represented as, $U(\vec{r} \cdot \vec{\sigma})U^{-1} = \vec{r}' \cdot \vec{\sigma}$, where

$$\vec{r} \cdot \vec{\sigma} = x\sigma_1 + y\sigma_2 + z\sigma_3 = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}$$

So the counterpart of 'g' is to the pair of matrices

$$U = \pm \begin{pmatrix} e^{i\gamma/2} & 0 \\ 0 & e^{-i\gamma/2} \end{pmatrix}$$

Similarly; The No transformation (i.e. No rotation at all) would corresponds to ($\gamma = 0$)

$$U = \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \pm I$$

$$U(\vec{r} \cdot \vec{\sigma})U^{-1} = I(\vec{r} \cdot \vec{\sigma})I^{-1} = \vec{r} \cdot \vec{\sigma} = \vec{r}' \cdot \vec{\sigma}$$

Unitary and unimodular condition

The requirement that $|r'| = |r|$ suffices that these matrices (U) must be unitary and unimodular.

$$\begin{aligned} \text{Unitary : - } & UU^\dagger = I = U^\dagger U \quad \text{or} \quad (U^\dagger = U^{-1}) \\ \text{unimodular : - } & |U| = +1 \end{aligned}$$

Important : All 2×2 matrices, which satisfies these conditions would represent physical rotations in 3-D space.

A general such 2×2 matrix has $\begin{pmatrix} a + bi & c + di \\ e + fi & g + ih \end{pmatrix}$ 8 reel elements (a, b, c, d, e, f, g, h).

$UU^\dagger = I$ provides 4 conditions and further $|U| = +1$ provides one more conditions. (So parameter left = $8 - 5 = 3$)

so we need ($8 - 5 = 3$) 3 adependent parameters to specify rotation in physical 3-D space which is exactly what we need in $SO(3)$, $R(\hat{n}, \psi)$, $R(\theta, \phi, \psi)$.

upon solving one gets

$$\begin{aligned} U &= \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \quad (|\alpha|^2 + |\beta|^2 = 1) \quad \text{or} \quad \alpha\alpha^* + \beta\beta^* = +1 \\ &(\alpha_1^2 + \alpha_2^2 + \beta_1^2 + \beta_2^2 = 1), \quad \alpha = \alpha_1 + i\alpha_2, \quad \beta = \beta_1 + i\beta_2 \end{aligned}$$